AIRFRAME PNEUMATIC DEICING SYSTEMS – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–11–00–710–801
Operational Check of the Deicing System Air Supply (CMR#301100–101)

1. General

   A. The maintenance procedure that follows is for the operational check of the deicing system air supply.

2. Job Set–Up Information

Subtask 30–11–00–946–005

   A. Reference Information

REFERENCE	DESIGNATION
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown

3. Procedure

Refer to Figure 501.

Subtask 30–11–00–710–002

   A. Do the operational check of the deicing system air supply as follows:

      (1) On the ICE PROTECTION panel, make sure that the BOOT AIR switch is set to the NORM position and the AIRFRAME MANUAL SELECTOR to OFF.

      (2) Start the left engine (Refer to TASK 71–00–00–868–801).

      (3) Make sure the AC generator toggle switch located on the AC control panel is set to GEN 1.

      (4) Move the condition lever to the MIN 850 detent.

      (5) Move the power lever to the FI position detent.

      (6) Set the two PACKS switches on the air conditioning panel to the AUTO position.

      (7) Set the BLEED 1 switch to the ON position, the BLEED 2 switch to the OFF position and the BLEED selector to the MAX position.

      (8) Let the system stabilize for three minutes.

      (9) On the copilot side panel, make sure that pressure indications 1 and 2 on the DEICE PRESS gauge increase (to approximately 18 psi).

      (10) Shutdown the left engine (Refer to TASK 71–00–00–868–802).

(11) Start the right engine (Refer to TASK 71–00–00–868–801.

(12) Move the condition lever to the MIN 850 detent.

(13) Move the power lever to the FI position detent.

(14) Set the two PACKS switches on the air conditioning panel to the AUTO position.

(15) Set the BLEED 2 switch to the ON position, the BLEED 1 switch to the OFF position and the BLEED selector to the MAX position.

(16) Let the system stabilize for three minutes.

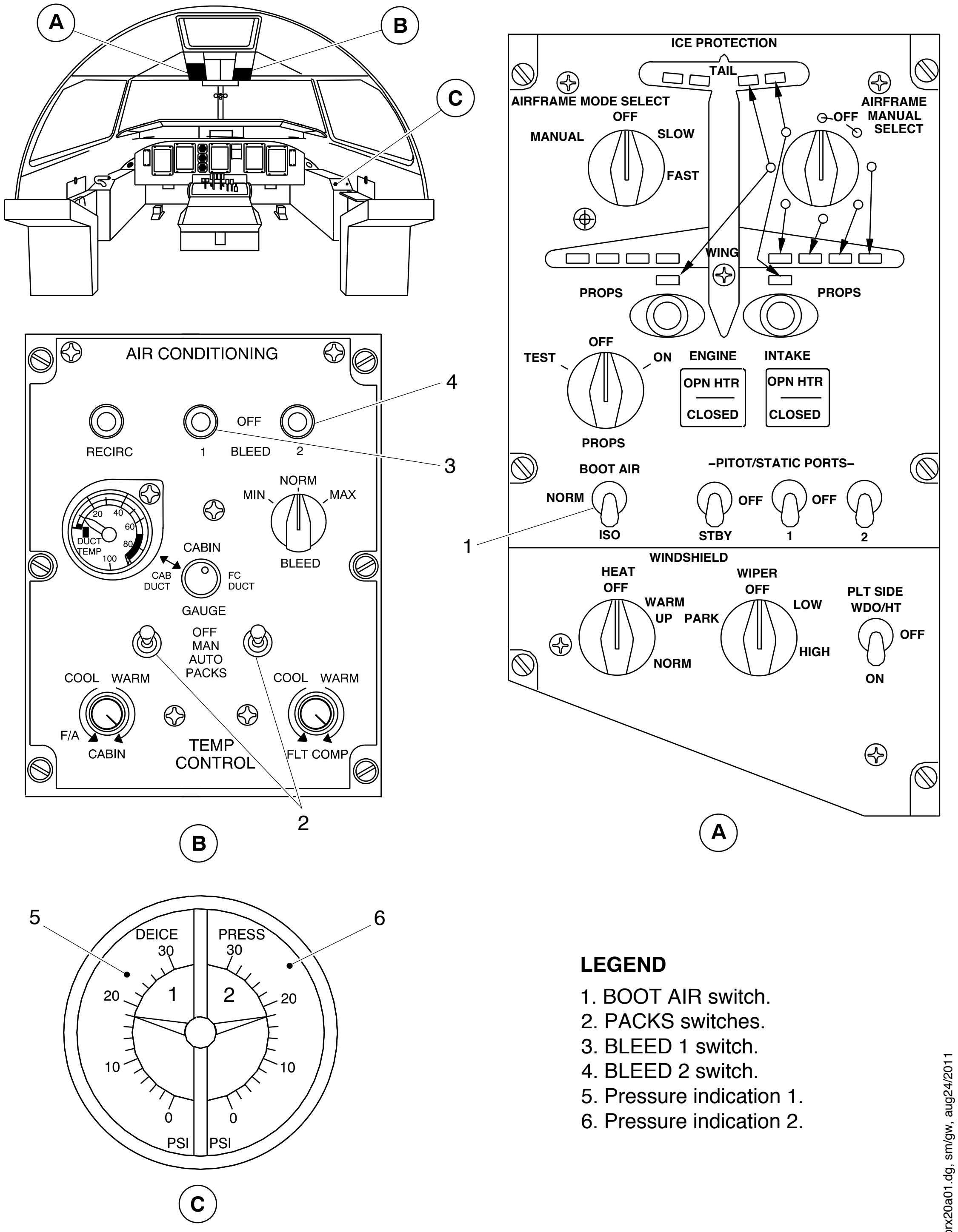
(17) On the copilot side panel, make sure that pressure indications 1 and 2 on the DEICE PRESS gauge increase (to approximately 18 psi).

(18) Shutdown the right engine (Refer to TASK 71–00–00–868–802).

4. Close Out

Subtask 30–11–00–941–001

A. Remove all tools, equipment, and unwanted materials from the work area.



LEGEND

1. BOOT AIR switch.
2. PACKS switches.
3. BLEED 1 switch.
4. BLEED 2 switch.
5. Pressure indication 1.
6. Pressure indication 2.

Deicing System – Operational Check
Figure 501

**ON A/C ALL

TASK 30–11–00–710–802
Operational Check of the Deicing System Rear Fuselage Heated Check Valve (MRB# 301100–203)

1. General

   A. The maintenance procedure that follows is for the operational check of the deicing system heated check valves. There are two heated check valves. They are installed in the rear fuselage.

2. Job Set–Up Information

Subtask 30–11–00–946–002

   A. Reference Information

REFERENCE	DESIGNATION
TASK 71–00–00–868–801	Engine start
TASK 71–00–00–868–802	Engine shutdown

3. Procedure

Refer to Figure 502.

Subtask 30–11–00–710–001

   A. Do the operational check of the heated check valves as follows:

      (1) On the ICE PROTECTION panel, set the BOOT AIR switch to the ISO position.

      (2) Start the left engine (Refer to TASK 71–00–00–868–801).

      (3) Make sure the AC generator toggle switch located on the AC control panel is set to GEN 1.

      (4) Move the condition lever to the MIN 850 detent.

      (5) On the copilot side panel, make sure that pressure indication 2 does not increase on the DEICE PRESS gauge.

      (6) On the ICE PROTECTION panel, set the AIRFRAME MODE SELECT rotary switch to the MANUAL position.

      (7) On the ICE PROTECTION panel, set the AIRFRAME MANUAL SELECT rotary switch to select the inner horizontal stabilizer boot set.

      NOTE: Make sure that the AIRFRAME MANUAL SELECT rotary switch stays in a position for a minimum of 6 seconds before you set it to a new position.

      - Make sure that the related set of boots inflate.
      - On the ICE PROTECTION panel, make sure that the related green advisory lights come on.

(8) On the ICE PROTECTION panel, set the AIRFRAME MANUAL SELECT rotary switch to the OFF position.

(9) Shutdown the left engine (Refer to TASK 71–00–00–868–802).

(10) Start the right engine (Refer to TASK 71–00–00–868–801).

(11) Make sure the AC generator toggle switch located on the AC control panel is set to GEN 2.

(12) Move the condition lever to the MIN 850 detent.

(13) On the copilot side panel, make sure that pressure indication 1 does not increase on the DEICE PRESS gauge.

(14) On the ICE PROTECTION panel, set the AIRFRAME MANUAL SELECT rotary switch to select the outer horizontal stabilizer boot set.

NOTE: Make sure that the AIRFRAME MANUAL SELECT rotary switch stays in a position for a minimum of 6 seconds before you set it to a new position.

- Make sure that the related set of boots inflate.
- On the ICE PROTECTION panel, make sure that the related green advisory lights come on.

(15) On the ICE PROTECTION panel, set the AIRFRAME MANUAL SELECT rotary switch to the OFF position.

(16) On the ICE PROTECTION panel, set the AIRFRAME MODE SELECT rotary switch to the OFF position.

(17) On the ICE PROTECTION panel, set the BOOT AIR switch to the NORM position.

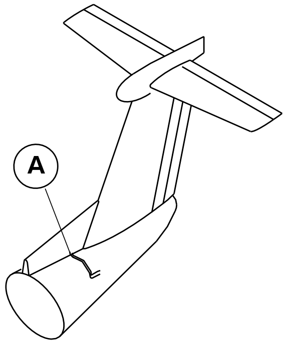
4. Close Out

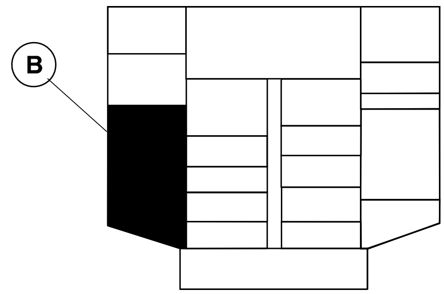
Subtask 30–11–00–861–005

A. Shutdown the right engine (Refer to TASK 71–00–00–868–802).

Subtask 30–11–00–941–007

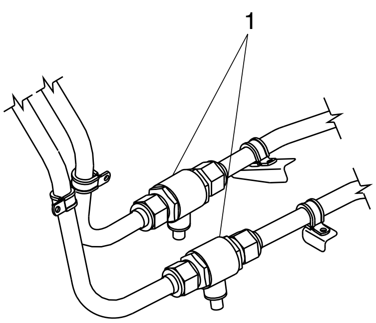
B. Remove all tools, equipment, and unwanted materials from the work area.





OVERHEAD CONSOLE

LEGEND
1. Rear fuselage heated check valves.
2. Boot air switch.
3. Airframe mode select rotary switch.
4. Airframe manual select rotary switch.



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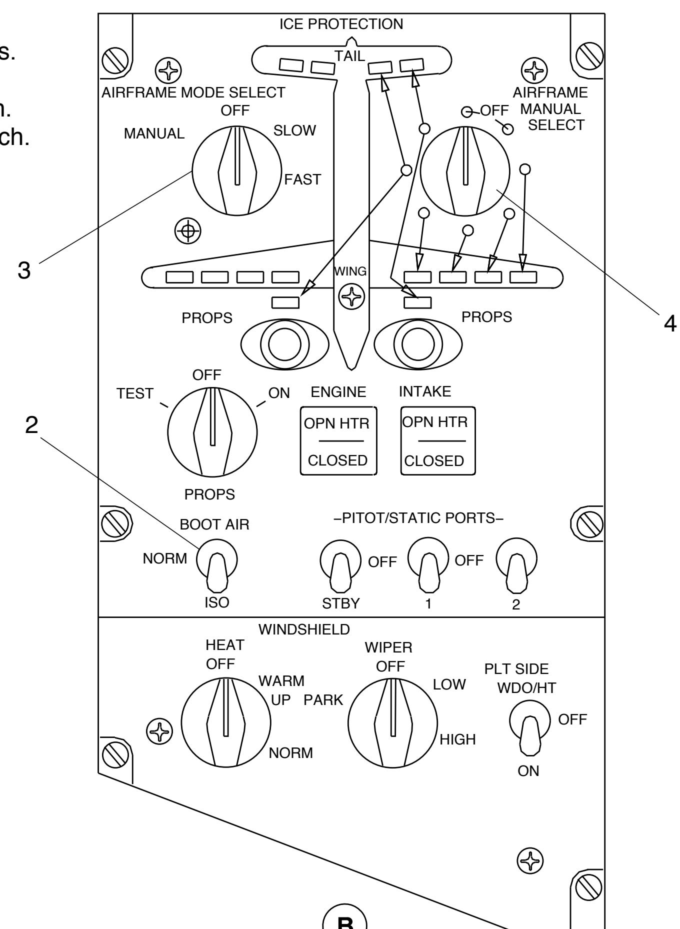

B

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De–icing System Rear Fuselage Heated Check Valve – Operational Check
Figure 502

**ON A/C ALL

TASK 30–11–00–710–803
Operational Check of the Deicing System Isolation Shut–off Valve (MRB# 301100–204)

1. General

   A. The maintenance procedure that follows is for the operational check of the deicing system isolation shut–off valve.

2. Job Set–Up Information

Subtask 30–11–00–946–015

   A. Reference Information

REFERENCE	DESIGNATION
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown

3. Procedure

Refer to Figure 503.

Subtask 30–11–00–710–007

   A. Do the operational check of the deicing system isolation shut–off valve as follows:

      (1) On the ICE PROTECTION panel, set the BOOT AIR switch to the ISO position.

      (2) Start the right engine (Refer to TASK 71–00–00–868–801).

      (3) Make sure the AC generator toggle switch located on the AC control panel is set to GEN 2.

      (4) Move the conditon lever to the MIN 850 detent.

      (5) On the copilot side panel, make sure that pressure indication 2 increases and pressure indication 1 does not increase on the DEICE PRESS gauge.

      NOTE: It is correct for the DEICE PRESS caution light to come on during this step.

      (6) On the ICE PROTECTION panel, set the BOOT AIR switch to the NORM position. On the copilot side panel, make sure that pressure indication 1 increases on the DEICE PRESS gauge.

      NOTE: It is correct for the DEICE PRESS caution light to go out during this step.

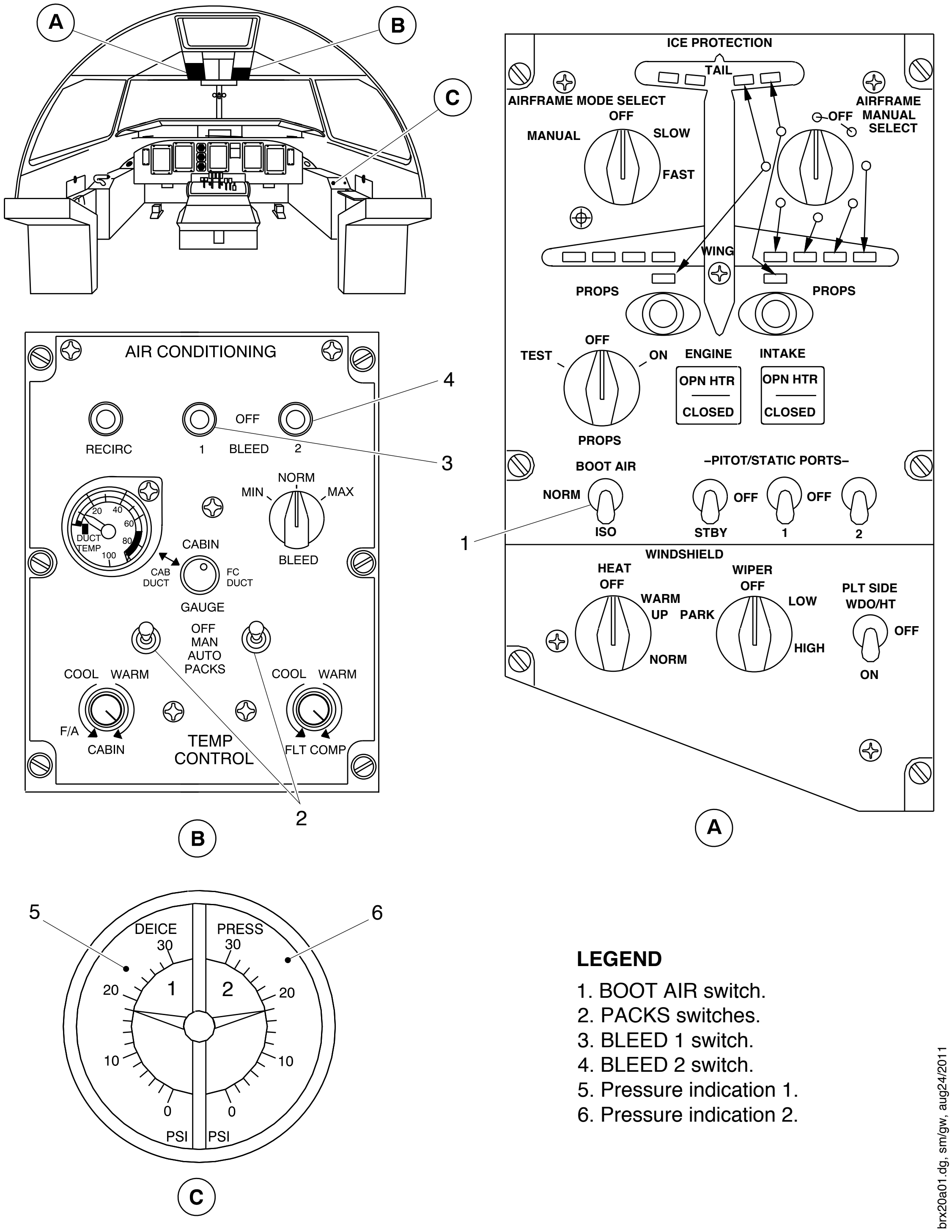
4. Close Out

Subtask 30–11–00–861–039

A. Shutdown the right engine (Refer to TASK 71–00–00–868–802).

Subtask 30–11–00–941–006

B. Remove all tools, equipment, and unwanted materials from the work area.



LEGEND

1. BOOT AIR switch.
2. PACKS switches.
3. BLEED 1 switch.
4. BLEED 2 switch.
5. Pressure indication 1.
6. Pressure indication 2.

Deicing System – Operational Check
Figure 503

**ON A/C ALL

TASK 30–11–00–710–804
Operational Check of the Deicing System in the Manual Mode (MRB# 301100–205)

1. General

   A. The maintenance procedure that follows is for the operational check of the deicing system in the manual mode. This task also incorporates the operational check of the airframe deicing equipment heating in manual mode (MRB #301100–206) and the operational check of the deicing hold down system (MRB #301100–202).

   B. All deicing boots can be pressurized by connecting an external regulated supply of dry air to either the left or the right engine P3.0 HP to high–pressure shut–off valve (HPSOV) duct (also referred to as the P3.0 duct in these procedures). The deicing system pressurization in this procedure is done by connecting the external air supply to the P3.0 duct on the left engine. Differences for the right engine are identified.

   C. An alternate procedure to pressurize the boots can be done by disconnecting the inlet line to the Pressure Regulating Valve (PRV) and connecting an external regulated supply of dry air to either the left or right engine PRV inlet line. The deicing system pressurization in this procedure is done by connecting the external air supply to the left engine PRV. Differences for the right engine are identified.

   D. The external pneumatic supply can be connected to either the P3.0 duct or the PRV inlet for this inspection procedure.

2. Job Set–Up Information

Subtask 30–11–00–943–016

   A. Tools & Equipment

(1)	Commercially Available	Wrench, Torque 0 to 500 lbf in (0 to 56.5 N·m)
(2)	Commercially Available from Innovative Aircraft Solutions LLC. USA	Alternate Bleed Air Source (ABAS) Kit 63605–001 (or equivalent)
(3)	Commercially Available	Dry air/shop air supply
(4)	Commercially Available	Maintenance stands

Subtask 30–11–00–946–001

B. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize the Electrical System
TASK 24–00–00–861–802	De–energize the Electrical System
TASK 24–00–00–910–801	Electrical/Electronic Safety Precautions
TASK 24–00–00–910–802	Electrostatic Discharge Safety Precautions
TASK 36–11–00–710–802	Operational Test of the Bleed Air System
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown
TASK 71–10–06–010–801	Opening of the Forward Side Door
TASK 71–10–06–410–801	Closing of the Forward Side Door

3. Job Set–Up

Subtask 30–11–00–910–001

A. Obey all the electrical/electronic safety precautions (Refer to TASK 24–00–00–910–801).

Subtask 30–11–00–910–002

B. Obey all the electrostatic discharge safety precautions (Refer to TASK 24–00–00–910–802).

Subtask 30–11–00–861–016

WARNING: MAKE SURE YOU TELL ALL PERSONS IN THE AREA OF THE AIRCRAFT THAT YOU WILL ENERGIZE THE AIRCRAFT ELECTRICAL SYSTEM, BEFORE YOU TURN ON THE ELECTRICAL SYSTEM. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

C. Energize the electrical system (Refer to TASK 24–00–00–861–801).

Subtask 30–11–00–010–001

D. Open the access door that follows:

REFERENCE	DESIGNATION
311AB	Aft Fuselage Door

Subtask 30–11–00–010–013

E. If the air supply is to be connected to the P3.0 duct on the left engine, open the access door that follows:

REFERENCE	DESIGNATION
412BR	Door

Subtask 30–11–00–010–014

F. If the air supply is to be connected to the P3.0 duct on the right engine, open the access door that follows:

REFERENCE	DESIGNATION
422BR	Door

Subtask 30–11–00–010–015

G. If the air supply is to be connected to the PRV on the left engine, open the access door that follows (Refer to TASK 71–10–06–010–801):

REFERENCE	DESIGNATION
412AR	Door

H. If the air supply is to be connected to the PRV on the left engine, open the access door that follows:

REFERENCE	DESIGNATION
412CT	Fairing

Subtask 30–11–00–010–016

I. If the air supply is to be connected to the PRV on the right engine, open the access door that follows (Refer to TASK 71–10–06–010–801):

REFERENCE	DESIGNATION
422AR	Door

J. If the air supply is to be connected to the PRV on the right engine, open the access door that follows:

REFERENCE	DESIGNATION
422CT	Fairing

Subtask 30–11–00–868–001

K. Option 1: If you do the deicing system pressurization with the engine, start the right engine (Refer to TASK 71–00–00–868–801).

Subtask 30–11–00–010–017

Refer to Figure 506.

L. Option 2: If you connect the air supply to the P3.0 duct on the left nacelle, do as follows:

NOTE: The procedures are identical for the P3.0 duct on the left and right engine.

(1) Disconnect the inlet end (1) of the P3.0 duct (2) from the engine. Retain the coupling (3) for adapter installation.

(2) If required, loosen the coupling (4) as necessary for access to install the air supply adapter.

(3) Install the air supply adapter P/N 63605–104 onto the P3.0 duct (2) using the coupling (3). Torque the coupling 70 to 90 lbf in (7.9 to 10.17 N·m).

(4) If loosened for access, tighten and torque the coupling (4) 70 to 90 lbf in (7.9 to 10.17 N·m).

(5) Connect the regulator P/N 63605–101 to the air supply adapter 63605–104 using the kit–supplied air hose P/N 63605–105.

(6) Connect the shop air supply to the regulator P/N 63605–101.

(7) Set the ABAS kit regulated pressure to 75.0±1.0 psig.

Subtask 30–11–00–010–018

Refer to Figure 507.

M. Option 3: If you connect the air supply to the PRV inlet on the left nacelle, do as follows:

NOTE: The procedures are identical for the PRV on the left and right engine.

(1) Disconnect the end of line (1) at the PRV inlet.

(2) For access to the PRV inlet, rotate the end of line (1) away from the PRV inlet. If required, loosen the clamps (2) as necessary.

(3) Install a blank cap to the open end of line (1).

(4) Set the ABAS kit regulated pressure to 65.0 psig.

(5) Connect the shop air supply to the ABAS kit inlet port.

(6) Connect the ABAS kit outlet line to the PRV inlet and tighten.

Subtask 30–11–00–865–014

N. Make sure the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT DC (MAIN), A7	DE–ICE CONT
LEFT DC (ESSENTIAL), M5	DE–ICE CONT
LEFT DC (SECONDARY), N8	AIRFRAME DEICE–PRESS IND 1
RIGHT DC (MAIN), S5	DE–ICE CONT
RIGHT DC (SECONDARY), A8	AIRFRAME DEICE–AUTO DIST VLV
RIGHT DC (ESSENTIAL), K7	AIRFRAME MANUAL CONT
RIGHT DC (SECONDARY), A7	AIRFRAME DEICE–PRESS IND 2
RIGHT DC (ESSENTIAL), M4	CAUT/WRN LTS 1
RIGHT DC (MAIN), N4	CAUT/WRN LTS 2

4. Procedure

Subtask 30–11–00–710–003

Refer to Figure 504 and to Figure 505.

A. Do an operational check of the deicing system in manual mode as follows:

(1) Make sure the BOOT AIR switch is in the NORM position.

(2) Set the AIRFRAME MODE SELECT switch to the MANUAL position.

NOTE: Do not operate the APU at the same time that the deice system test is done.

(3) If you used Option 2 or 3, do as follows:

(a) Open the air supply.

(b) In the flight compartment, make sure that the two DEICE PRESS indicators show 18 + 1.5/–1 psi.

(4) Operate the deicing system as follows:

(a) Rotate the AIRFRAME MANUAL SELECT switch sequentially clockwise, and hold for 6 seconds at each position. Make sure that the related deicer boots inflate correctly, by a visual inspection, and the associated advisory light illuminates.

(b) Make sure that each boot deflates in 10 seconds or less and the advisory light extinguishes after you move the switch to the next position.

(5) Visually examine the deicer boots. Make sure that they are deflated and are held smoothly to the airframe.

(6) Select the AIRFRAME MANUAL SELECT switch and the AIRFRAME MODE SELECT switch to the OFF position.

(7) If you used Option 1, shut down the engine (Refer to TASK 71–00–00–868–802).

(8) Make sure that the following components located in the rear fuselage are hot:

- ❖ The two heated check valves
- ❖ The two dual distributing valves.

5. Close Out

Subtask 30–11–00–861–017

A. De–energize the electrical system (Refer to TASK 24–00–00–861–802)

Subtask 30–11–00–410–014

Refer to Figure 506.

B. If the air supply was connected to the P3.0 duct, disconnect the air supply from the P3.0 duct on the left nacelle, as follows:

(1) Shut off the shop air supply.

(2) Release the pressure and disconnect the shop air supply from the regulator P/N 63605–101.

(3) Remove the air–supply adapter P/N 63605–104 from the P3.0 duct.

(4) Reconnect the inlet end (1) of the P3.0 duct (2) to the left engine with the coupling (3). Torque the coupling 70 to 90 lbf in (7.9 to 10.17 N·m).

(5) If loosened, tighten and torque the coupling (4) 70 to 90 lbf in (7.9 to 10.17 N·m).

Subtask 30–11–00–410–015

Refer to Figure 507.

C. If the air supply was connected to the PRV inlet, disconnect the air supply from the PRV inlet on the left nacelle, as follows:

(1) Shut off the shop air supply.

(2) Close the ABAS kit shutoff valve. Make sure the output pressure gage reading drops to 0 psi.

(3) Loosen and remove the ABAS kit outlet line from the PRV inlet.

(4) Remove the blank cap from the end of line (1).

(5) Connect the end of line (1) to the PRV inlet.

(6) Torque the end of line (1) nut 300 to 360 lbf in (33.30 to 40.67 N·m).

(7) If loosened, tighten the clamps (2).

Subtask 30–11–00–410–001

D. Close the access door that follows:

REFERENCE	DESIGNATION
311AB	Aft Fuselage Door

Subtask 30–11–00–410–016

E. If the air supply was connected to the P3.0 duct on the left engine, close the access door that follows:

REFERENCE	DESIGNATION
412BR	Door

Subtask 30–11–00–410–017

F. If the air supply was connected to the P3.0 duct on the right engine, close the access door that follows:

REFERENCE	DESIGNATION
422BR	Door

Subtask 30–11–00–410–018

G. If the air supply was connected to the PRV on the left engine, close the access door that follows (Refer to TASK 71–10–06–410–801):

REFERENCE	DESIGNATION
412AR	Door

H. If the air supply was connected to the PRV on the left engine, close the access door that follows:

REFERENCE	DESIGNATION
412CT	Fairing

Subtask 30–11–00–410–019

I. If the air supply was connected to the PRV on the right engine, close the access door that follows (Refer to TASK 71–10–06–410–801):

REFERENCE	DESIGNATION
422AR	Door

J. If the air supply was connected to the PRV on the right engine, close the access door that follows:

REFERENCE	DESIGNATION
422CT	Fairing

Subtask 30–11–00–941–008

K. Remove all tools, equipment, and unwanted materials from the work area.

Subtask 30–11–00–710–006

L. If the pneumatic air supply was connected to the P3.0 duct, do an operational test of the bleed air system (Refer to TASK 36–11–00–710–802).

Subtask 30–11–00–710–008

M. If the pneumatic air supply was connected to the PRV inlet, do an operational check of the deicing system air supply as follows:

(1) On the ICE PROTECTION panel, make sure that the BOOT AIR switch is set to the NORM position.

(2) Start the left engine (Refer to TASK 71–00–00–868–801).

(3) Make sure the AC generator toggle switch located on the AC control panel is set to GEN 1.

(4) Move the condition lever to the MIN 850 detent

(5) Allow the bleed air system to stabilize for 2 minutes (minimum).

(6) On the copilot side panel, make sure that pressure indication 1 and pressure indication 2 increases to 18±1.5 psi on the DEICE PRESS gauge.

(7) Shutdown the left engine (Refer to TASK 71–00–00–868–802).

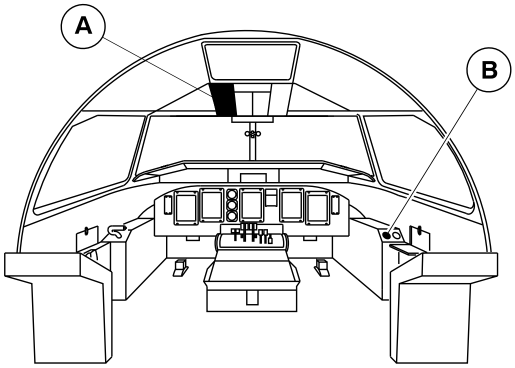
(8) Start the right engine (Refer to TASK 71–00–00–868–801).

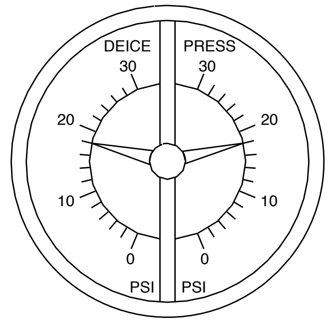
(9) Move the condition lever to the MIN 850 detent

(10) Allow the bleed air system to stabilize for 2 minutes (minimum).

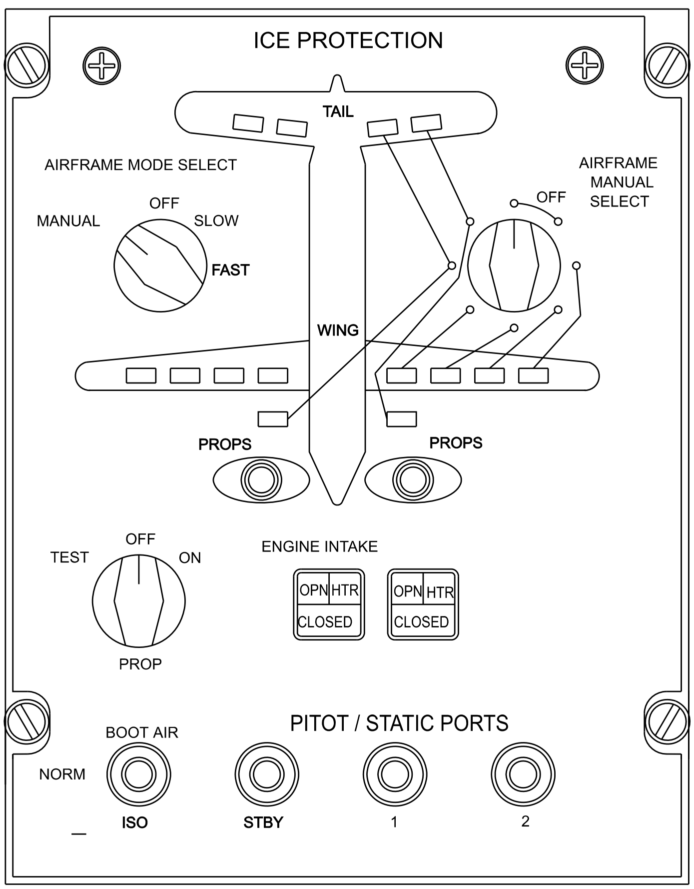
(11) On the copilot side panel, make sure that pressure indication 1 and pressure indication 2 increases to 18±1.5 psi on the DEICE PRESS gauge.

(12) Shutdown the right engine (Refer to TASK 71–00–00–868–802).





B
DEICE DUAL PRESSURE INDICATOR



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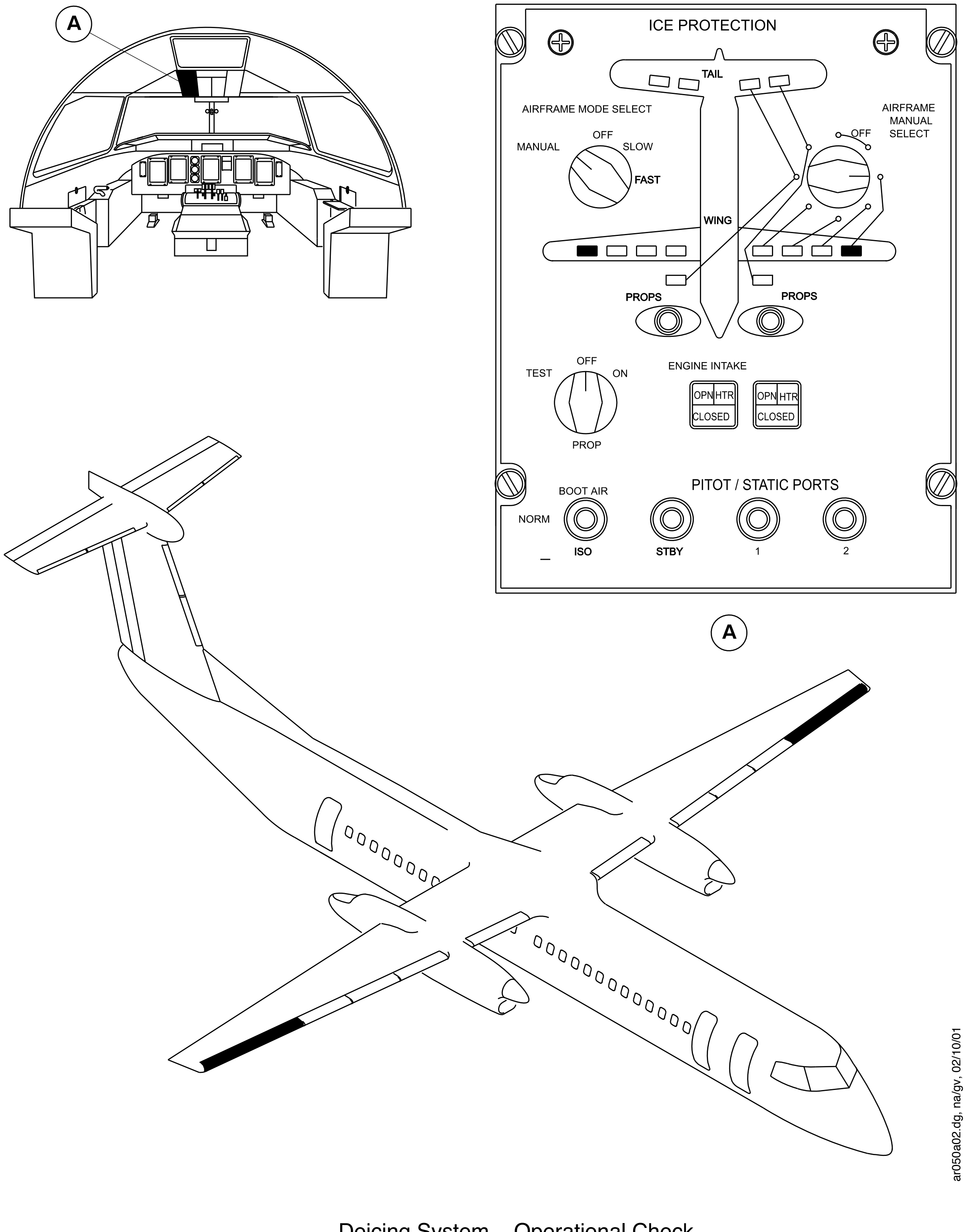
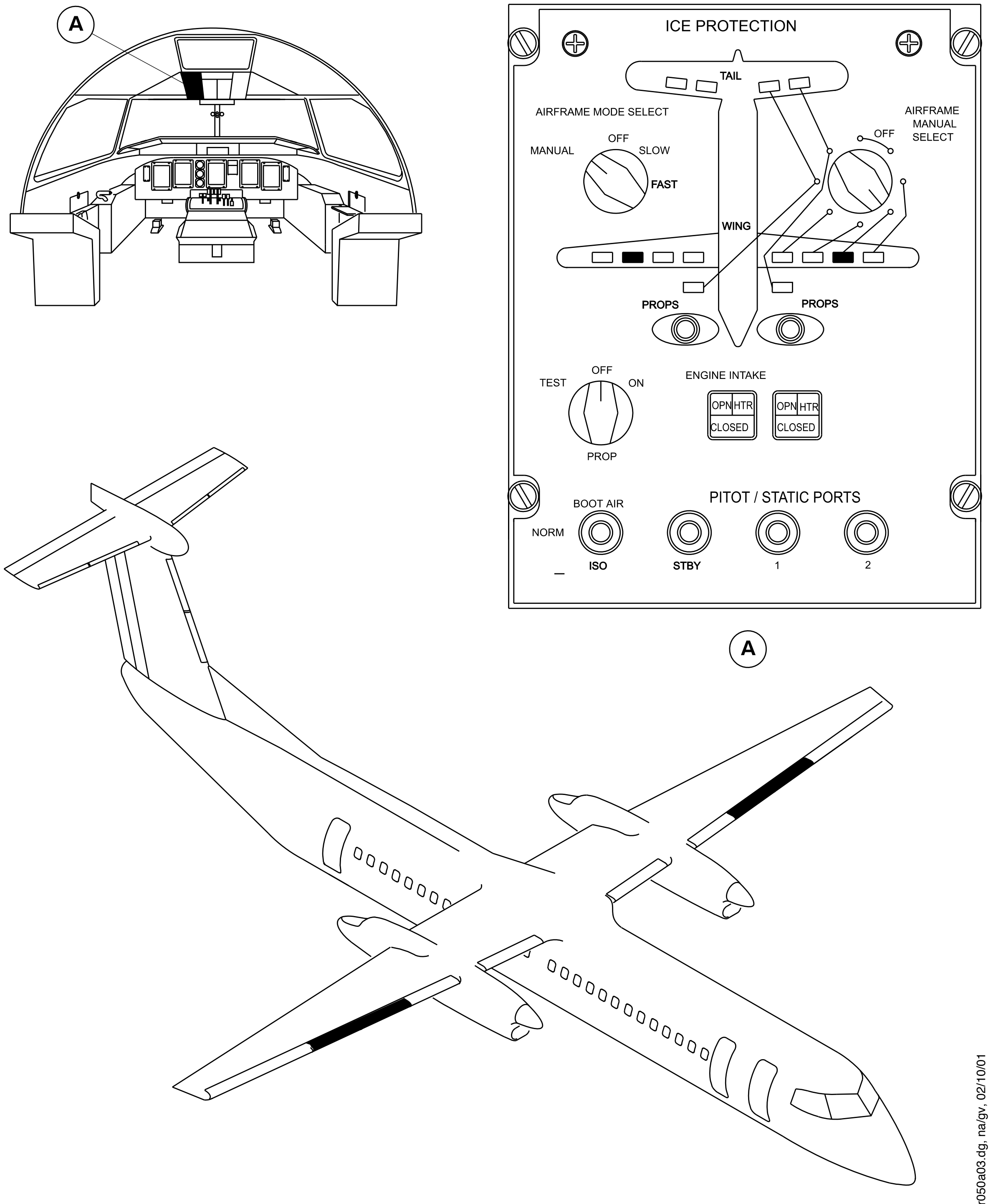
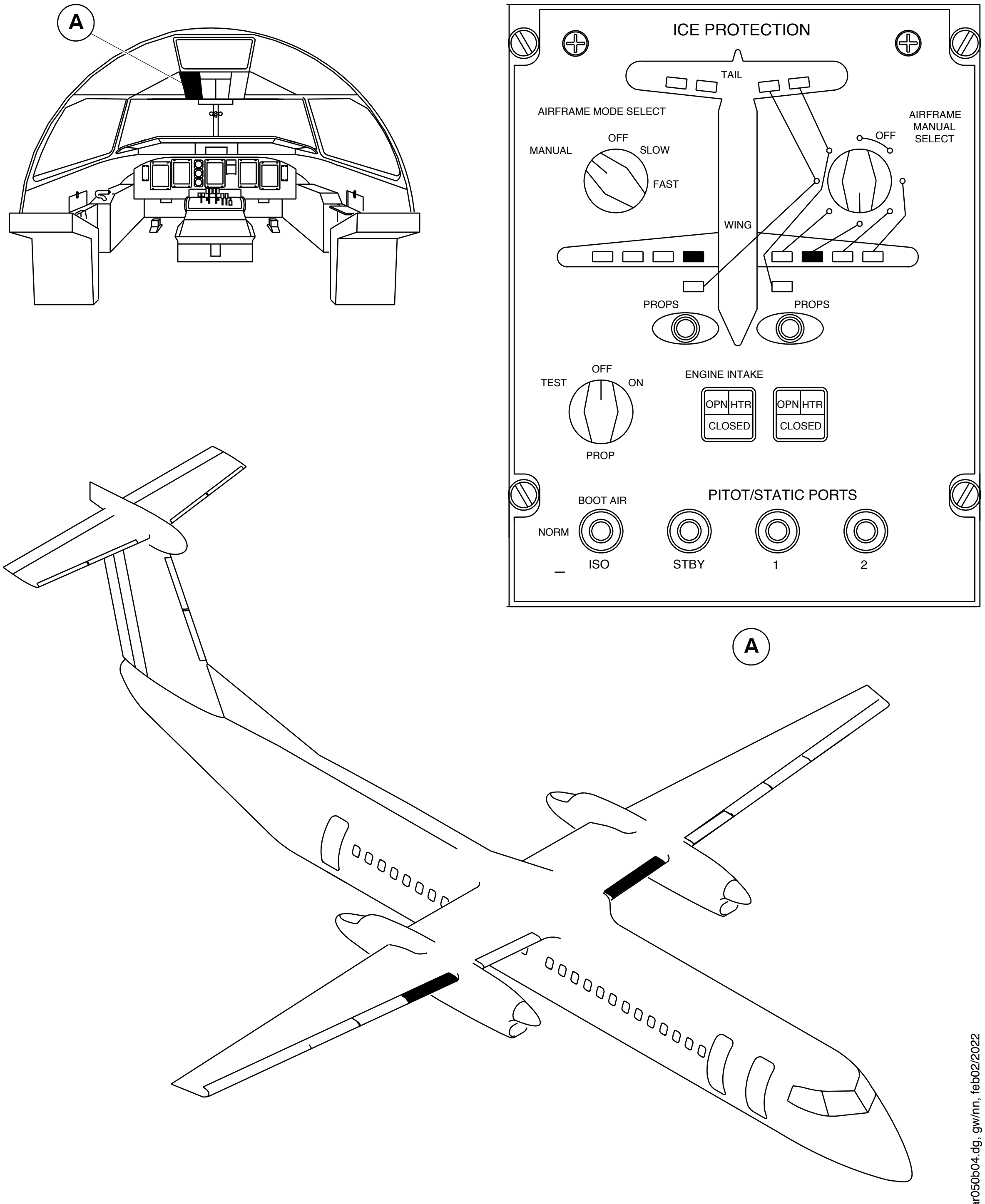
Deicing System – Operational Check
Figure 504 (Sheet 1 of 7)



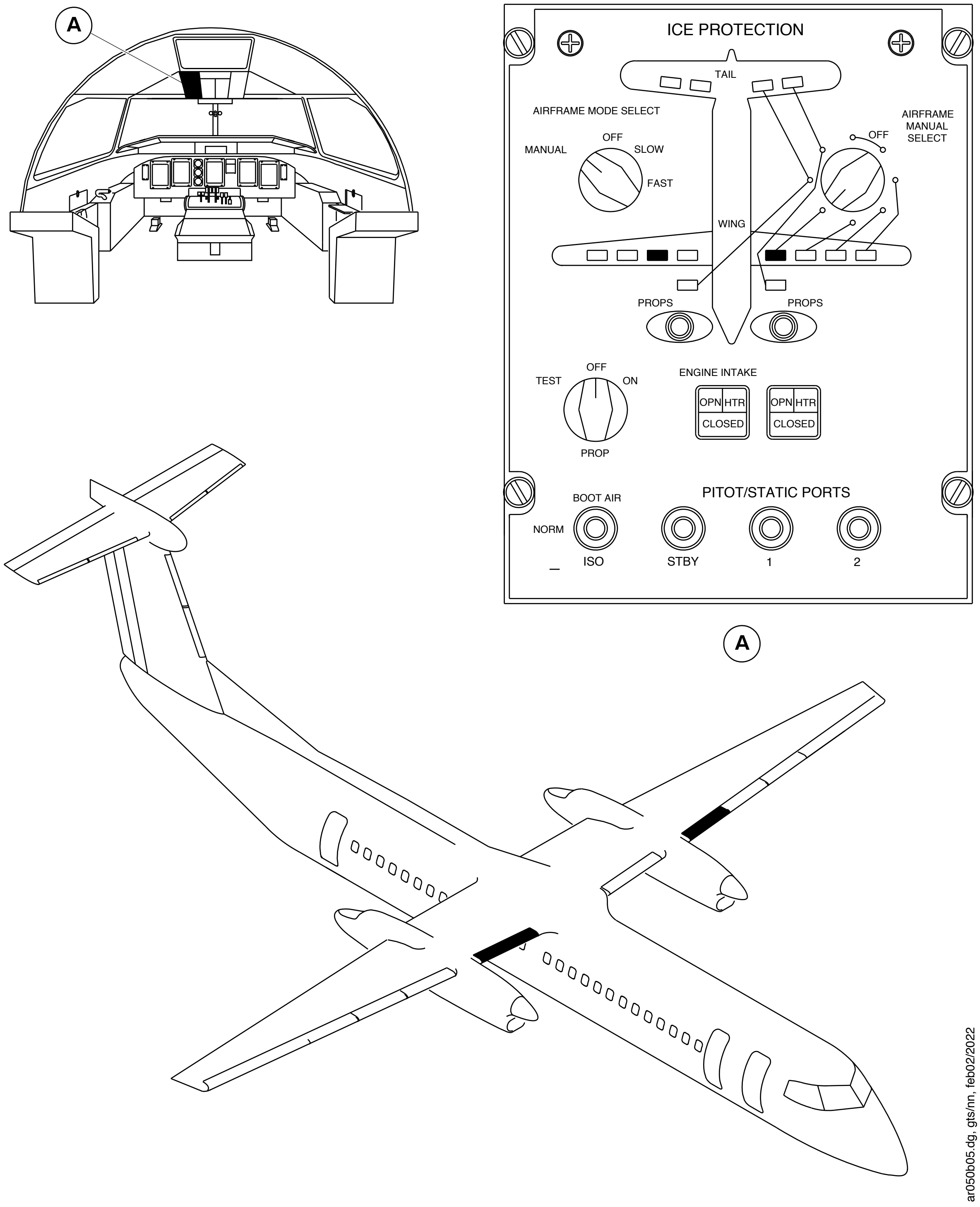
Deicing System – Operational Check
Figure 504 (Sheet 2 of 7)



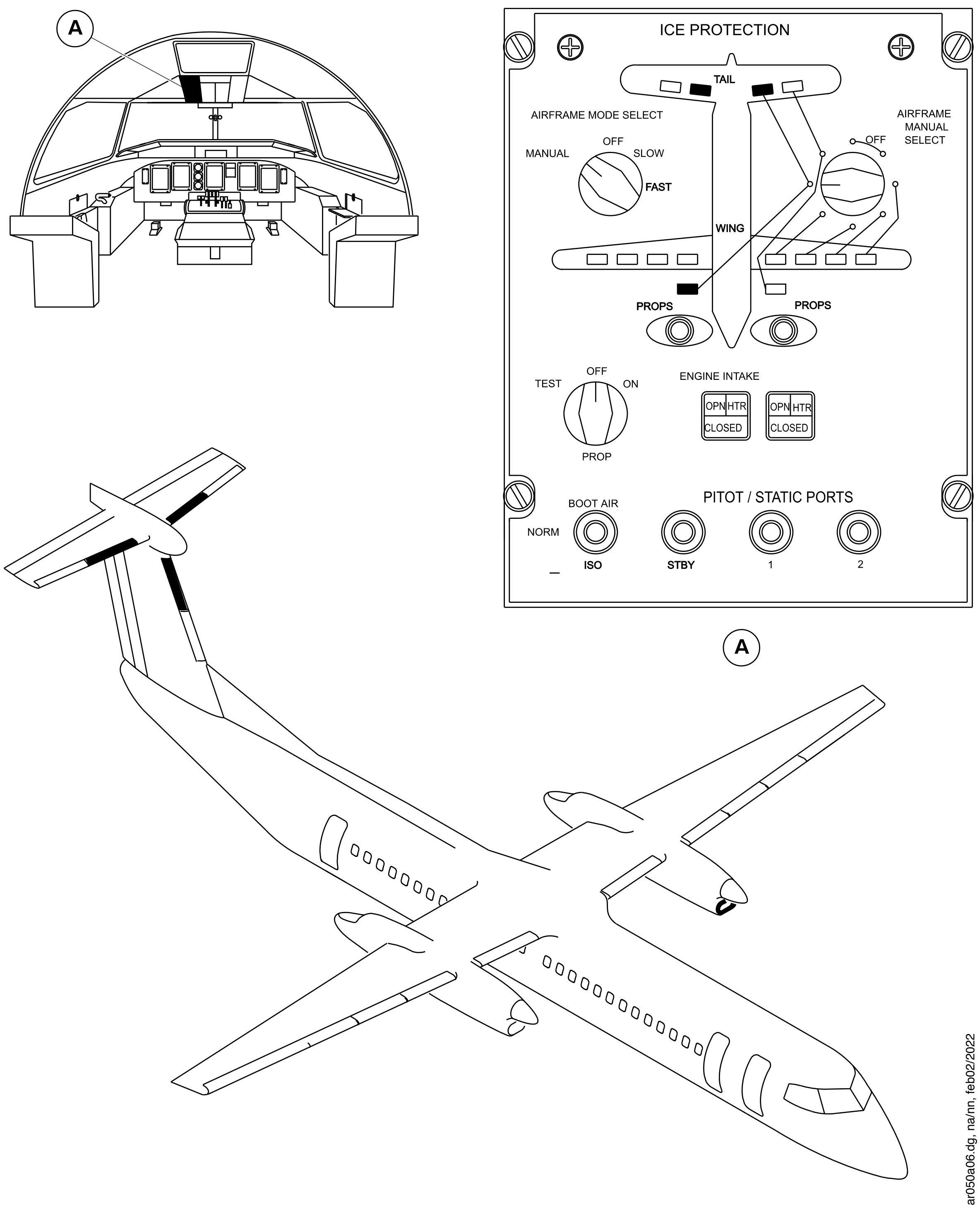
Deicing System – Operational Check
Figure 504 (Sheet 3 of 7)



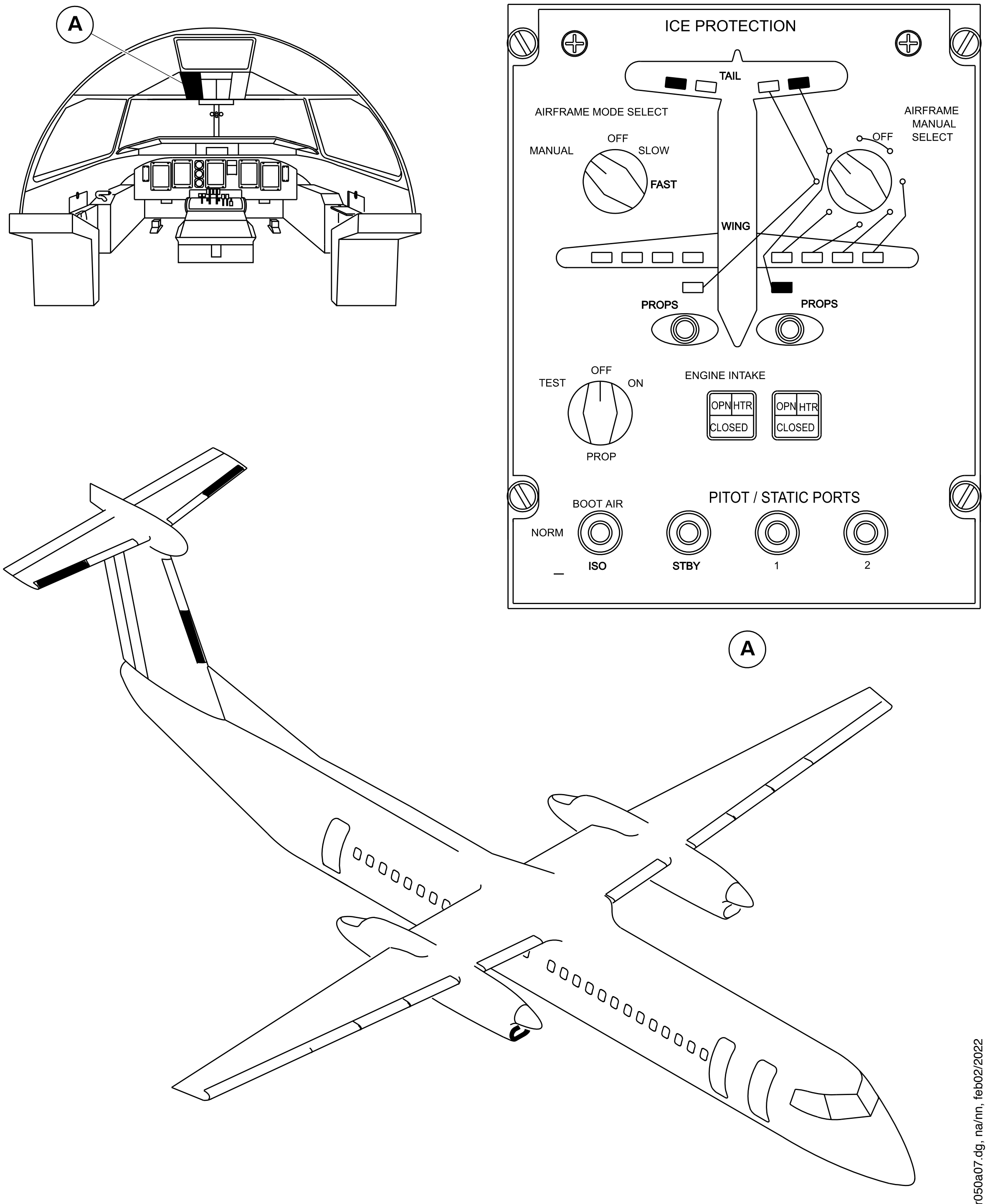
Deicing System – Operational Check
Figure 504 (Sheet 4 of 7)



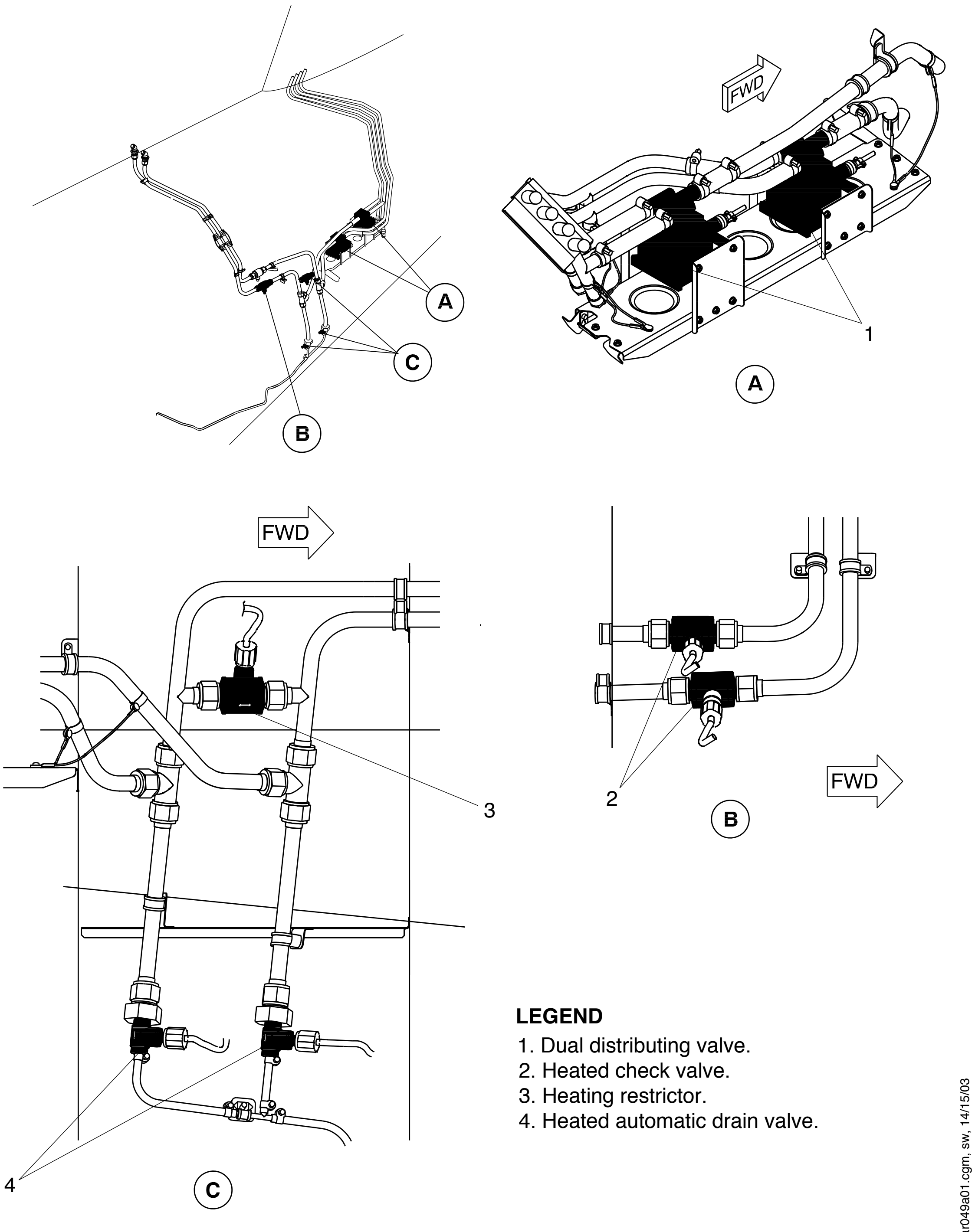
Deicing System – Operational Check
Figure 504 (Sheet 5 of 7)



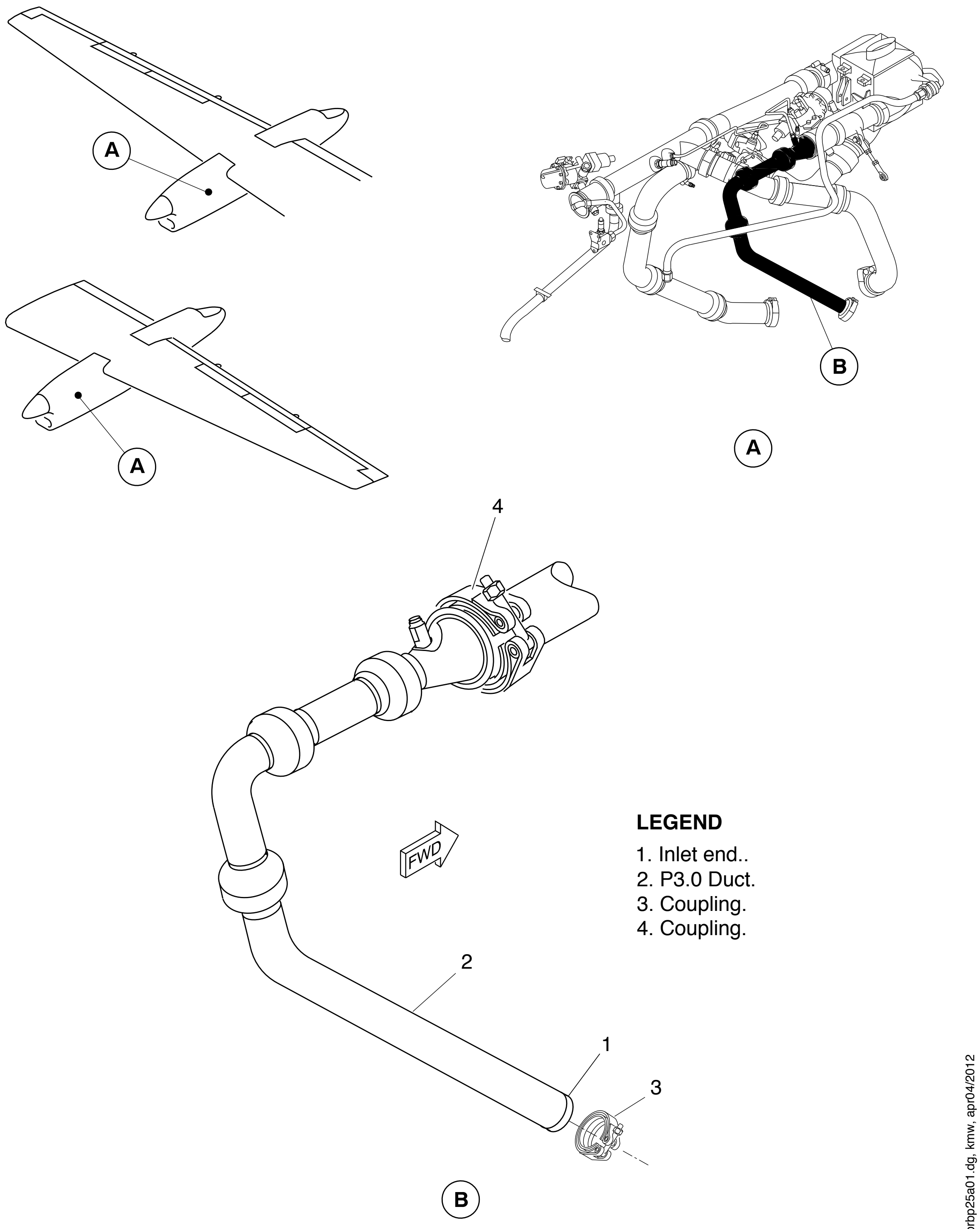
Deicing System – Operational Check
Figure 504 (Sheet 6 of 7)



Deicing System – Operational Check
Figure 504 (Sheet 7 of 7)

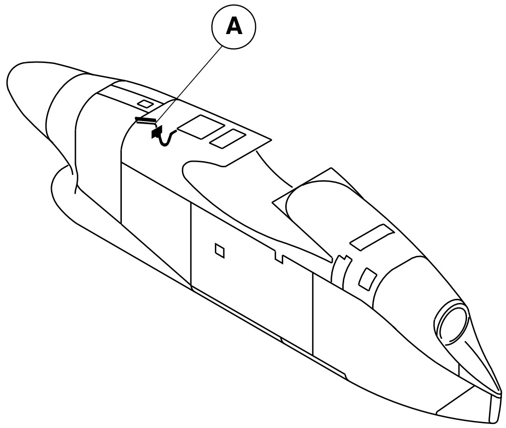


Deicing System – Operational Check
Figure 505



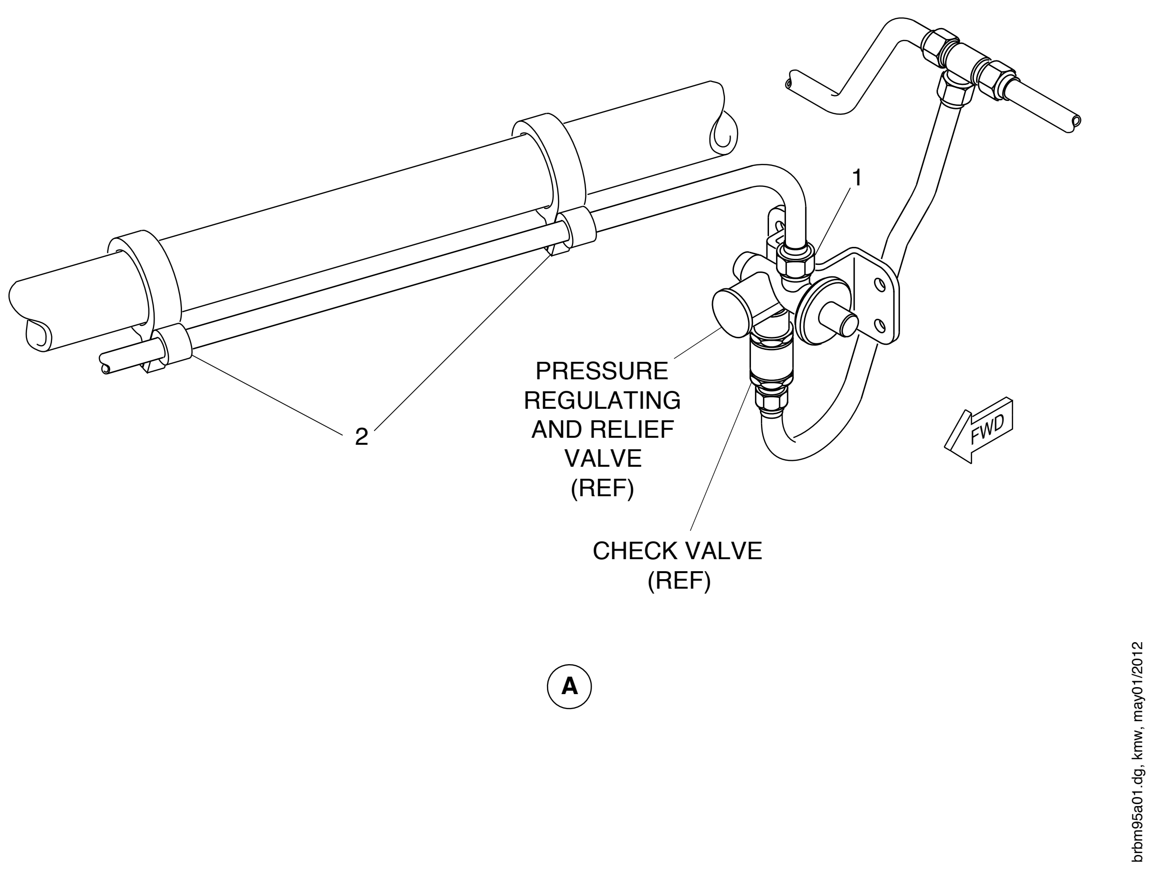
LEGEND
1. Inlet end..
2. P3.0 Duct.
3. Coupling.
4. Coupling.

Air Supply to the P3.0 Duct
Figure 506



LEGEND

1. End of line.
2. Clamps.



Deicing Line in Nacelle at PRV
Figure 507

**ON A/C ALL

TASK 30–11–00–710–806
Operational Check of the Deicing System Air Supply

1. General

   A. The maintenance procedure that follows is for the operational check of the deicing system air supply.

2. Job Set–Up Information

Subtask 30–11–00–943–017

   A. Tools & Equipment

      (1) None specified — Torque Wrench 0 – 500 lbf·in (0 – 56.5 N·m)

      (2) Commercially Available from Innovative Aircraft Solutions LLC. USA — Alternate Bleed Air Source (ABAS) Kit 63605–001 (or equivalent)

         NOTE: Do not increase the supply pressure more than 120 psi (827.37 kPa) at the ABAS kit inlet.

      (3) Commercially Available — Dry air/shop air supply

      (4) Commercially Available — Maintenance stands

Subtask 30–11–00–946–020

   B. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize the Electrical System
TASK 24–00–00–861–802	De–Energize the Electrical System
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown
TASK 71–10–06–010–801	Opening of the Forward Side Door
TASK 71–10–06–410–801	Closing of the Forward Side Door

3. Job Set–Up

Subtask 30–11–00–010–019

A. If the air supply is to be connected to the PRV on the left engine, open the access door that follows (Refer to TASK 71–10–06–010–801):

REFERENCE	DESIGNATION
412AR	Door

B. If the air supply is to be connected to the PRV on the left engine, open the access door that follows:

REFERENCE	DESIGNATION
412CT	Fairing

C. If the air supply is to be connected to the PRV on the right engine, open the access door that follows (Refer to TASK 71–10–06–010–801):

REFERENCE	DESIGNATION
422AR	Door

D. If the air supply is to be connected to the PRV on the right engine, open the access door that follows:

REFERENCE	DESIGNATION
422CT	Fairing

Subtask 30–11–00–861–026

E. Energize the electrical system (Refer to TASK 24–00–00–861–801).

4. Procedure

Subtask 30–11–00–710–009

Refer to Figure 508.

A. Do the operational check of the deicing system air supply as follows:

NOTE: There are two procedures to do the operational test. Procedure 1 uses compressed air from the engine to do the test and the Procedure 2 uses compressed shop air to do the test. Do one of the procedures.

(1) Procedure 1:

(a) Make sure that the AC generator toggle switch located on the AC control panel is set to GEN 1.

(b) On the ICE PROTECTION panel, make sure that the BOOT AIR switch (1) is set to the NORM position and the AIRFRAME MANUAL SELECTOR is set to OFF position.

(c) Start the left engine (Refer to TASK 71–00–00–868–801).

(d) Move the condition lever to the MIN 850 detent.

(e) Move the power lever to the FI position detent.

(f) Set the two PACKS switches (2) on the air conditioning panel to the AUTO position.

(g) Set the BLEED 1 switch (3) to the BLEED position, the BLEED 2 switch (4) to the OFF position and the BLEED selector to the MAX position.

(h) Let the system stabilize for three minutes.

(i) On the copilot side panel, make sure that the pressure indication 1 (5) and the pressure indication 2 (6) on the DEICE PRESS gauge increase to approximately 18 psi (124.1 kPa).

(j) Shutdown the left engine (Refer to TASK 71–00–00–868–802).

(k) Start the right engine (Refer to TASK 71–00–00–868–801).

(l) Move the condition lever to the MIN 850 detent.

(m) Move the power lever to the FI position detent.

(n) Set the two PACKS switches (2) on the air conditioning panel to the AUTO position.

(o) Set the BLEED 2 switch (4) to the BLEED position, the BLEED 1 switch (3) to the OFF position and the BLEED selector to the MAX position.

(p) Let the system stabilize for three minutes.

(q) On the copilot side panel, make sure that the pressure indication 1 (5) and the pressure indication 2 (6) on the DEICE PRESS gauge increase to approximately 18 psi (124.1 kPa).

(r) Shutdown the right engine (Refer to TASK 71–00–00–868–802).

(2) Procedure 2:

Refer to Figure 509.

(a) Connect the air supply to the Pressure Regulating Valve (PRV) inlet on the left nacelle, as follows:

NOTE: The procedures are same for the PRV on the left and right engine.

1. Disconnect the deicing line (1) at the PRV inlet.

2. For access to the PRV inlet, rotate the deicing line (1) away from the PRV inlet. If necessary, loosen the clamps (2) as required.

3. Install a blank cap to the open end of the deicing line (1).

4. Set the ABAS kit regulated pressure to 65.0 psi (448.2 kPa).
5. Connect the shop air supply to the ABAS kit inlet port.
6. Connect the ABAS kit outlet line to the PRV inlet and tighten.
7. Supply the shop air.

(b) On the ICE PROTECTION panel:

1. Set the airframe mode selector to the OFF position.
2. Set the airframe manual selector to the OFF position.
3. Set the BOOT AIR switch to the NORM position.

(c) Examine the dual de–icing pressure indicator in the flight compartment, make sure that the No. 1 and the No. 2 systems indicate 18±1.5 psi (124.1 ±10.34 kPa).

(d) On the ICE PROTECTION panel:

1. Set the BOOT AIR switch to the ISO position.

(e) Examine the dual de–icing pressure indicator:

1. Make sure that the No. 1 system indicates 18±1.5 psi (124.1 ±10.34 kPa).
2. Make sure that the No. 2 system indicates zero pressure.

(f) On the ICE PROTECTION panel:

1. Set the BOOT AIR switch to the NORM position.

(g) Disconnect the air supply from the PRV inlet on the left nacelle, as follows:

1. Shut off the shop air supply.
2. Close the ABAS kit shutoff valve.
3. Make sure the output pressure gage reading drops to 0 psi.
4. Loosen and remove the ABAS kit outlet line from the PRV inlet.
5. Remove the blank cap from the deicing line (1).
6. Connect the deicing line (1) to the PRV inlet.
7. Torque the deicing line (1) nut to 300 to 360 lbf·in (33.9 to 40.7 N·m).
8. If loosened, tighten the clamps (2).

5. Close Out

Subtask 30–11–00–861–027

A. De–energize the electrical system (Refer to TASK 24–00–00–861–802).

Subtask 30–11–00–941–023

B. Remove all tools, equipment, and unwanted materials from the work area.

Subtask 30–11–00–410–020

C. If the air supply was connected to the PRV on the left engine, close the access door that follows (Refer to TASK 71–10–06–410–801):

REFERENCE	DESIGNATION
412AR	Door

D. If the air supply was connected to the PRV on the left engine, close the access door that follows:

REFERENCE	DESIGNATION
412CT	Fairing

E. If the air supply was connected to the PRV on the right engine, close the access door that follows (Refer to TASK 71–10–06–410–801):

REFERENCE	DESIGNATION
422AR	Door

F. If the air supply was connected to the PRV on the right engine, close the access door that follows:

REFERENCE	DESIGNATION
422CT	Fairing

Subtask 30–11–00–710–010

G. If the pneumatic air supply was connected to the PRV, do an operational check of the deicing system air supply as follows:

(1) On the ICE PROTECTION panel, make sure that the BOOT AIR switch is set to the NORM position.

(2) Start the left engine (Refer to TASK 71–00–00–868–801).

(3) Make sure the AC generator toggle switch located on the AC control panel is set to GEN 1.

(4) Move the condition lever to the MIN 850 detent.

(5) Allow the bleed air system to stabilize for two minutes (minimum).

(6) On the copilot side panel, make sure that pressure indication 1 and pressure indication 2 increases to 18±1.5 psi (124.1 ±10.34 kPa) on the DEICE PRESS gauge.

(7) Shutdown the left engine (Refer to TASK 71–00–00–868–802).

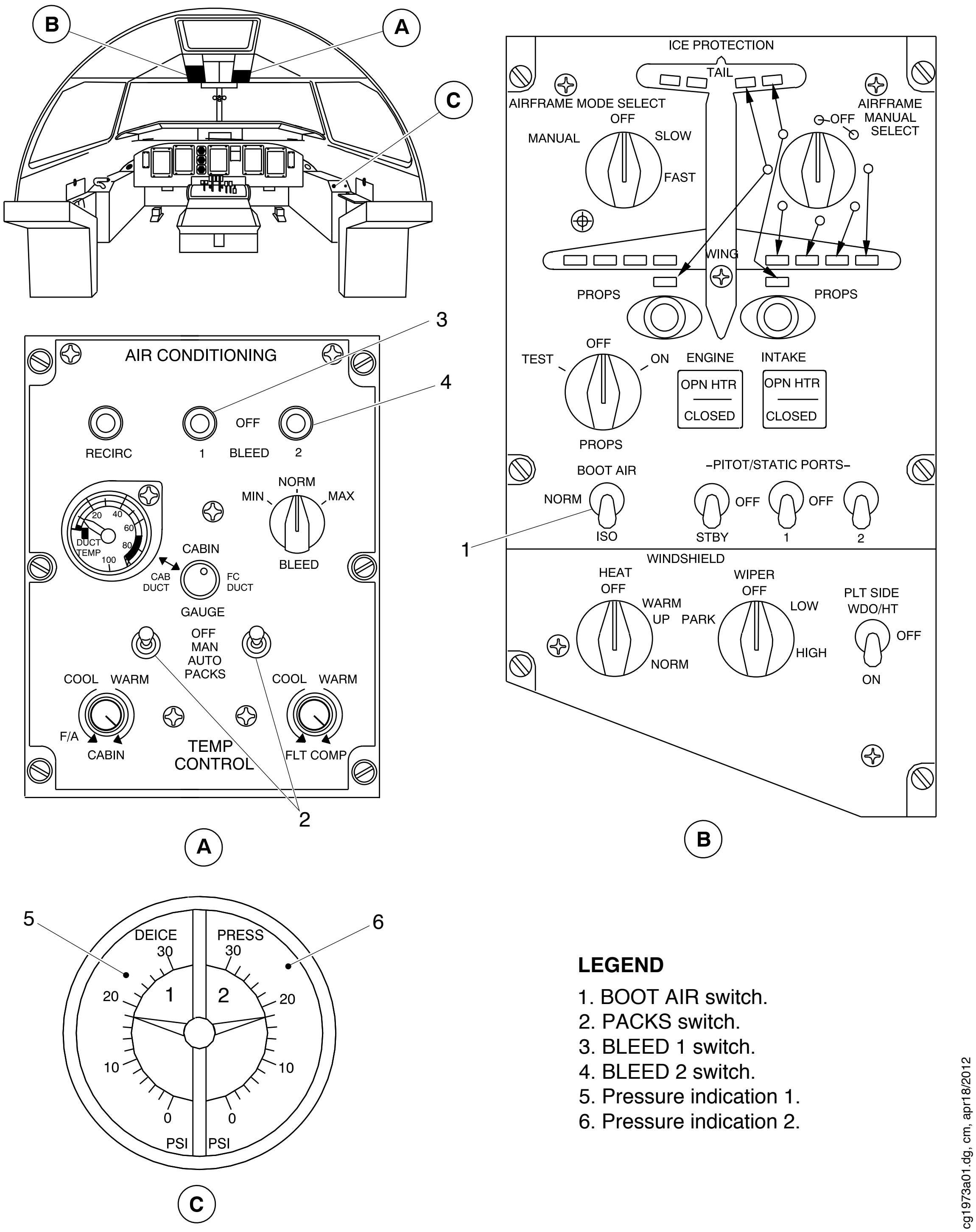
(8) Start the right engine (Refer to TASK 71−00−00−868−801).

(9) Move the condition lever to the MIN 850 detent.

(10) Allow the bleed air system to stabilize for two minutes (minimum).

(11) On the copilot side panel, make sure that pressure indication 1 and pressure indication 2 increases to 18±1.5 psi (124.1 ±10.34 kPa) on the DEICE PRESS gauge.

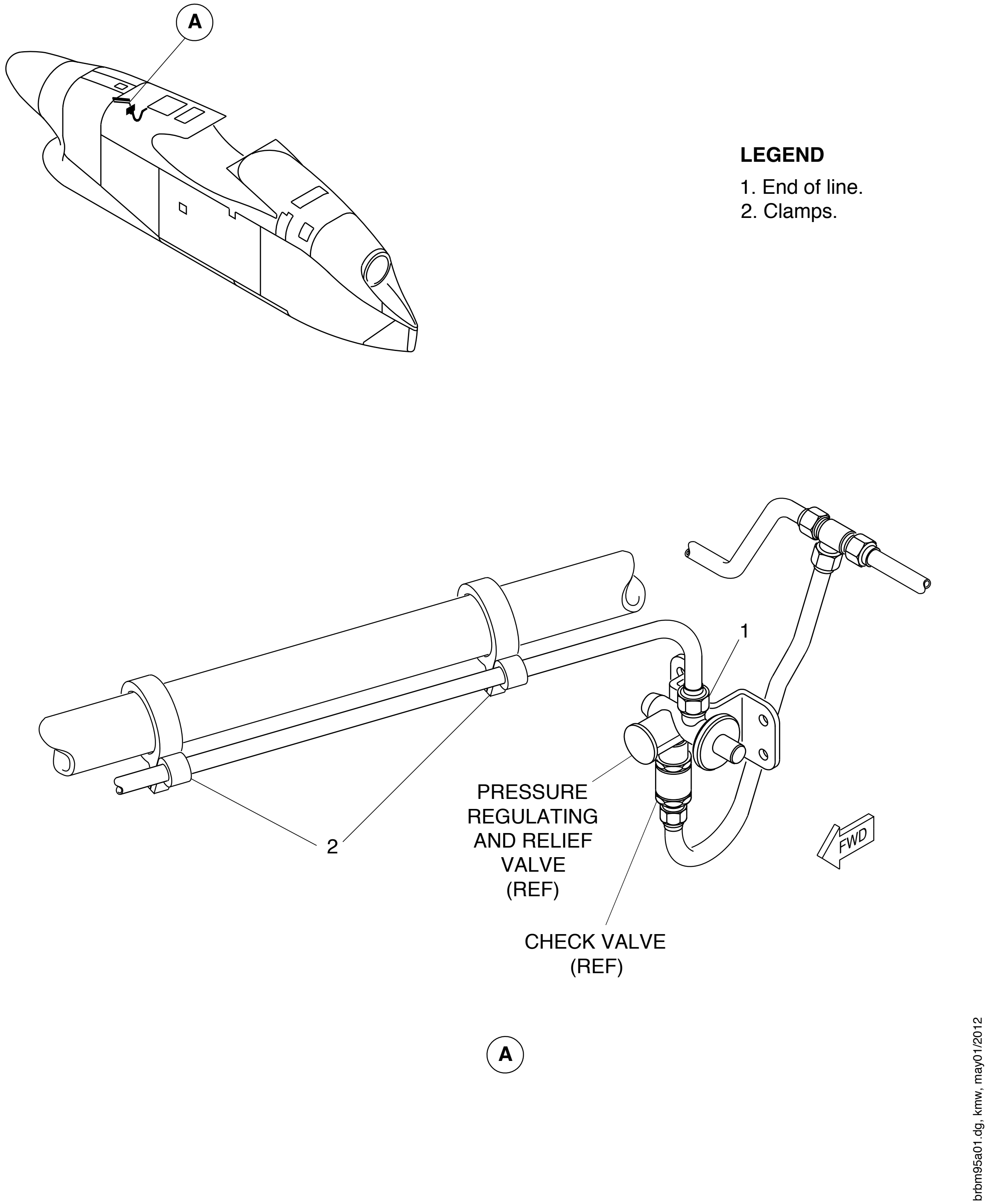
(12) Shutdown the right engine (Refer to TASK 71−00−00−868−802).



LEGEND

1. BOOT AIR switch.
2. PACKS switch.
3. BLEED 1 switch.
4. BLEED 2 switch.
5. Pressure indication 1.
6. Pressure indication 2.

Deicing System – Operational Check
Figure 508



LEGEND

1. End of line.
2. Clamps.

Deicing Line in Nacelle at PRV
Figure 509

**ON A/C ALL

TASK 30–11–00–710–807
Operational Check of the Deicing System in the Manual Mode

1. General

   A. The maintenance procedure that follows is for the operational check of the deicing system in the manual mode. This task also incorporates the operational check of the airframe deicing equipment heating in manual mode and the operational check of the deicing hold down system.

   B. All deicing boots can be pressurized by running the right engine.

   C. All deicing boots can be pressurized by connecting an external regulated supply of dry air to either the left or the right engine P3.0 HP to high–pressure shut–off valve (HPSOV) duct (also referred to as the P3.0 duct in these procedures). The deicing system pressurization in this procedure is done by connecting the external air supply to the P3.0 duct on the left engine. Differences for the right engine are identified.

   D. An alternate procedure to pressurize the boots can be done by disconnecting the inlet line to the Pressure Regulating Valve (PRV) and connecting an external regulated supply of dry air to either the left or right engine PRV inlet line. The deicing system pressurization in this procedure is done by connecting the external air supply to the left engine PRV. Differences for the right engine are identified.

   E. The external pneumatic supply can be connected to either the P3.0 duct or the PRV inlet for this inspection procedure.

2. Job Set–Up Information

Subtask 30–11–00–943–020

   A. Tools & Equipment

(1)	Commercially Available	Wrench, Torque 0 to 500 lbf·in (0 to 56.5 N·m)
(2)	Commercially Available from Innovative Aircraft Solutions LLC. USA	Alternate Bleed Air Source (ABAS) Kit 63605–001 (or equivalent)
(3)	Commercially Available	Dry air/shop air supply
(4)	Commercially Available	Maintenance stands

Subtask 30–11–00–946–023

B. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize the Electrical System
TASK 24–00–00–861–802	De–energize the Electrical System
TASK 24–00–00–910–801	Electrical/Electronic Safety Precautions
TASK 24–00–00–910–802	Electrostatic Discharge Safety Precautions
TASK 36–11–00–710–802	Operational Test of the Bleed Air System
TASK 36–11–01–000–801	Removal of the P3.0 HP to HPSOV Duct
TASK 36–11–01–400–801	Installation of the P3.0 HP to HPSOV Duct
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown
TASK 71–10–06–010–801	Opening of the Forward Side Door
TASK 71–10–06–410–801	Closing of the Forward Side Door

3. Job Set–Up

Subtask 30–11–00–910–021

A. Obey all the electrical/electronic safety precautions (Refer to TASK 24–00–00–910–801).

Subtask 30–11–00–910–022

B. Obey all the electrostatic discharge safety precautions (Refer to TASK 24–00–00–910–802).

Subtask 30–11–00–865–019

C. Make sure the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT DC (MAIN), A7	DE–ICE CONT
LEFT DC (ESSENTIAL), M5	DE–ICE CONT
LEFT DC (SECONDARY), N8	AIRFRAME DEICE–PRESS IND 1
RIGHT DC (MAIN), S5	DE–ICE CONT
RIGHT DC (SECONDARY), A8	AIRFRAME DEICE–AUTO DIST VLV
RIGHT DC (ESSENTIAL), K7	AIRFRAME MANUAL CONT
RIGHT DC (SECONDARY), A7	AIRFRAME DEICE–PRESS IND 2

CB PANEL AND CB NO	NAME
RIGHT DC (ESSENTIAL), M4	CAUT/WRN LTS 1
RIGHT DC (MAIN), N4	CAUT/WRN LTS 2

Subtask 30–11–00–861–030

WARNING: MAKE SURE YOU TELL ALL PERSONS IN THE AREA OF THE AIRCRAFT THAT YOU WILL ENERGIZE THE AIRCRAFT ELECTRICAL SYSTEM, BEFORE YOU TURN ON THE ELECTRICAL SYSTEM. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

D. Energize the electrical system (Refer to TASK 24–00–00–861–801).

Subtask 30–11–00–010–024

E. Open the access door that follows:

REFERENCE	DESIGNATION
311AB	Aft Fuselage Door

Subtask 30–11–00–010–025

F. If the air supply is to be connected to the P3.0 duct on the left engine, open the access door that follows:

REFERENCE	DESIGNATION
412BR	Door

Subtask 30–11–00–010–026

G. If the air supply is to be connected to the P3.0 duct on the right engine, open the access door that follows:

REFERENCE	DESIGNATION
422BR	Door

Subtask 30–11–00–010–027

H. If the air supply is to be connected to the PRV on the left engine, open the access door that follows (Refer to TASK 71–10–06–010–801):

REFERENCE	DESIGNATION
412AR	Door

I. If the air supply is to be connected to the PRV on the left engine, open the access door that follows:

REFERENCE	DESIGNATION
412CT	Fairing

Subtask 30–11–00–010–028

J. If the air supply is to be connected to the PRV on the right engine, open the access door that follows (Refer to TASK 71–10–06–010–801):

REFERENCE	DESIGNATION
422AR	Door

K. If the air supply is to be connected to the PRV on the right engine, open the access door that follows:

REFERENCE	DESIGNATION
422CT	Fairing

Subtask 30–11–00–868–002

L. Option 1: If you do the deicing system pressurization with the engine, start the right engine (Refer to TASK 71–00–00–868–801).

Subtask 30–11–00–010–029

Refer to Figure 512.

M. Option 2: If you connect the air supply to the P3.0 duct (2) on the left nacelle, do as follows:

NOTE: The procedures are identical for the P3.0 duct (2) on the left and right engine.

(1) Disconnect the inlet end (1) of the P3.0 duct (2) from the engine (Refer to TASK 36–11–01–000–801). Retain the coupling (3) for adapter installation.

(2) If required, loosen the coupling (4) as necessary for access to install the air supply adapter.

(3) Install the air supply adapter P/N 63605–104 onto the P3.0 duct (2) using the coupling (3). Torque the coupling (3) 70 to 90 lbf·in (7.9 to 10.17 N·m).

(4) If loosened for access, tighten and torque the coupling (4) 70 to 90 lbf·in (7.9 to 10.17 N·m).

(5) Connect the regulator P/N 63605–101 to the air supply adapter 63605–104 using the kit–supplied air hose P/N 63605–105.

(6) Connect the shop air supply to the regulator P/N 63605–101.

(7) Set the ABAS kit regulated pressure to 75.0 ±1.0 psig (517.10 ±6.9 kPa).

Subtask 30–11–00–010–030

Refer to Figure 513.

N. Option 3: If you connect the air supply to the PRV inlet on the left nacelle, do as follows:

NOTE: The procedures are identical for the PRV on the left and right engine.

(1) Disconnect the deicing line (1) at the PRV inlet.

(2) For access to the PRV inlet, rotate the deicing line (1) away from the PRV inlet. If required, loosen the clamps (2) as necessary.

(3) Install a blanking cap to the open end of the deicing line (1).

(4) Set the ABAS kit regulated pressure to 65.0 psig (448.2 kPa).

(5) Connect the shop air supply to the ABAS kit inlet port.

(6) Connect the ABAS kit outlet line to the PRV inlet and tighten.

4. Procedure

Subtask 30–11–00–710–015

Refer to Figure 510 and to Figure 511.

A. Do an operational check of the deicing system in manual mode as follows:

(1) Make sure the BOOT AIR switch is in the NORM position.

(2) Set the AIRFRAME MODE SELECT switch to the MANUAL position.

NOTE: Do not operate the APU at the same time that the deice system test is done.

(3) If you used Option 2 or 3, do as follows:

(a) Open the air supply.

(b) In the flight compartment, make sure that the two DEICE PRESS indicators show 18.0 +1.5/–1.0 psi (124.10 +10.3/–6.9 kPa).

(4) Operate the deicing system as follows:

(a) Rotate the AIRFRAME MANUAL SELECT switch sequentially clockwise, and hold for six seconds at each position. Make sure that the related deicer boots inflate correctly.

(b) Make sure that each boot deflates in 10 seconds or less after you move the switch to the next position.

(5) Visually examine the deicer boots. Make sure that they are deflated and are held smoothly to the airframe.

(6) Select the AIRFRAME MANUAL SELECT switch and the AIRFRAME MODE SELECT switch to the OFF position.

(7) If you used Option 1, shut down the engine (Refer to TASK 71–00–00–868–802).

(8) Make sure that the following components located in the rear fuselage are hot:

- The two heated check valves (2)
- The two dual distributing valves (1).

5. <u>Close Out</u>

Subtask 30–11–00–861–031

A. De–energize the electrical system (Refer to TASK 24–00–00–861–802).

Subtask 30–11–00–410–025

Refer to Figure 512.

B. If the air supply was connected to the P3.0 duct (2), disconnect the air supply from the P3.0 duct (2) on the left nacelle, as follows:

(1) Shut off the shop air supply.

(2) Release the pressure and disconnect the shop air supply from the regulator P/N 63605–101.

(3) Remove the air–supply adapter P/N 63605–104 from the P3.0 duct (2).

(4) Reconnect the inlet end (1) of the P3.0 duct (2) to the left engine with the coupling (3) (Refer to TASK 36–11–01–400–801). Torque the coupling (3) 70 to 90 lbf·in (7.9 to 10.17 N·m).

(5) If loosened, tighten and torque the coupling (4) 70 to 90 lbf·in (7.9 to 10.17 N·m).

Subtask 30–11–00–410–026

Refer to Figure 513.

C. If the air supply was connected to the PRV inlet, disconnect the air supply from the PRV inlet on the left nacelle, as follows:

(1) Shut off the shop air supply.

(2) Close the ABAS kit shutoff valve. Make sure the output pressure gage reading drops to 0 psi (0 kPa).

(3) Loosen and remove the ABAS kit outlet line from the PRV inlet.

(4) Remove the blanking cap from the deicing line (1).

(5) Connect the deicing line (1) to the PRV inlet.

(6) Torque the deicing line (1) nut 300 to 360 lbf·in (33.89 to 40.67 N·m).

(7) If loosened, tighten the clamps (2).

Subtask 30–11–00–941–030

D. Remove all tools, equipment and unwanted materials from the work area.

Subtask 30–11–00–410–027

E. Close the access door that follows:

REFERENCE	DESIGNATION
311AB	Aft Fuselage Door

Subtask 30–11–00–410–028

F. If the air supply was connected to the P3.0 duct on the left engine, close the access door that follows:

REFERENCE	DESIGNATION
412BR	Door

Subtask 30–11–00–410–029

G. If the air supply was connected to the P3.0 duct on the right engine, close the access door that follows:

REFERENCE	DESIGNATION
422BR	Door

Subtask 30–11–00–410–030

H. If the air supply was connected to the PRV on the left engine, close the access door that follows (Refer to TASK 71–10–06–410–801):

REFERENCE	DESIGNATION
412AR	Door

I. If the air supply was connected to the PRV on the left engine, close the access door that follows:

REFERENCE	DESIGNATION
412CT	Fairing

Subtask 30–11–00–410–031

J. If the air supply was connected to the PRV on the right engine, close the access door that follows (Refer to TASK 71–10–06–410–801):

REFERENCE	DESIGNATION
422AR	Door

K. If the air supply was connected to the PRV on the right engine, close the access door that follows:

REFERENCE	DESIGNATION
422CT	Fairing

Subtask 30–11–00–710–016

L. If the pneumatic air supply was connected to the P3.0 duct, do an operational test of the bleed air system (Refer to TASK 36–11–00–710–802).

Subtask 30–11–00–710–017

M. If the pneumatic air supply was connected to the PRV inlet, do an operational check of the deicing system air supply as follows:

(1) On the ICE PROTECTION panel, make sure that the BOOT AIR switch is set to the NORM position.

(2) Start the left engine (Refer toTASK 71–00–00–868–801).

(3) Make sure the AC generator toggle switch located on the AC control panel is set to GEN 1.

(4) Move the condition lever to the MIN 850 detent.

(5) Allow the bleed air system to stabilize for two minutes (minimum).

(6) On the copilot side panel, make sure that pressure indication 1 and pressure indication 2 increases to 18.0 ±1.5 psi (124.10 ±10.3 kPa) on the DEICE PRESS gauge.

(7) Shutdown the left engine (Refer to TASK 71–00–00–868–802).

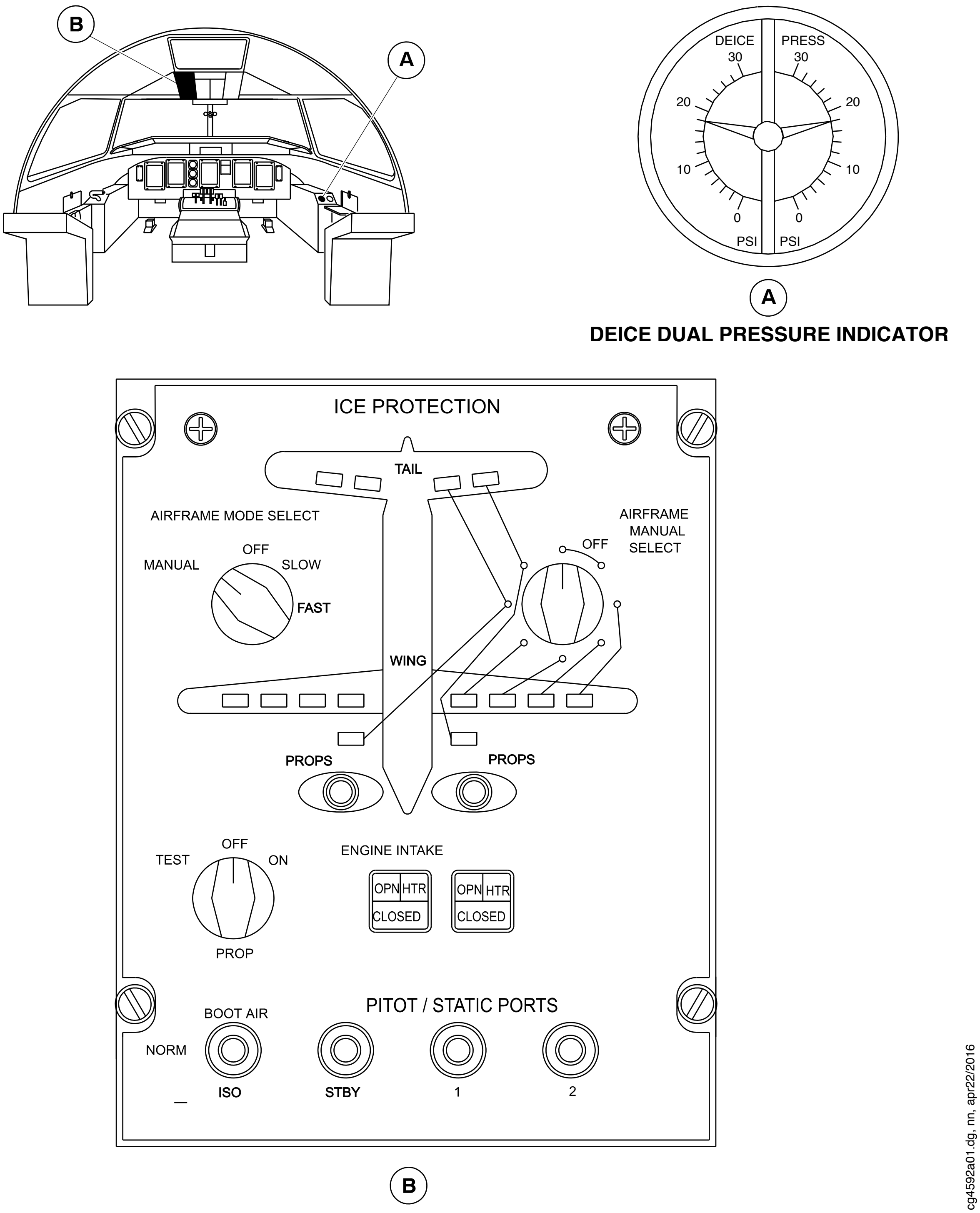
(8) Start the right engine (Refer to TASK 71–00–00–868–801).

(9) Move the condition lever to the MIN 850 detent.

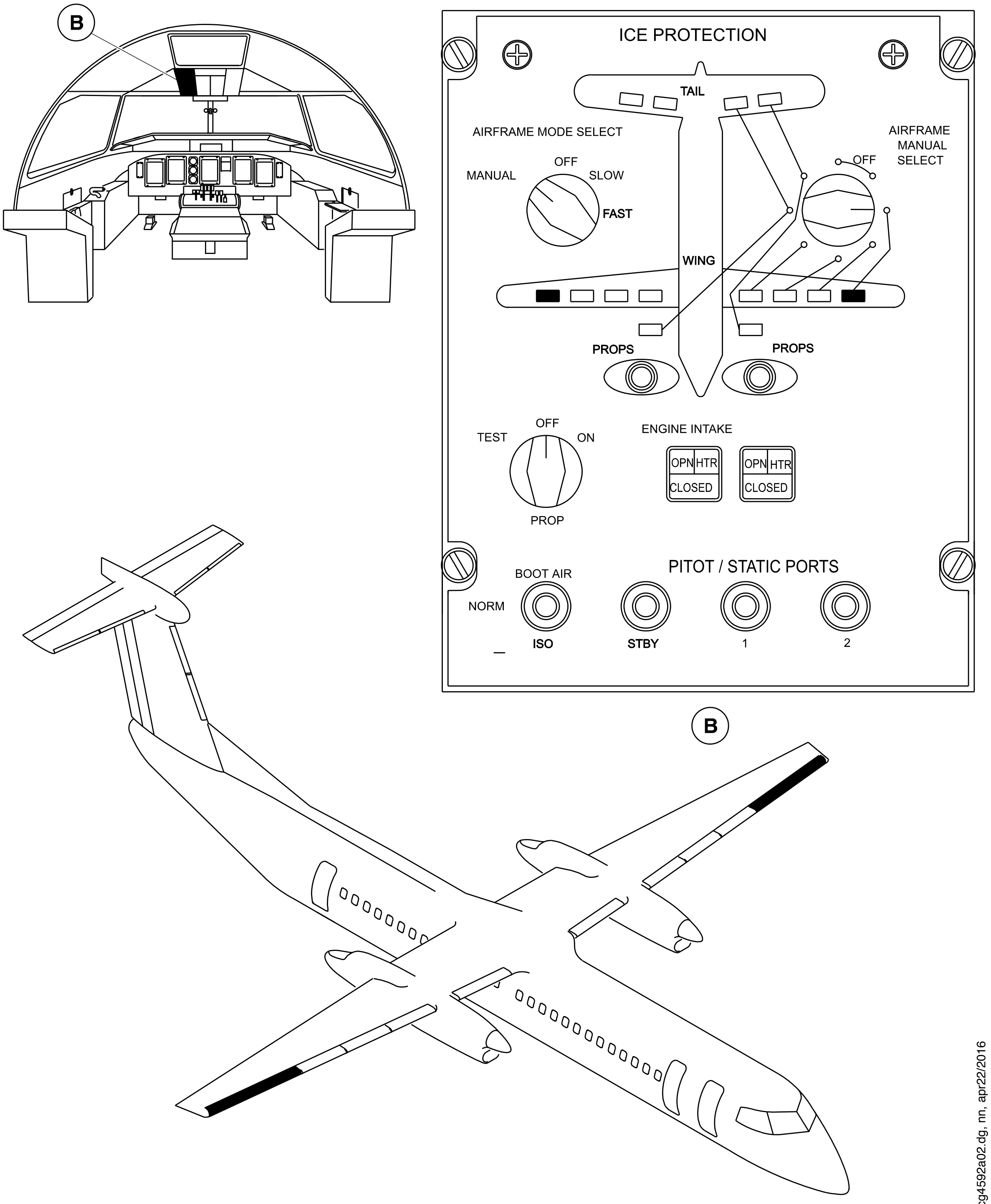
(10) Allow the bleed air system to stabilize for two minutes (minimum).

(11) On the copilot side panel, make sure that pressure indication 1 and pressure indication 2 increases to 18.0 ±1.5 psi (124.10 ±10.3 kPa) on the DEICE PRESS gauge.

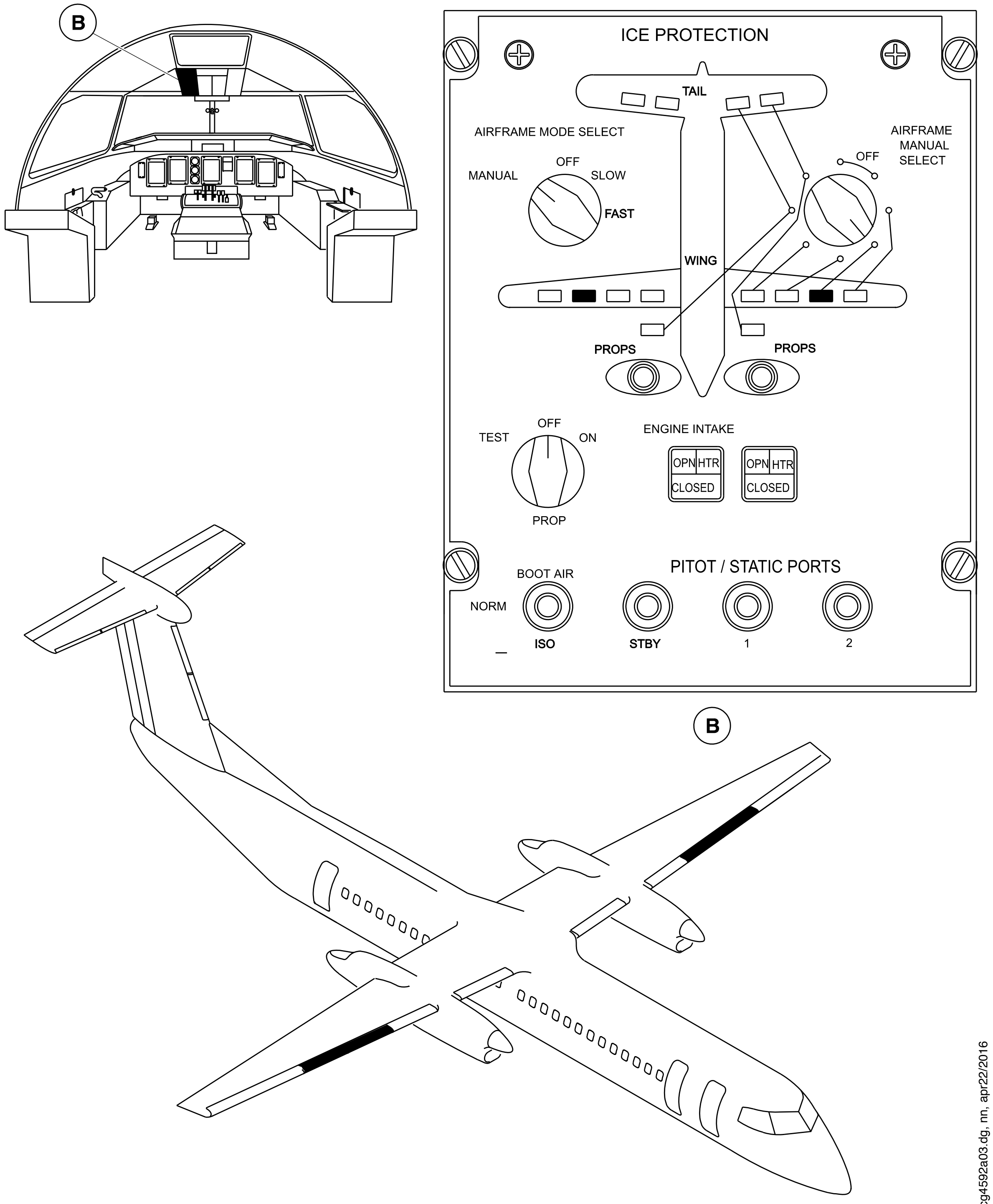
(12) Shutdown the right engine (Refer to TASK 71–00–00–868–802).



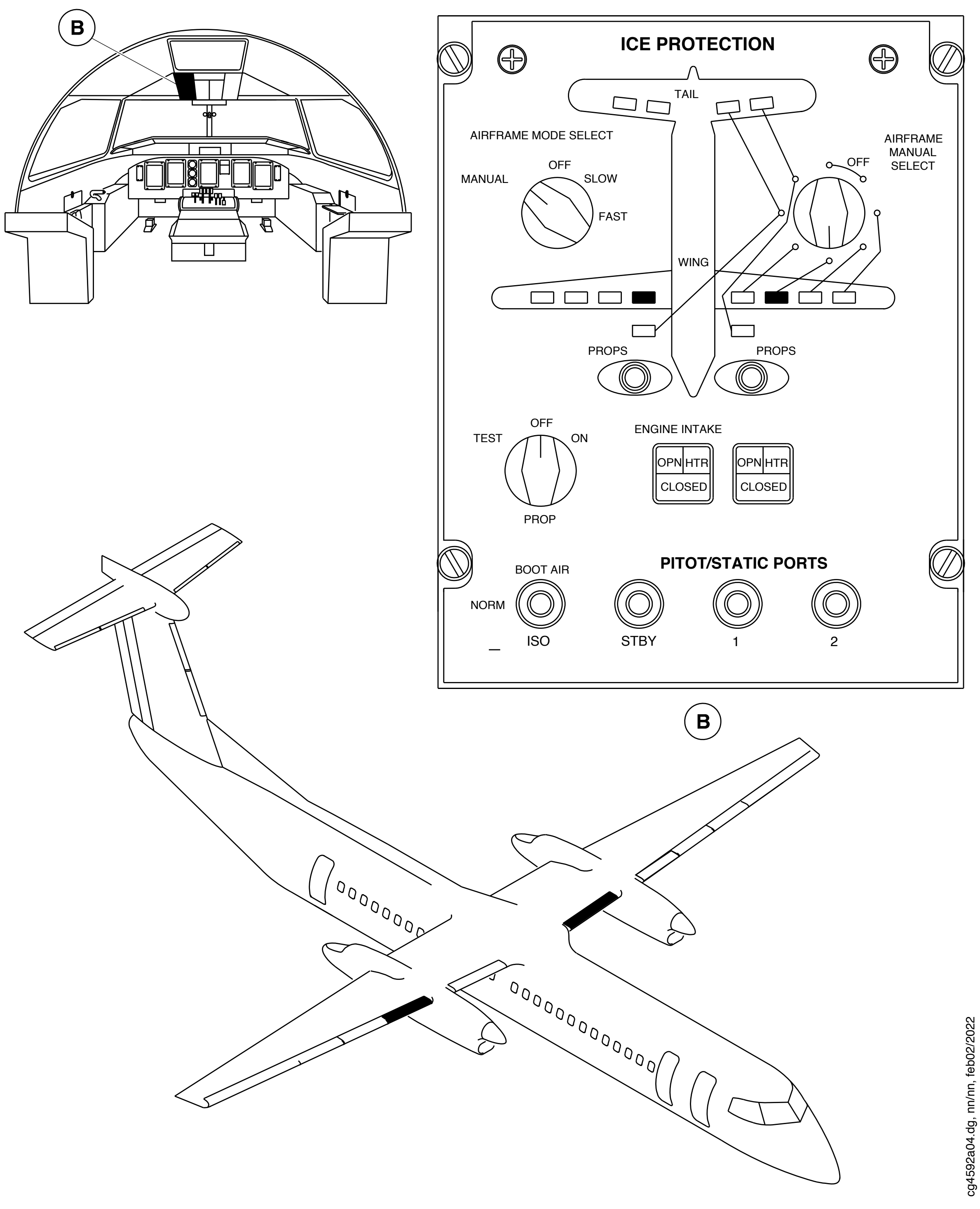
Deicing System – Operational Check
Figure 510 (Sheet 1 of 7)



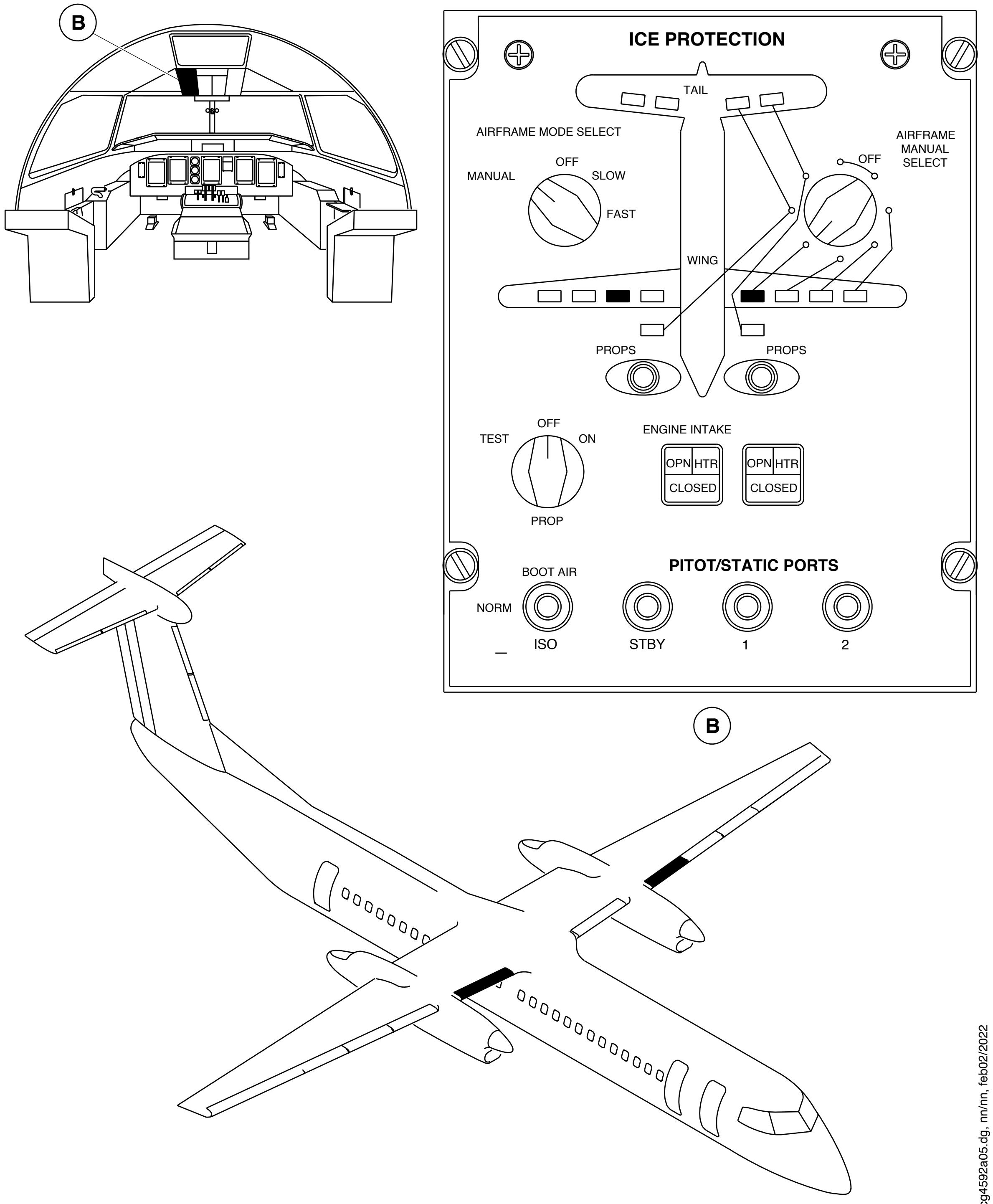
Deicing System – Operational Check
Figure 510 (Sheet 2 of 7)



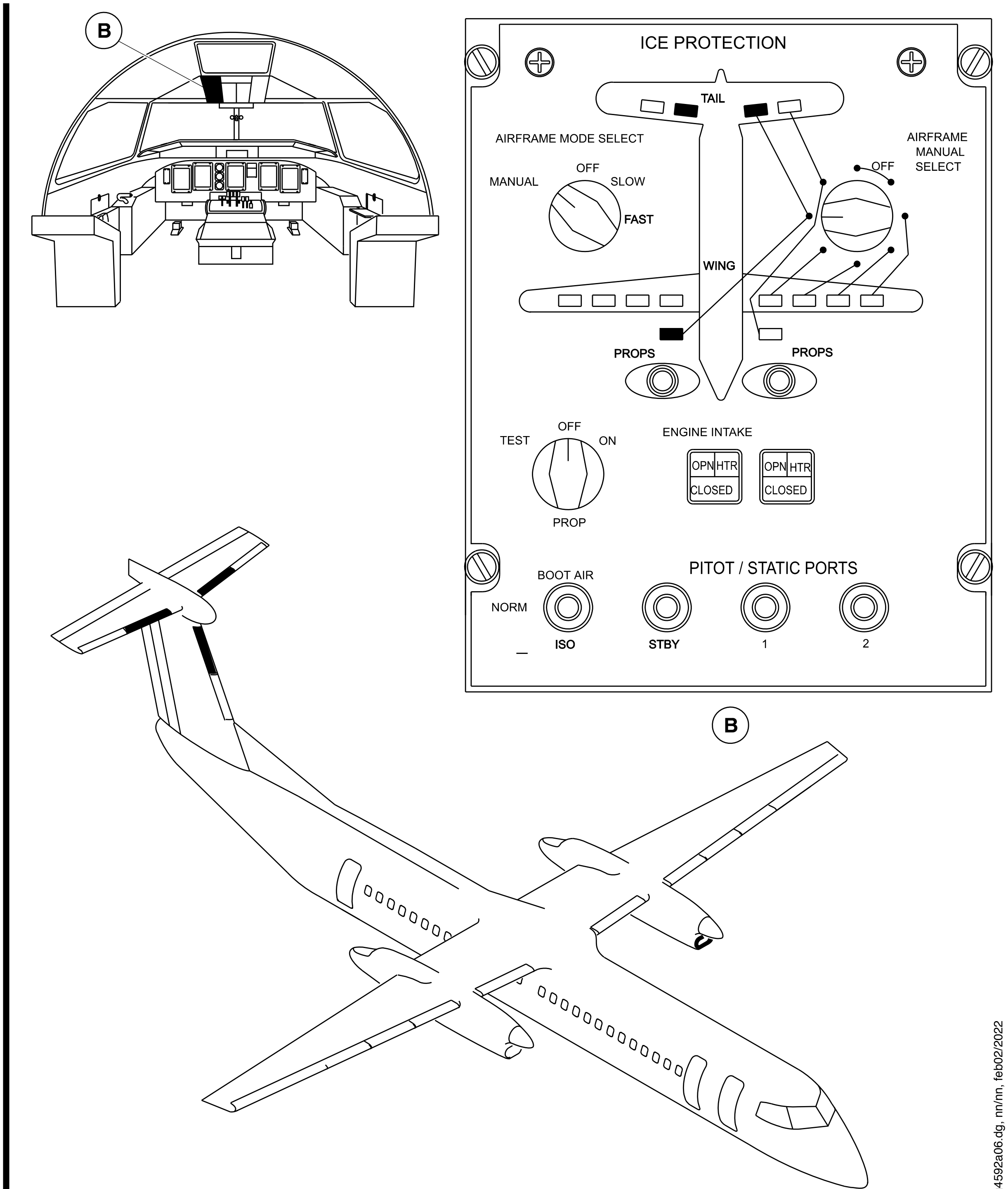
Deicing System – Operational Check
Figure 510 (Sheet 3 of 7)



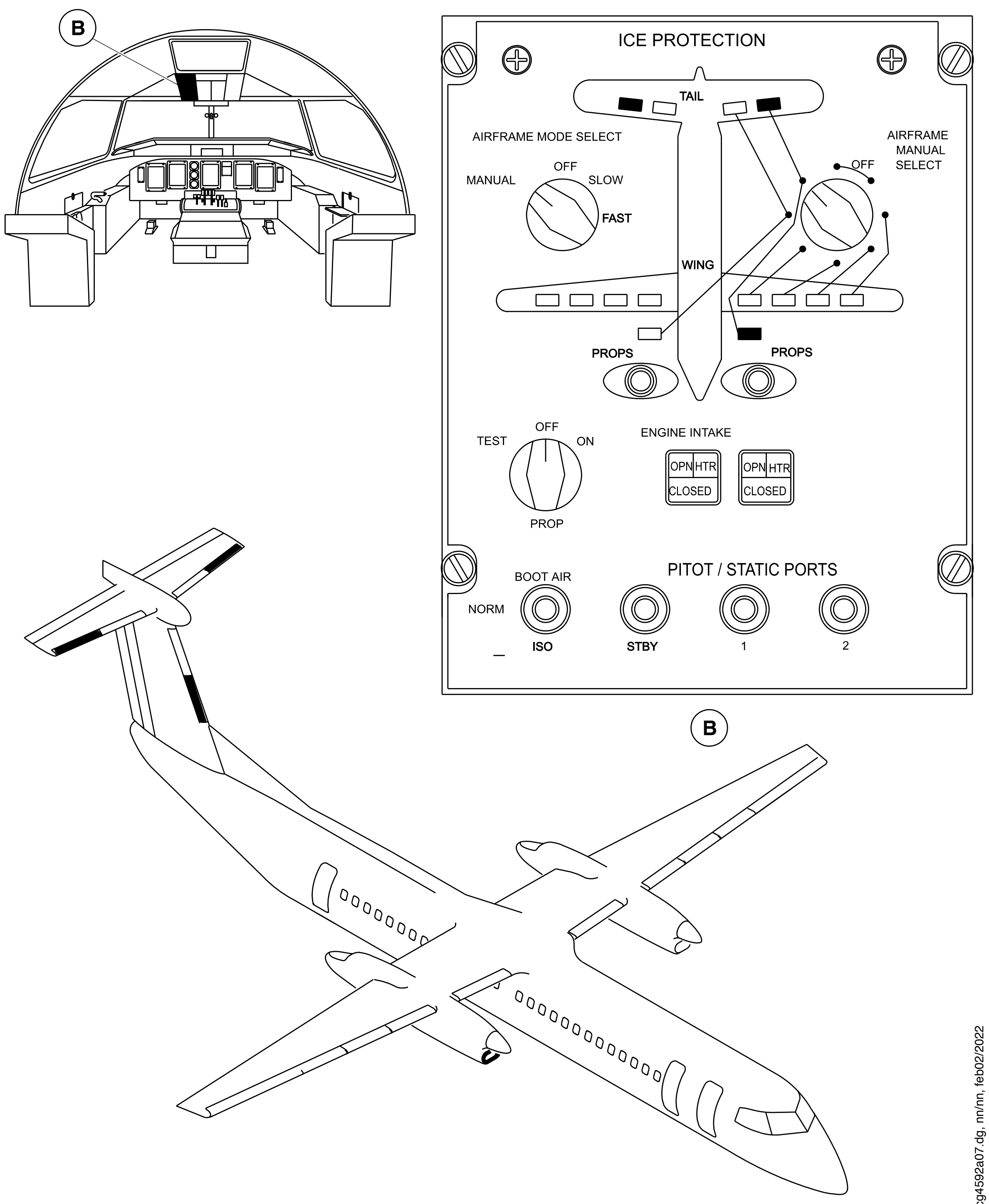
Deicing System – Operational Check
Figure 510 (Sheet 4 of 7)



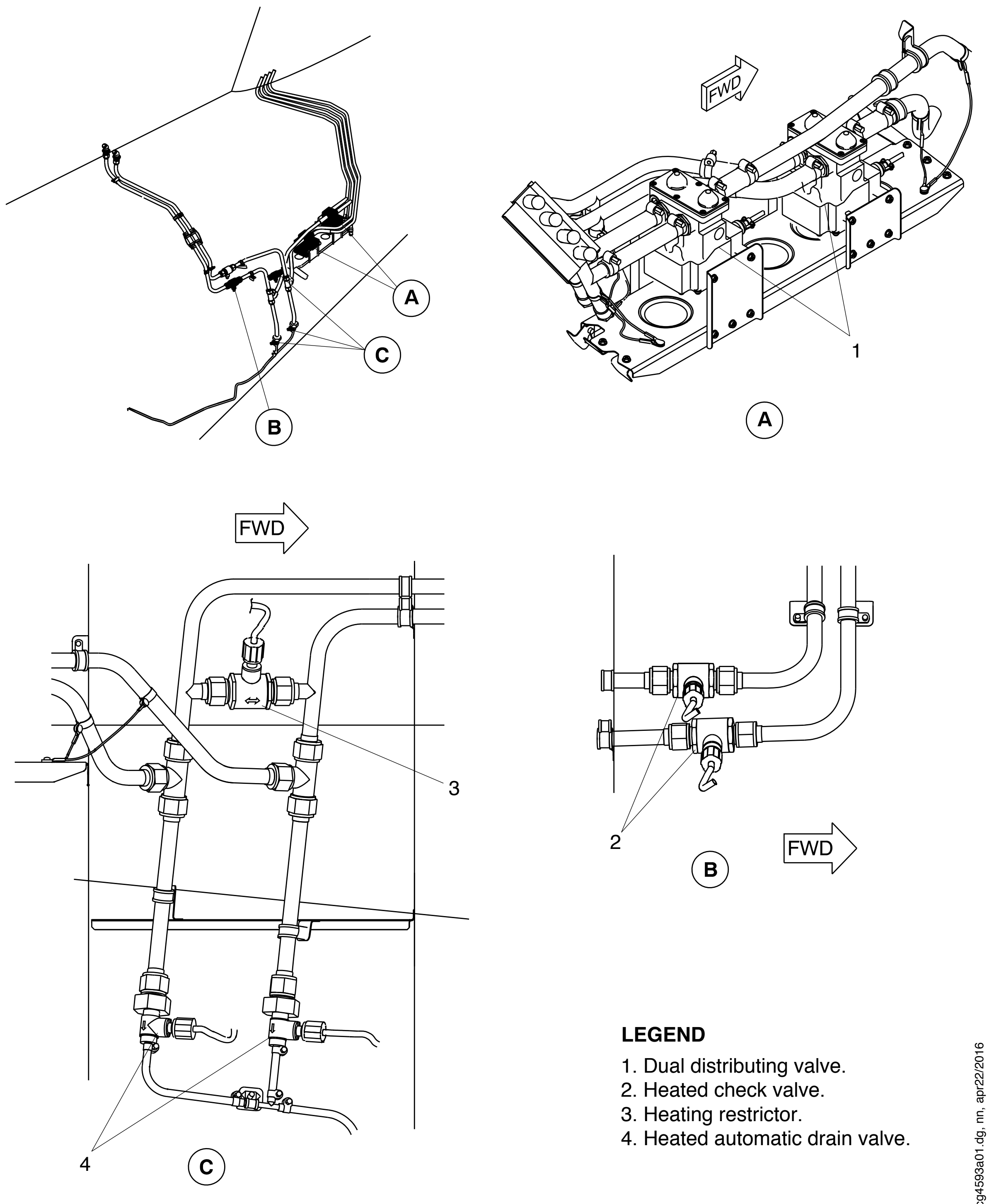
Deicing System – Operational Check
Figure 510 (Sheet 5 of 7)



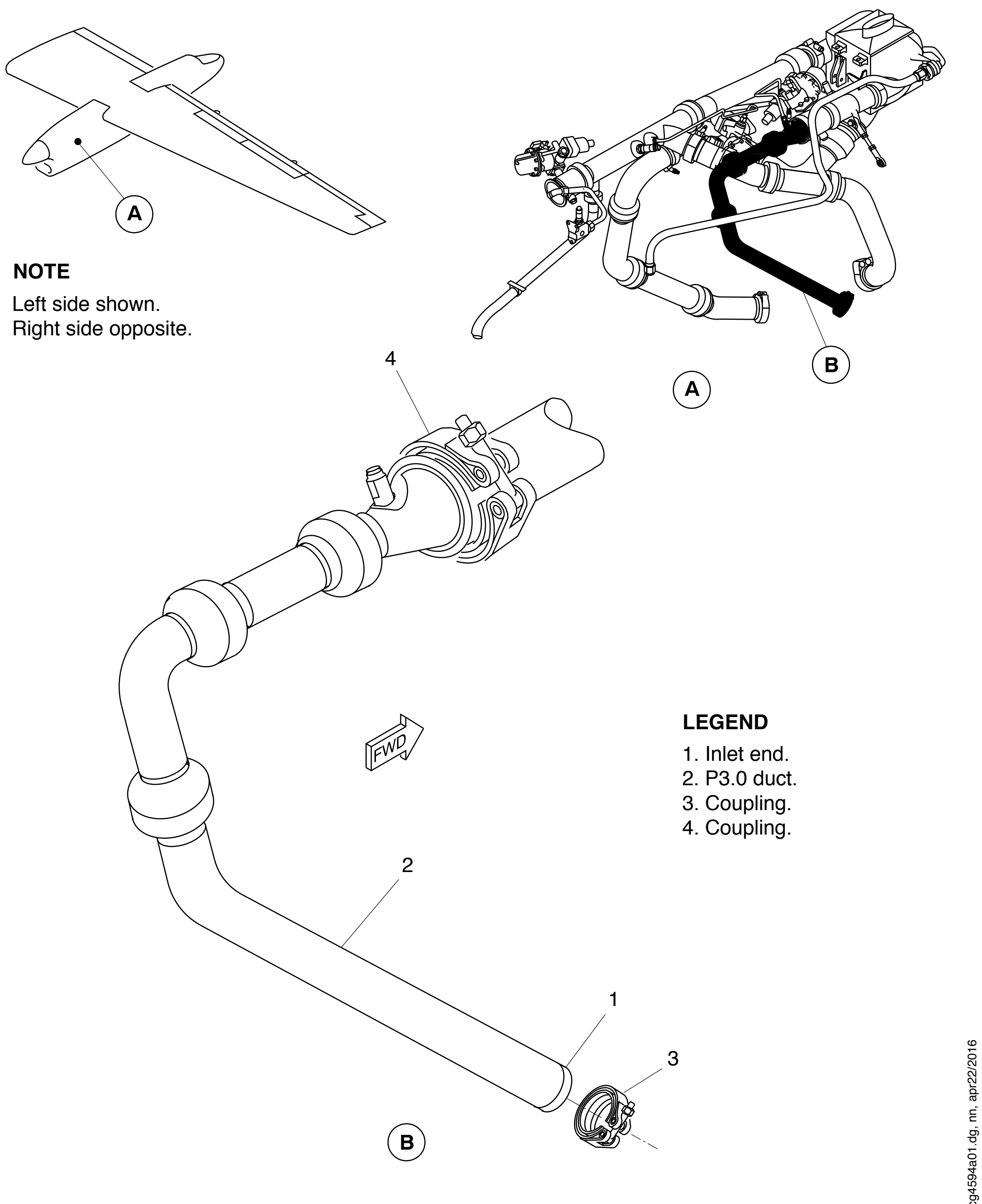
Deicing System – Operational Check
Figure 510 (Sheet 6 of 7)



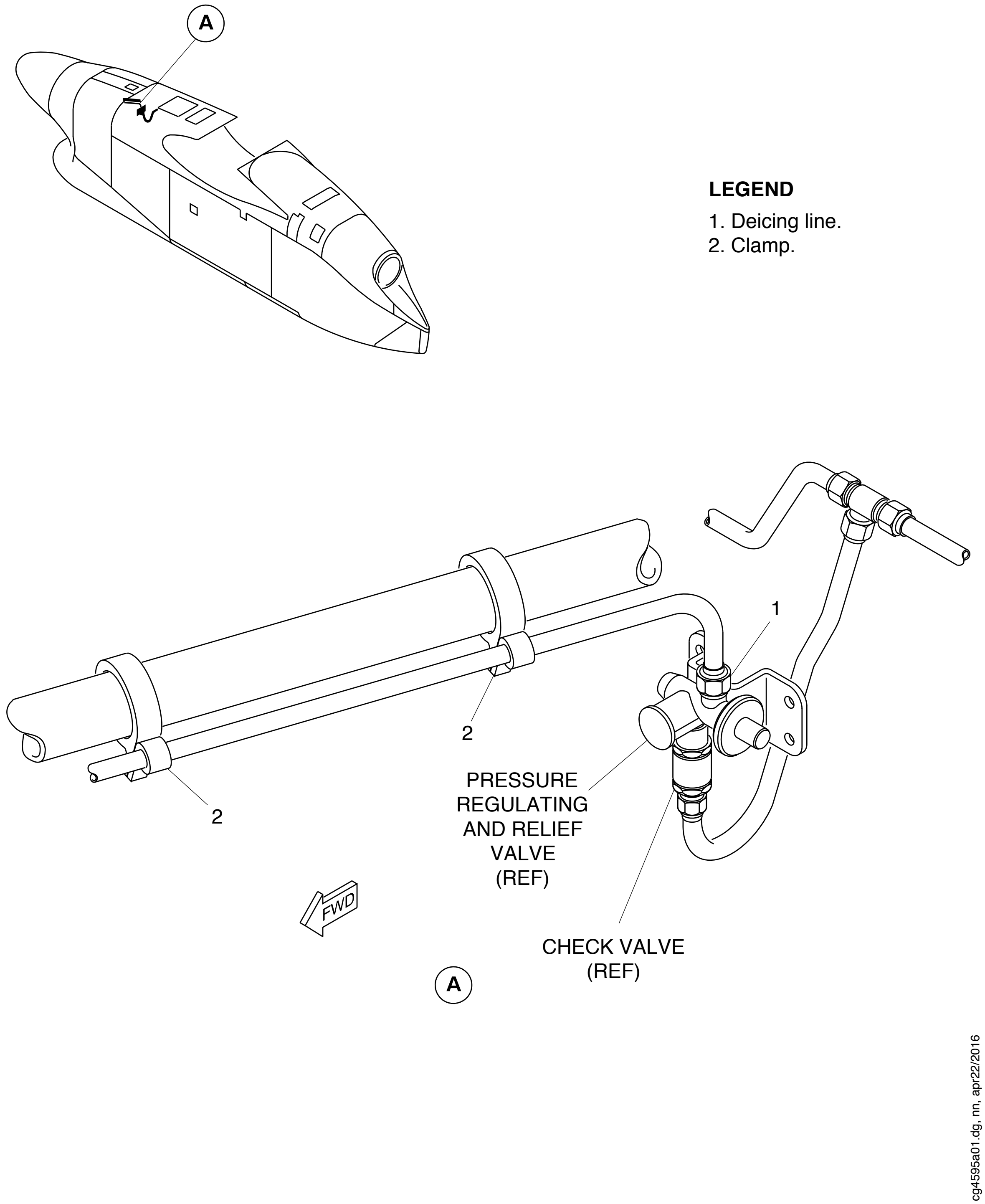
Deicing System – Operational Check
Figure 510 (Sheet 7 of 7)



Deicing System − Operational Check
Figure 511



Air Supply to the P3.0 Duct
Figure 512



LEGEND

1. Deicing line.
2. Clamp.

Deicing Line in Nacelle at PRV
Figure 513

VALVES, PRESSURE REGULATING AND RELIEF – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–11–06–710–801
Operational Check of the Deicing System Air Supply

1. General

   A. The maintenance procedure that follows is for the operational check of the deicing system air supply.

2. Job Set–Up Information

Subtask 30–11–06–946–005

   A. Reference Information

REFERENCE	DESIGNATION
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown

3. Procedure

Subtask 30–11–06–710–001

Refer to Figure 501.

   A. Do the operational check of the deicing system air supply as follows:

      (1) On the ICE PROTECTION panel, make sure that the BOOT AIR switch is set to the NORM position and the AIRFRAME MANUAL SELECTOR to OFF.

      (2) Start the left engine (Refer to TASK 71–00–00–868–801).

      (3) Make sure the AC generator toggle switch located on the AC control panel is set to GEN 1.

      (4) Move the condition lever to the MIN 850 detent.

      (5) Move the power lever to the FLIGHT IDLE position detent.

      (6) Set the two PACKS switches on the air conditioning panel to the AUTO position.

      (7) Set the BLEED 1 switch to the ON position, the BLEED 2 switch to the OFF position and the BLEED selector to the MAX position.

      (8) Let the system stabilize for three minutes.

      (9) On the copilot side panel, make sure that pressure indications 1 and 2 on the DEICE PRESS gauge increase (to approximately 18 psi).

      (10) Shutdown the left engine (Refer to TASK 71–00–00–868–802).

(11) Start the right engine (Refer to TASK 71–00–00–868–801).

(12) Move the condition lever to the MIN 850 detent.

(13) Move the power lever to the FLIGHT IDLE position detent.

(14) Set the two PACKS switches on the air conditioning panel to the AUTO position.

(15) Set the BLEED 2 switch to the ON position, the BLEED 1 switch to the OFF position and the BLEED selector to the MAX position.

(16) Let the system stabilize for three minutes.

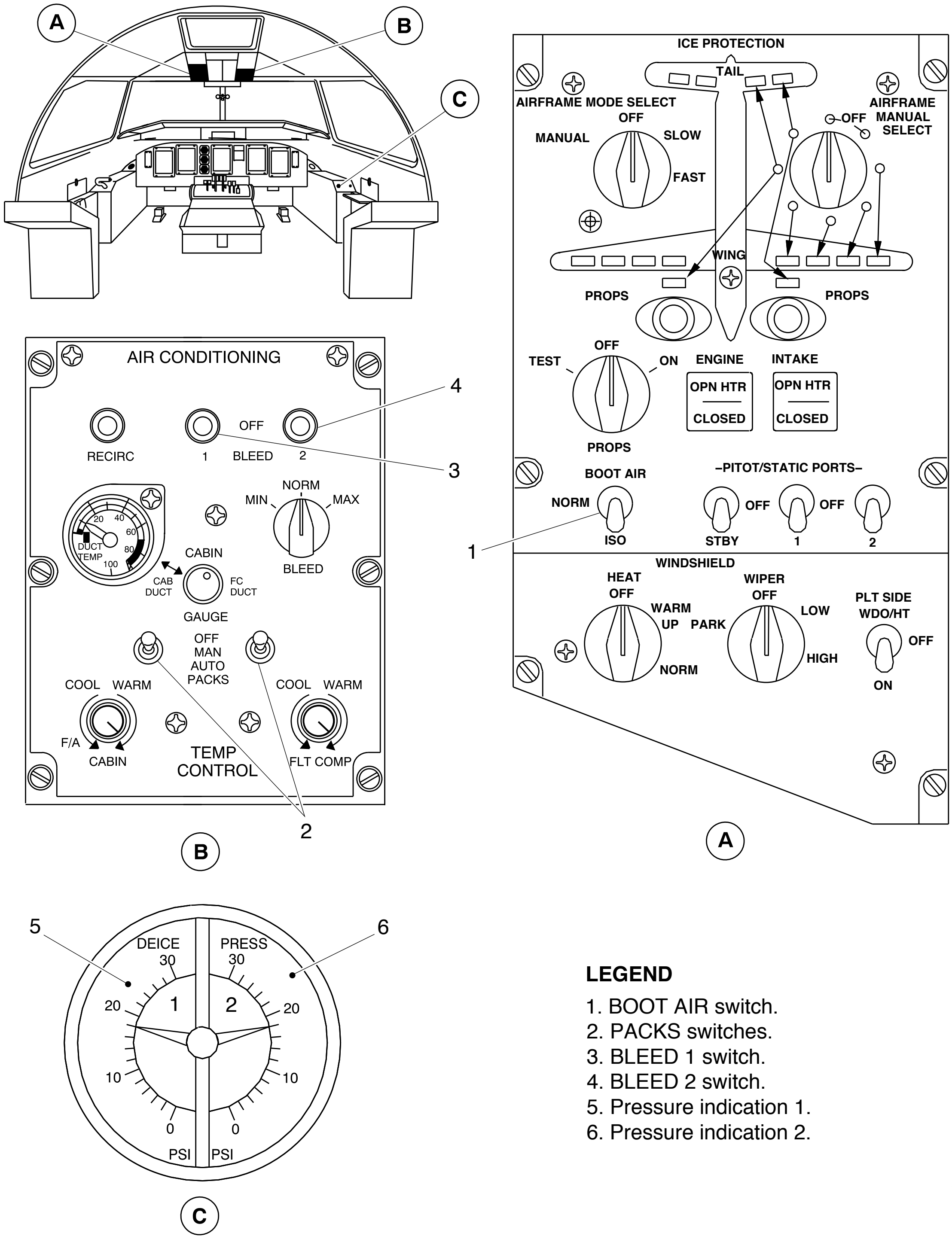
(17) On the copilot side panel, make sure that pressure indications 1 and 2 on the DEICE PRESS gauge increase (to approximately 18 psi).

(18) Shutdown the right engine (Refer to TASK 71–00–00–868–802).

4. Close Out

Subtask 30–11–06–941–003

A. Remove all tools, equipment, and unwanted materials from the work area.



LEGEND

1. BOOT AIR switch.
2. PACKS switches.
3. BLEED 1 switch.
4. BLEED 2 switch.
5. Pressure indication 1.
6. Pressure indication 2.

brx20a01.dg, sm/gw, aug24/2011

Deicing System – Operational Check
Figure 501

VALVE, ISOLATION SHUT–OFF – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–11–11–710–801
Operational Test of the Isolation Shut–off Valve

1. General

   A. The maintenance procedure that follows is for the operational test of the isolation shut–off valve (ISOV). The ISOV is installed behind the fuselage to wing fairing panels.

2. Job Set–Up Information

Subtask 30–11–11–946–003

   A. Reference Information

REFERENCE	DESIGNATION
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown

3. Procedure

Subtask 30–11–11–710–001

   A. Do an operational test of the ISOV as follows:

   NOTE: Do not operate the APU when you do the ISOV test procedures.

   (1) Make sure that BOOT AIR switch on the Ice Protection Panel is selected to ISO.

   (2) Start engine No. 2 (Refer to TASK 71–00–00–868–801).

   (3) Check right DEICE PRESS indicator.

   (4) The right DEICE PRESS indicator indicates about 18 +1.5/–1 psi and the left one indicates 0 psi.

   (5) Select BOOT AIR switch to NORM.

   (6) Both DEICE PRESS indicators indicate about 18 +1.5/–1 psi

4. Close Out

Subtask 30–11–11–868–001

   A. Shut down the engine (Refer to TASK 71–00–00–868–802).

Subtask 30–11–11–941–002

B. Remove all tools, equipment, and unwanted materials from the work area.

VALVES, DUAL DISTRIBUTING – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–11–16–710–801
Operational Test of the Dual Distributing Valves

1. General

   A. The maintenance procedure that follows is for the operational test of the dual distributing valve (written as the "DDV" in these procedures). The No. 1 to No. 4 DDVs are in the wings forward of the wing front spars. The No. 5 and No. 6 DDVs are in the aft fuselage. These procedures are applicable to all DDVs. The procedures for the left side system are given. Differences for the right side system or aft fuselage DDVs are identified.

2. Job Set–Up Information

Subtask 30–11–16–946–003

   A. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–910–801	Electrical/Electronic Safety Precautions
TASK 71–00–00–868–801	Engine start
TASK 71–00–00–868–802	Engine shutdown

3. Job Set–Up

Subtask 30–11–16–910–003

   A. Obey all the electrical/electronic safety precautions (Refer to TASK 24–00–00–910–801).

4. Procedure

Subtask 30–11–16–710–002

   A. Do an operational test of a dual distributing valve as follows:

      (1)

         (a) Start one engine (Refer to TASK 71–00–00–868–801).

         (b) On the ICE PROTECTION panel, make sure that BOOT AIR switch is set to NORM.

         (c) On the right side panel, make sure the two DEICE PRESS indicators show 18 +1.5/–1 psi.

         (d) Set the AIRFRAME MODE SELECT switch to MANUAL.

(e) Set the AIRFRAME MANUAL SELECT switch to a de–icer boot location.

1. Make sure the chosen de–icer boot indication light comes on.

2. Make sure the chosen deicer boots operate on the two wings.

(f) Set the AIRFRAME MANUAL SELECT switch to OFF.

(g) Set the AIRFRAME MODE SELECT switch to OFF.

(h) Set the AIRFRAME MODE SELECT switch to FAST, and do a complete cycle.

NOTE: When FAST mode is selected:

- ❖ The complete cycle should last no longer than 42 seconds.
- ❖ The next cycle shall start 63 seconds (plus or minus 2.5 seconds) after the first cycle has started.

(i) Shut down the engine (Refer to TASK 71–00–00–868–802).

5. Close Out

Subtask 30–11–16–941–002

A. Remove all tools, equipment, and unwanted materials from the work area.

CHECK VALVES, NON–HEATED – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–11–21–710–801
Operational Test of the Non–heated Check Valves

1. General

   A. The maintenance procedure that follows is for the operational test of the non–heated check valves (written as the “NHCVs” in these procedures). The NHCVs are installed in the nacelles.

2. Job Set–Up Information

Subtask 30–11–21–946–002

   A. Reference Information

REFERENCE	DESIGNATION
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown

3. Procedure

Subtask 30–11–21–710–002

   A. Do an operational test of the non–heated check valves as follows:

      NOTE: Do not operate the APU when you do these test procedures.

      (1) Check that the BOOT AIR toggle switch is in the NORM position on the ICE PROTECTION panel.

      (2) Start the left engine (Refer to TASK 71–00–00–868–801).

      (3) Pressure increase, check as follows:

         (a) DE–ICE PRESS caution light shall extinguish.

         (b) The Dual Pressure Indicator (DPI) shall indicate 18 psig on both sides.

         (c) DE–ICE PRESS caution light remains off.

      (4) Shut down the left engine (Refer to TASK 71–00–00–868–802).

      (5) Check as follows:

         (a) DE–ICE PRESS caution light shall illuminate

         (b) DPI shall indicate 0 psig on both sides.

      (6) Start the right engine (Refer to TASK 71–00–00–868–801).

      (7) Pressure increase, check as follows:

(a) DE–ICE PRESS caution light shall extinguish.

(b) The Dual Pressure Indicator (DPI) shall indicate 18 psig on both sides.

(c) DE–ICE PRESS caution light remains off.

4. Close Out

Subtask 30–11–21–868–001

A. Shut down the right engine (Refer to TASK 71–00–00–868–802).

SWITCHES, LOW PRESSURE WARNING – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–11–31–710–801
Operational Test of the Low Pressure Warning Switches

1. General

   A. The maintenance procedure that follows is for the functional test of the two Low Pressure Warning Switches (LPWS). The LPWS are installed in the fuselage to wing fairing.

2. Job Set–Up Information

Subtask 30–11–31–946–003

   A. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize the Electrical System
TASK 24–00–00–861–802	De–energize the Electrical System
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown

3. Job Set–Up

Subtask 30–11–31–861–001

   A. Energize the electrical system (Refer to TASK 24–00–00–861–801).

Subtask 30–11–31–865–003

   B. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT DC, M5	DE–ICE CONT
LEFT DC, A7	DE–ICE CONT
RIGHT DC, S5	DE–ICE CONT
LEFT DC, N8	AIRFRAME DEICE PRESS IND 1
RIGHT DC, A7	AIRFRAME DEICE PRESS IND 2
RIGHT DC, K7	AIRFRAME MANUAL CONT
RIGHT DC, M4	CAUT/WRN LTS 1
RIGHT DC, N4	CAUT/WRN LTS 2

4. Procedure

Subtask 30–11–31–710–002

A. Do an operational test of the Low Pressure Warning Switch as follows:

NOTE: Aircraft must not have the APU operating when performing Isolation Low Pressure Warning Switch test procedures.

(1) Make sure that BOOT AIR is selected to NORM.

(2) Check the CAUTION PANEL:

(a) DE–ICE PRESS caution light is illuminated.

(3) Start an engine (Refer to TASK 71–00–00–868–801).

(4) Check as follows:

(a) Pressure increase and the DE–ICE PRESS caution light shall extinguish after 30 seconds maximum.

(b) The Dual Pressure Indicator (DPI) shall indicate 18 psig on both sides.

(5) Shut down the engine (Refer to TASK 71–00–00–868–802).

(6) Check as follows:

(a) DE–ICE PRESS caution light shall illuminate.

(b) DPI shall indicate 0 psig on both sides.

5. Close Out

Subtask 30–11–31–861–002

A. De–energize the electrical system (Refer to TASK 24–00–00–861–802).

CHECK VALVES, HEATED – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–11–46–710–801
Operational Test of the Heated Check Valves

1. General

   A. The maintenance procedure that follows is for the operational test of the Heated Check Valves (HCV). The HCV are installed in the Rear Fuselage.

2. Job Set–Up Information

Subtask 30–11–46–946–003

   A. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize the Electrical System
TASK 24–00–00–861–802	De–energize the Electrical System
TASK 71–00–00–868–801	Engine start
TASK 71–00–00–868–802	Engine shutdown

3. Job Set–Up

Subtask 30–11–46–861–001

   A. Energize the electrical system (Refer to TASK 24–00–00–861–801).

Subtask 30–11–46–865–003

   B. Make sure that the circuit breakers that follow are closed :

CB PANEL AND CB NO	NAME
RIGHT DC, S6	AFR DEICE BOOT LTS
RIGHT DC, K7	AIRFRAME MANUAL CONT
RIGHT DC, A7	AIRFRAME DEICE–PRESS IND 1
RIGHT DC, B8	AIRFRAME DEICE–CV/DRAIN HTRS
RIGHT DC, S5	DEICE CONT
LEFT DC, A7	DEICE CONT
LEFT DC, M5	DEICE CONT
LEFTDC, N8	AIRFRAME DEICE–PRESS IND 2
LEFTDC, R8	AIRFRAME DEICE–CV/DRAIN HTRS

4. Procedure

Subtask 30–11–46–710–001

A. Do an operational test of the Heated Check Valve as follows:

(1) Set the BOOT AIR switch located on the ICE PROTECTION panel to ISO.

(2) Start the left engine (Refer to TASK 71–00–00–868–801).

(3) Check as follows:

(a) Pressure increase and the DE–ICE PRESS caution light stay illuminated.

(b) The Dual Pressure Indicator (DPI) shall indicate 18 psig on the left side and 0 psig on the right side.

(4) Visually check that all boots are held on the fin and on the horizontal stabilizer and maintain their airfoil shape.

(5) On the ICE PROTECTION panel, select the MANUAL MODE with the Airframe Mode Selector switch.

(6) Turn the Airframe Manual Selector clockwise to select both positions allowing boot inflation on the horizontal stabilizer. Allow 10 seconds between both positions. After that select OFF.

(7) Check as follows:

(a) The advisory lights on the ICE PROTECTION panel associated to the position shall illuminate.

(b) Boots inflation shall be in accordance with selection.

(8) On the ICE PROTECTION panel select the Airframe Mode Selector to OFF.

(9) Shut down the left engine (Refer to TASK 71–00–00–868–802).

(10) Start the right engine (Refer to TASK 71–00–00–868–801).

(11) Check as follows:

(a) Pressure increase and the DE–ICE PRESS caution light stay illuminated.

(b) The Dual Pressure Indicator (DPI) shall indicate 0 psig on the left side and 18 psig on the right side.

(12) Visually check that all boots are held on the fin and on the horizontal stabilizer and maintain their airfoil shape.

(13) On the ICE PROTECTION panel, select the MANUAL MODE with the Airframe Mode Selector switch.

(14) Turn the Airframe Manual Selector clockwise to select both positions allowing boot inflation on the horizontal stabilizer. Allow 10 seconds between both positions. After that select OFF.

(15) Check as follows:

(a) The advisory lights on the ICE PROTECTION panel associated to the position shall illuminate.

(b) Boots inflation shall be in accordance with selection.

(16) On the ICE PROTECTION panel select the Airframe Mode Selector to OFF.

(17) Set the BOOT AIR switch on the ICE PROTECTION panel to NORM.

(18) Shut down the right engine (Refer to TASK 71–00–00–868–802).

5. Close Out

Subtask 30–11–46–861–003

A. De–energize the electrical system (Refer to TASK 24–00–00–861–802).

ENGINE AIR INTAKE ANTI–ICING – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–21–00–710–801
Operational Test of the Engine Air Intake Anti–Icing

1. General

   A. The maintenance procedure that follows is for the operational test of the Engine Air Intake Anti–icing. The procedure for the left side system is given. The right side system is the same.

2. Job Set–Up Information

Subtask 30–21–00–943–001

   A. Consumable Materials

      (1) 13–04, Freeze mist

Subtask 30–21–00–946–001

   B. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–910–801	Electrical/Electronic Safety Precautions
TASK 24–00–00–910–802	Electrostatic Discharge Safety Precautions
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown
TASK 45–00–30–742–802	Retrieval of Data from the Central Diagnostic System (CDS) – Ice Protection System (ICE PROTECTION)
TASK 45–00–30–743–802	Erase the Data from the Central Diagnostic System (CDS) – Ice Protection System (ICE PROTECTION)

3. Job Set–Up

Subtask 30–21–00–910–001

   A. Obey all the electrical/electronic safety precautions (Refer to TASK 24–00–00–910–801).

   B. Obey all the electrostatic safety precautions (Refer to TASK 24–00–00–910–802).

Subtask 30–21–00–865–001

   C. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT DC, A7	DE–ICE CONT
LEFT DC, S5	DE–ICE CONT
LEFT DC, M5	DE–ICE CONT
RIGHT DC, D6	INTAKE DEFLECT ACT 2
LEFT DC, P6	INTAKE DEFLECT ACT 1
LEFT DC, H7	HYD SYS CONT
RIGHT DC, N4	CAUT/WRN LTS 2
RIGHT DC, M4	CAUT/WRN LTS 1
115V AC VARIABLE FREQUENCY,	INTK LIP HTR ENG 1
115V AC VARIABLE FREQUENCY,	INTK LIP HTR ENG 2

4. Procedure

**ON A/C 4001 and ON A/C 4003–4004, 4006, 4008, 4010, 4012–4017, 4019–4021, 4024–4029, 4034–4036, 4038–4045, 4052, 4054, 4056, 4058, 4063, 4065–4068, 4071–4080, 4082, 4084–4085, 4087–4118, 4120–4123, 4125–4128, 4130–4136, 4138–4140, 4142, 4144, 4146, 4148, 4151–4152, 4155, 4157–4159, 4162, 4165, 4168, 4170–4172, 4174, 4176–4177, 4179–4182, 4184 Pre SB84–30–10

Subtask 30–21–00–710–001

A. Do an operational test of the engine air intake anti–icing as follows:

(1) Start the applicable engine (Refer to TASK 71–00–00–868–801).

(a) Make sure that the two ENGINE INTAKE switch–lights are in the off position, before you start the test.

NOTE: Make sure there are no other electrical items are energized or other tests are done during this operational test. This will show the load changes on the anti–icing system.

(b) Make sure that the engine is at flight idle and the propeller is in the unfeathered position.

(c) On the AC CONTROL panel, set the applicable GEN switch to the on position.

(d) Open the electrical system page on the MFD.

(e) On the ARCDU retrieve the CDS fault indications (Refer to TASK 45–00–30–742–802) and note fault data related to the system.

(f) On the ARCDU erase the CDS fault indications (Refer to TASK 45–00–30–743–802)

(2) When the external air temperature is below 50° F (10° C), do the test that follows:

NOTE: The air intake heater thermostat comes on when the temperature is 59° F (15° C).

(a) Push the ENGINE INTAKE switch–light and check the respective generator Phase B shows an increase in electrical load.

(b) Make sure the HTR light of the ENGINE INTAKE switch–light comes on.

(c) Make sure the OPN light of the ENGINE INTAKE switch–light comes on.

(d) Make sure the air intake bypass–door opens.

NOTE: It will be necessary to have a second person monitor the position of the bypass–door from a safe location outside the aircraft.

(e) On the ARCDU retrieve the CDS fault indications (Refer to TASK 45–00–30–742–802) and make sure there is no related fault data.

(3) If the external air temperature is more than 50° F (10° C) or if there is no indication on first test, do the steps that follow:

(a) Use freeze mist to cool the airframe thermostat at Sta. –54.00 between stringer 32P and stringer 31P.

NOTE: The thermostat is installed behind the skin in this area.

(b) Push the ENGINE INTAKE switch–light and check the respective generator Phase B shows an increases in electrical load.

(c) Make sure the HTR light of the ENGINE INTAKE switch–light comes on.

(d) Make sure the OPN light of the ENGINE INTAKE switch–light comes on.

(e) Make sure the air intake bypass–door opens.

NOTE: It will be necessary to have a second person monitor the position of the bypass–door from a safe location outside the aircraft.

(f) On the ARCDU retrieve the CDS fault indications (Refer to TASK 45–00–30–742–802) and there is no related fault indication.

**ON A/C 4185–4199, 4201–4999 and ON A/C 4003–4004, 4006, 4008–4140, 4142–4184 Post SB84–30–10

Subtask 30–21–00–710–011

B. Do an operational test of the engine air intake anti–icing as follows:

(1) Start the applicable engine (Refer to TASK 71–00–00–868–801).

(a) Make sure that the two ENGINE INTAKE switch–lights are in the off position, before you start the test.

NOTE: Make sure there are no other electrical items are energized or other tests are done during this operational test. This will show the load changes on the anti–icing system.

(b) Make sure that the engine is at flight idle and the propeller is in the unfeathered position.

(c) On the AC CONTROL panel, set the applicable GEN switch to the on position.

(d) Open the electrical system page on the MFD.

(e) On the ARCDU retrieve the CDS fault indications (Refer to TASK 45–00–30–742–802) and note fault data related to the system.

(f) On the ARCDU erase the CDS fault indications (Refer to TASK 45–00–30–743–802)

(2) When the external air temperature is below 50° F (10° C), do the test that follows:

NOTE: The air intake heater thermostat comes on when the temperature is 59° F (15° C).

(a) Push the ENGINE INTAKE switch–light and check the respective generator Phase B shows an increase in electrical load.

(b) Make sure the HTR light of the ENGINE INTAKE switch–light comes on.

(c) Make sure the OPN light of the ENGINE INTAKE switch–light comes on.

(d) Make sure the air intake bypass–door opens.

NOTE: It will be necessary to have a second person monitor the position of the bypass–door from a safe location outside the aircraft.

(e) On the ARCDU retrieve the CDS fault indications (Refer to TASK 45–00–30–742–802) and make sure there is no related fault data.

(3) If the external air temperature is more than 50° F (10° C) or if there is no indication on first test, do the steps that follow:

(a) Use freeze mist to cool the airframe thermostat at Sta. –54.00 between stringer 26P and stringer 27P.

NOTE: The thermostat is installed behind the skin in this area.

(b) Push the ENGINE INTAKE switch–light and check the respective generator Phase B shows an increases in electrical load.

(c) Make sure the HTR light of the ENGINE INTAKE switch–light comes on.

(d) Make sure the OPN light of the ENGINE INTAKE switch–light comes on.

(e) Make sure the air intake bypass–door opens.

NOTE: It will be necessary to have a second person monitor the position of the bypass–door from a safe location outside the aircraft.

(f) On the ARCDU retrieve the CDS fault indications (Refer to TASK 45–00–30–742–802) and there is no related fault indication.

**ON A/C ALL

5. Close Out

Subtask 30–21–00–868–002

A. Shutdown the engine (Refer to TASK 71–00–00–868–802).

Subtask 30–21–00–941–001

B. Remove all tools, equipment, and unwanted materials from the work area.

**ON A/C ALL

TASK 30–21–00–710–802
Operational Check of the Engine Air Intake Adapter Heater (CMR#302100–101)

1. General

   A. The maintenance procedure that follows is for the operational check of the engine air–intake adapter–heater (written as the “adapter heater” in this procedure). The purpose of this procedure is to check the status of dormant failures of the heater circuit and the switching logic of the adapter heater system.

   B. This procedure is applicable to the left intake adapter heater and the right intake adapter heater. The procedure for the left intake adapter heater is given. Differences for the right intake adapter heater are identified.

   C. There is a main heater circuit and also a spare heater circuit for the adapter heater. The components that follow are tested in this procedure:

      – The main heater element

      – The spare heater element

      – The timer monitor unit (TMU) for the ice and rain protection system (IRPS)

      – The left engine relay K1

      – The right engine relay K2

      – The left selection relay K3

      – The right selection relay K4.

2. Job Set–Up Information

Subtask 30–21–00–943–002

   A. Tools and Equipment

(1)	GSB3021001 (or manufacture locally)	Test box (tester)

Subtask 30–21–00–944–001

   B. Consumable Materials

(1)	13–04	Mist, Freeze

Subtask 30–21–00–946–008

   C. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize the Electrical System
TASK 24–00–00–861–802	De–energize the Electrical System
TASK 24–00–00–910–801	Electrical/Electronic Safety Precautions
TASK 24–00–00–910–802	Electrostatic Discharge Safety Precautions
TASK 25–24–03–000–801	Removal of the Forward Lower Wardrobe
TASK 25–24–03–000–803	Removal of the Forward Lower Wardrobe with Wardrobe Insert
TASK 25–24–03–400–801	Installation of the Forward Lower Wardrobe
TASK 25–24–03–400–803	Installation of the Forward Lower Wardrobe with Wardrobe Insert
TASK 45–00–30–742–802	Retrieval of Data from the Central Diagnostic System (CDS) – Ice Protection System
TASK 45–00–30–743–802	Erase the Data from the Central Diagnostic System (CDS) – Ice Protection System
TASK 71–00–00–868–801	Engine Start
TASK 71–00–00–868–802	Engine Shutdown

3. Job Set–Up

Subtask 30–21–00–010–002

A. If necessary, remove the access panel that follows:

REFERENCE	DESIGNATION
121DZ	Flight Compartment Floor Panel

B. Remove the forward lower wardrobe as necessary to get access to the relay panel (Refer to TASK 25–24–03–000–801 or TASK 25–24–03–000–803 , as required).

Subtask 30–21–00–910–002

C. Obey all the electrical/electronic safety precautions (Refer to TASK 24–00–00–910–801).

D. Obey all the electrostatic safety precautions (Refer to TASK 24–00–00–910–802).

Subtask 30–21–00–861–007

E. Make sure that the electrical system is de–energized and that no other tests are done during this operational test.

Subtask 30–21–00–480–001

F. Attach the test box to the relay panel as follows:

(1) Remove connector 9811–P350 (for the left heater assembly) from the relay panel.

(2) Remove connector 9811–P354 (for the right heater assembly) from the relay panel.

(3) Attach connector D38999/26WF11SN to receptacle 9811–J350 on the relay panel.

(4) Attach connector D38999/26WF11SA to receptacle 9811–J354 on the relay panel.

(5) Attach connector 9811–P350 to connector D38999/24WF11PN on the test box.

(6) Attach connector 9811–P354 to connector D38999/24WF11PA on the test box.

(7) Make sure that the switches S1 and S2 on the test box are in the closed position.

(8) Attach the wire to an appropriate ground point with the alligator clip.

Subtask 30–21–00–865–002

G. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT DC, A7	DE–ICE CONT
LEFT DC, H5	ECIU A
LEFT DC, H7	HYD SYS CONT
LEFT DC, J4	ENG 1 FADEC A
LEFT DC, K4	ENG 2 FADEC A
LEFT DC, M5	DE–ICE CONT
LEFT DC, P6	INTAKE DEFLECT ACT 1
RIGHT DC, D6	INTAKE DEFLECT ACT 1
RIGHT DC, J4	ENG 1 FADEC B
RIGHT DC, K4	ENG 2 FADEC B
RIGHT DC, N4	CAUT/WRN LTS 2
RIGHT DC, M4	CAUT/WRN LTS 1
RIGHT DC, S5	DE–ICE CONT
115V AC VARIABLE FREQUENCY	INTK LIP HTR ENG 1
115V AC VARIABLE FREQUENCY	INTK LIP HTR ENG 2
115V AC VARIABLE FREQUENCY	L TRU
115V AC VARIABLE FREQUENCY	R TRU

4. Procedure

Refer to Figure 501, Figure 502, Figure 503, Figure 504 and to Figure 505.

Subtask 30–21–00–710–002

NOTE: This procedure is done in conjunction with an engine run.

A. Do the operational check of the left (No. 1 engine) intake adapter heater as follows:

(1) Start the No. 1 engine (Refer to TASK 71–00–00–868–801).

(2) Make sure that:

– The two ENGINE INTAKE switch–lights are not illuminated

– The NE 1, NE 2 and NE 3 lights on the test box are not illuminated.

(3) If the outside air temperature (OAT) is more than 55 °F (13 °C), use Freeze Mist (or equivalent) to cool the airframe thermostats (located behind the skin) at FS –54.00:

- Between stringer 32P and stringer 31P on Pre Modsum 4–126402 (SB 84–30–10) aircraft

  or

- Between stringer 27P and stringer 26P on Post Modsum 4–126402 (SB 84–30–10) aircraft.

NOTE: This will simulate a closed airframe thermostat for the test. The air–intake heater thermostat is closed when it senses the OAT is approximately 55 °F (13 °C) or less.

NOTE: As an alternative (if the Freeze Mist is not available), you can put a jumper wire between the splices that follow to bypass the applicable thermostat:

– Put a jumper wire between the splices 3000–E10 and 3000–E11 that are located below the flight compartment floor on the left side.

(4) Push the No. 1 ENGINE INTAKE switch–light. Make sure that:

– The HTR light of the ENGINE INTAKE switch–light comes on

– The NE 1 light on the test box comes on

– The NE 2 light on the test box comes on.

(5) On the test box, set the switch S1 to the open position (to simulate the failure of the main heater element). Make sure that:

– The NE 1 light remains on

– The NE 2 light goes out

– The NE 3 light comes on.

(6) Push the No. 1 ENGINE INTAKE switch–light again. Make sure that:

– The HTR light of the ENGINE INTAKE switch–light goes out

– The NE 1, NE 2 and NE 3 lights on the test box go out.

(7) Shut down the No. 1 engine (Refer to TASK 71–00–00–868–802).

B. Do the operational check of the right (No. 2 engine) intake adapter heater as follows:

(1) Start the No. 2 engine (Refer to TASK 71–00–00–868–801).

(2) Make sure that:

– The two ENGINE INTAKE switch–lights are not illuminated

– The NE 4, NE 5 and NE 6 lights on the test box are not illuminated.

(3) If the outside air temperature (OAT) is more than 55 °F (13 °C), use Freeze Mist (or equivalent) to cool the airframe thermostats (located behind the skin) at FS –54.00:

- Between stringer 32P and stringer 31P on Pre Modsum 4–126402 (SB 84–30–10) aircraft

  or

- Between stringer 27P and stringer 26P on Post Modsum 4–126402 (SB 84–30–10) aircraft.

NOTE: This will simulate a closed airframe thermostat for the test. The air–intake heater thermostat is closed when it senses the OAT is approximately 55 °F (13 °C) or less.

NOTE: As an alternative (if the Freeze Mist is not available), you can put a jumper wire between the splices that follow to bypass the applicable thermostat:

– Put a jumper wire between the splices 3000–E8 and 3000–E9 that are located below the flight compartment floor on the right side.

(4) Push the No. 2 ENGINE INTAKE switch–light. Make sure that:

– The HTR light of the ENGINE INTAKE switch–light comes on

– The NE 4 light on the test box comes on

– The NE 5 light on the test box comes on.

(5) On the test box, set the switch S2 to the open position (to simulate the failure of the main heater element). Make sure that:

– The NE 4 light remains on

– The NE 5 light goes out

– The NE 6 light comes on.

(6) Push the No. 2 ENGINE INTAKE switch–light again. Make sure that:

– The HTR light of the ENGINE INTAKE switch–light goes out

– The NE 4, NE 5 and NE 6 lights on the test box go out.

(7) Shut down the No. 2 engine (Refer to TASK 71–00–00–868–802).

5. Close Out

Subtask 30–21–00–080–001

A. Remove the test box as follows:

(1) Disconnect the wire from the ground point.

(2) Remove connector 9811–P350 from the test box.

(3) Remove connector 9811–P354 from the test box.

(4) Remove connector D38999/26WF11SN from the relay panel.

(5) Remove connector D38999/26WF11SA from the relay panel.

(6) Attach connector 9811–P350 to receptacle 9811–J350 on the relay panel.

(7) Attach connector 9811–P354 to receptacle 9811–J354 on the relay panel.

Subtask 30–21–00–743–002

B. Remove the Ice and Rain Protection System (IRPS) faults from the Central Diagnostic System (CDS) as follows:

(1) Energize the electrical system (Refer to TASK 24–00–00–861–801). DC power is required for this step.

NOTE: Use the auxiliary power unit (APU) or a ground power cart to energize the electrical system (apply DC electrical power).

(2) Erase the IRPS data (faults) from the CDS (Refer to TASK 45–00–30–743–802).

(3) De–energize the electrical system (Refer to TASK 24–00–00–861–802).

Subtask 30–21–00–941–008

C. Remove all tools, equipment, and unwanted materials from the work area.

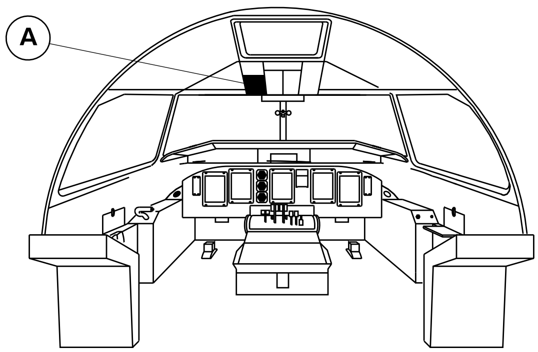
Subtask 30–21–00–410–004

D. Install the forward lower wardrobe (Refer to TASK 25–24–03–400–801 or TASK 25–24–03–400–803 , as required).

Subtask 30–21–00–410–002

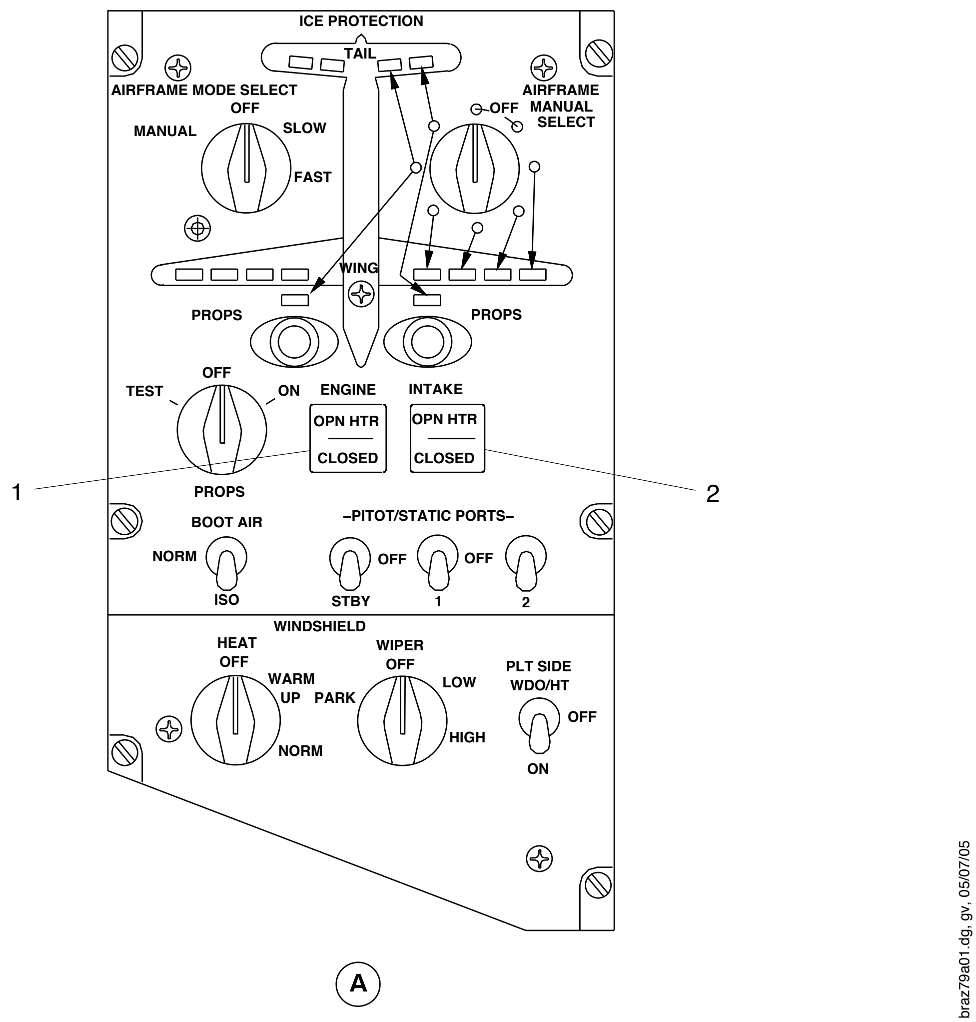
E. If removed for access, install the floor panel that follows:

REFERENCE	DESIGNATION
121DZ	Flight Compartment Floor Panel

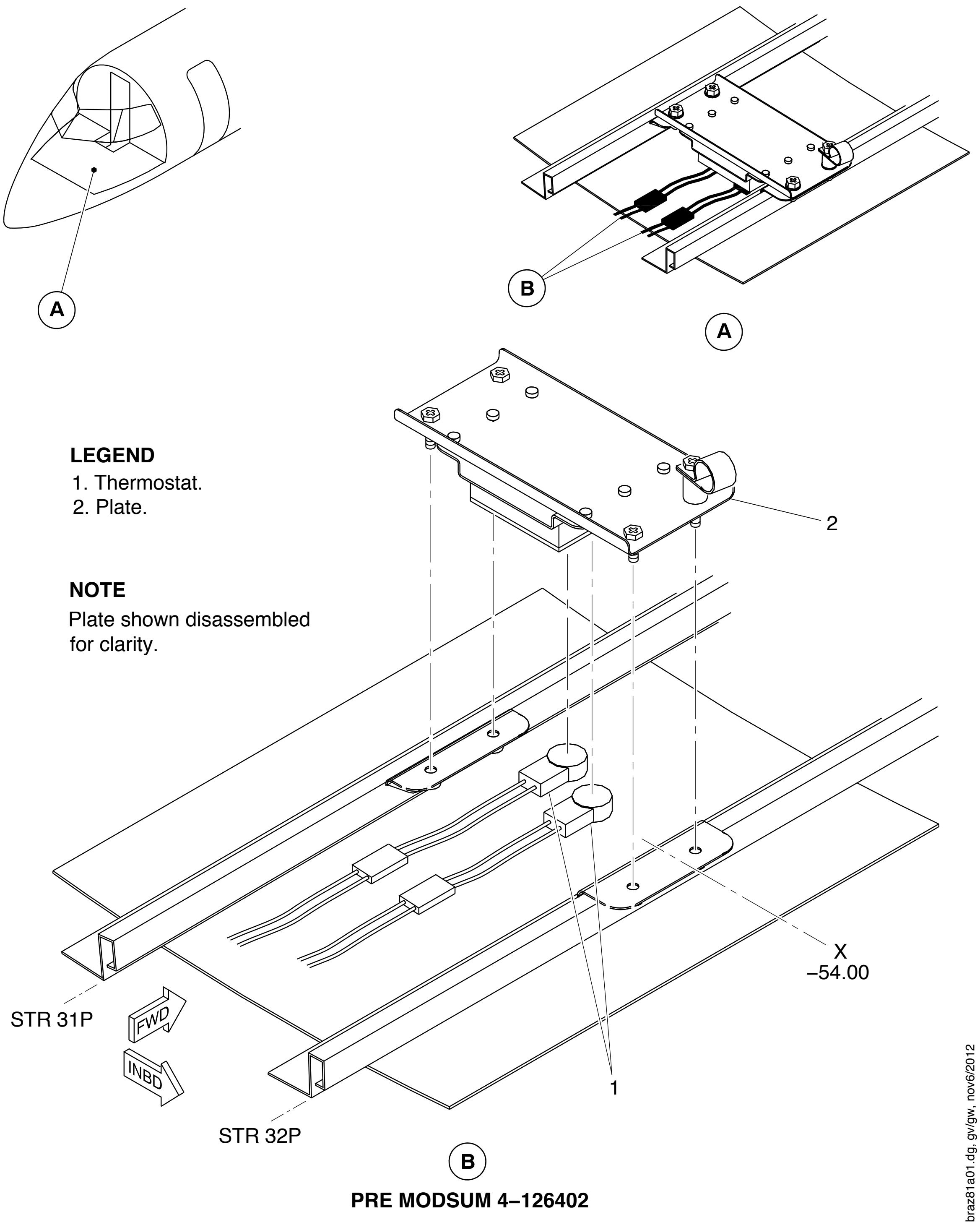


LEGEND

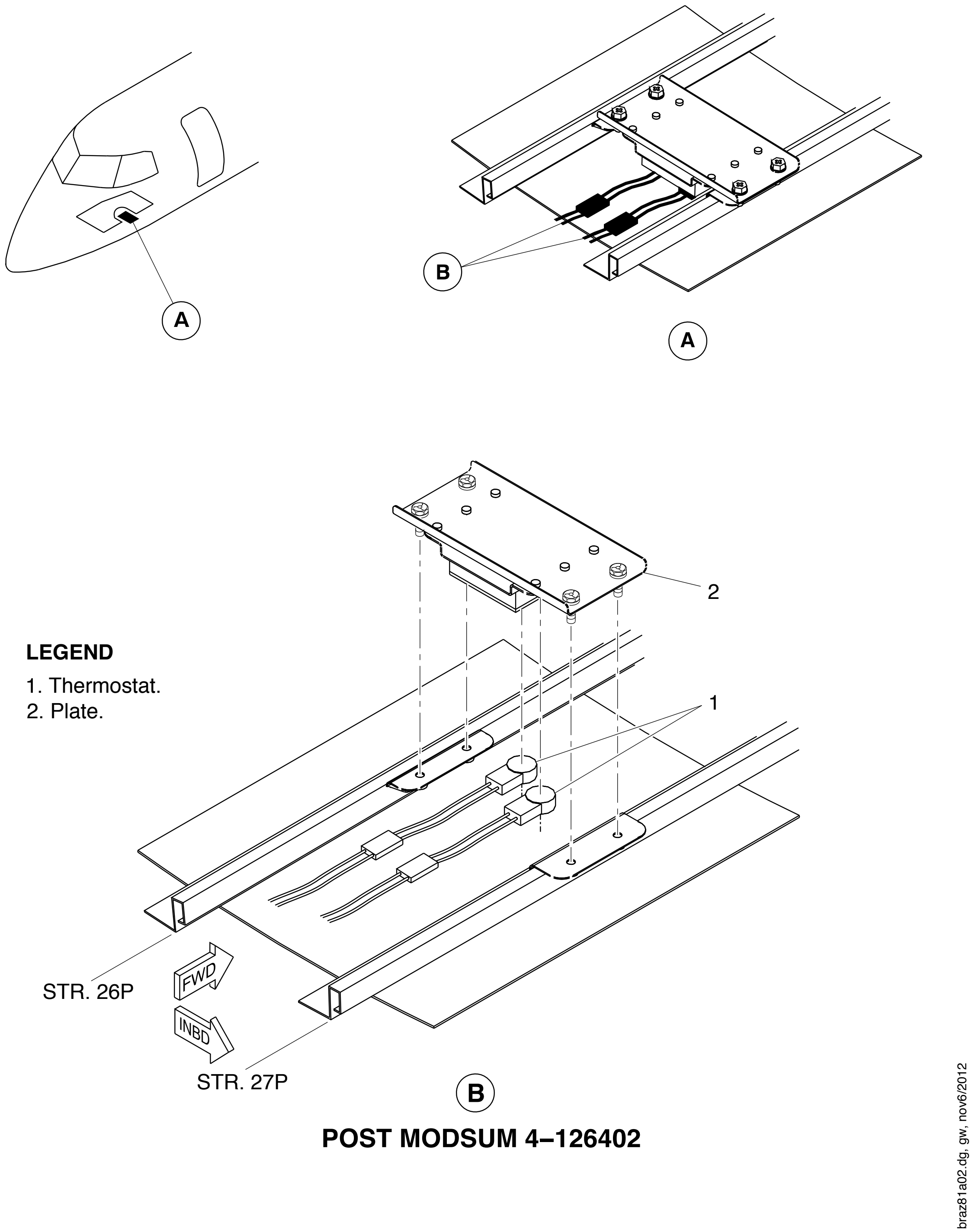
1. Left engine intake switch–light.
2. Right engine intake switch–light.



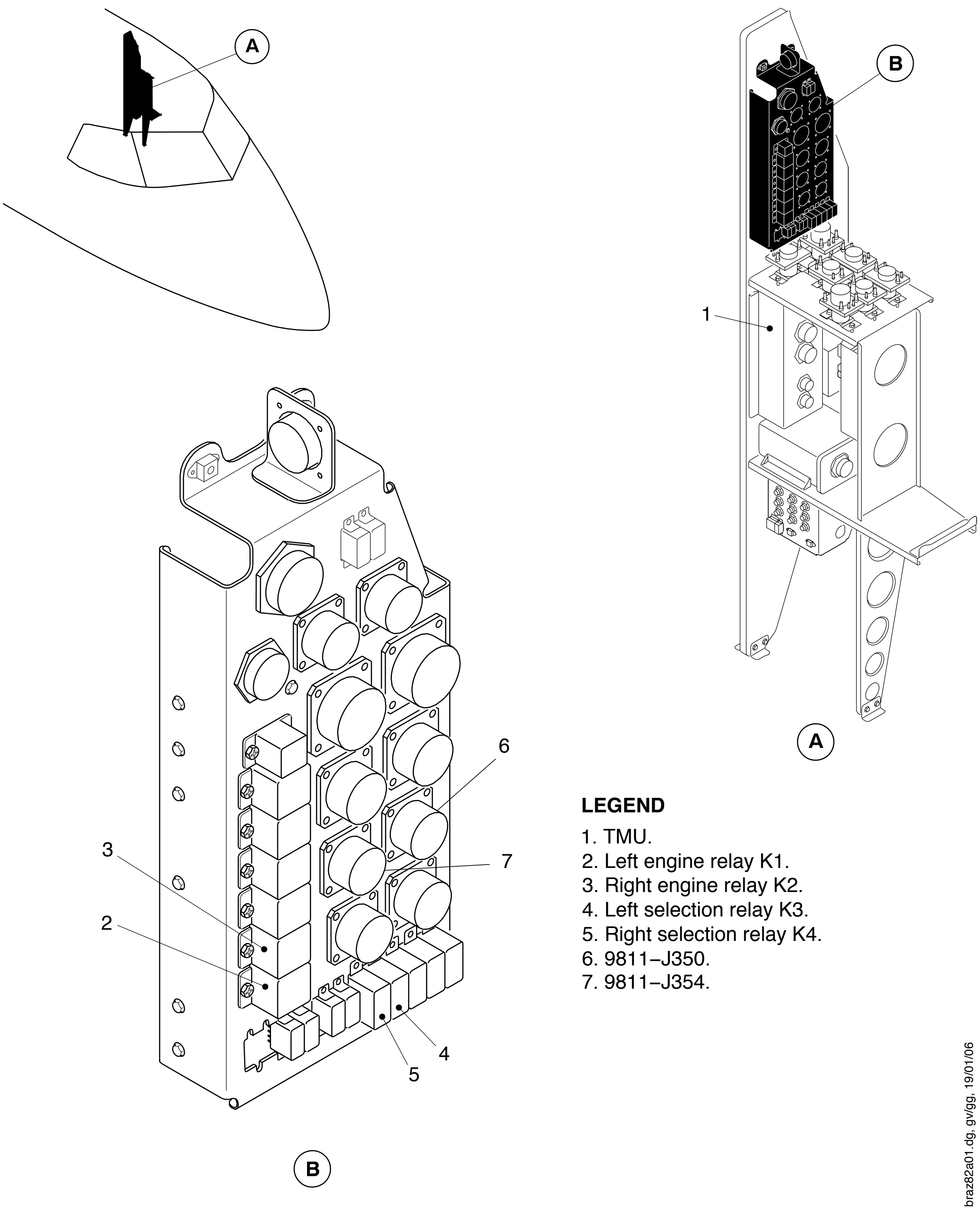
Operational Check of the Intake Adapter Heater
Figure 501



Operational Check of the Intake Adapter Heater
Figure 502 (Sheet 1 of 2)



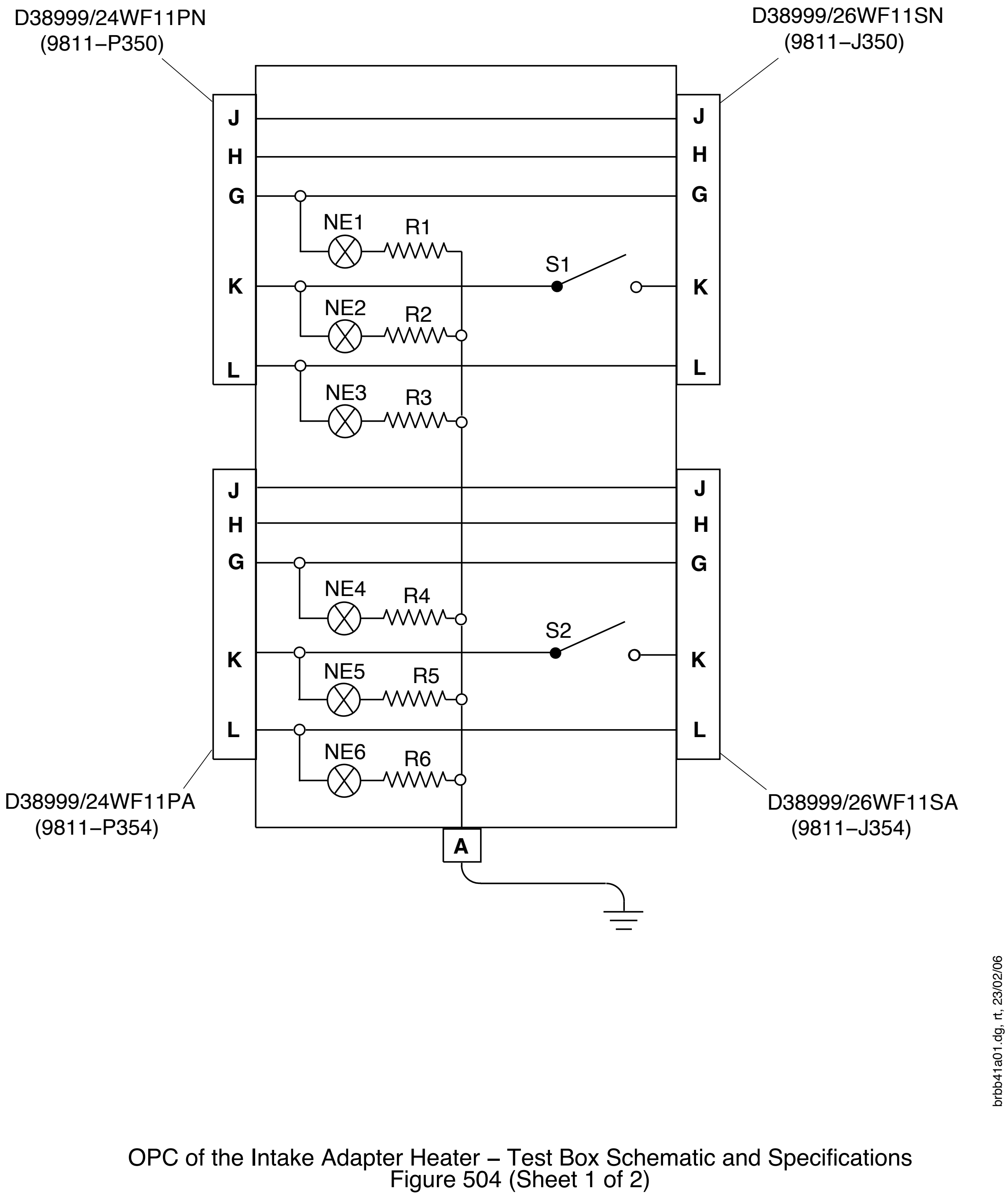
Operational Check of the Intake Adapter Heater
Figure 502 (Sheet 2 of 2)



LEGEND

1. TMU.
2. Left engine relay K1.
3. Right engine relay K2.
4. Left selection relay K3.
5. Right selection relay K4.
6. 9811–J350.
7. 9811–J354.

Operational Check of the Intake Adapter Heater
Figure 503



OPC of the Intake Adapter Heater – Test Box Schematic and Specifications
Figure 504 (Sheet 1 of 2)

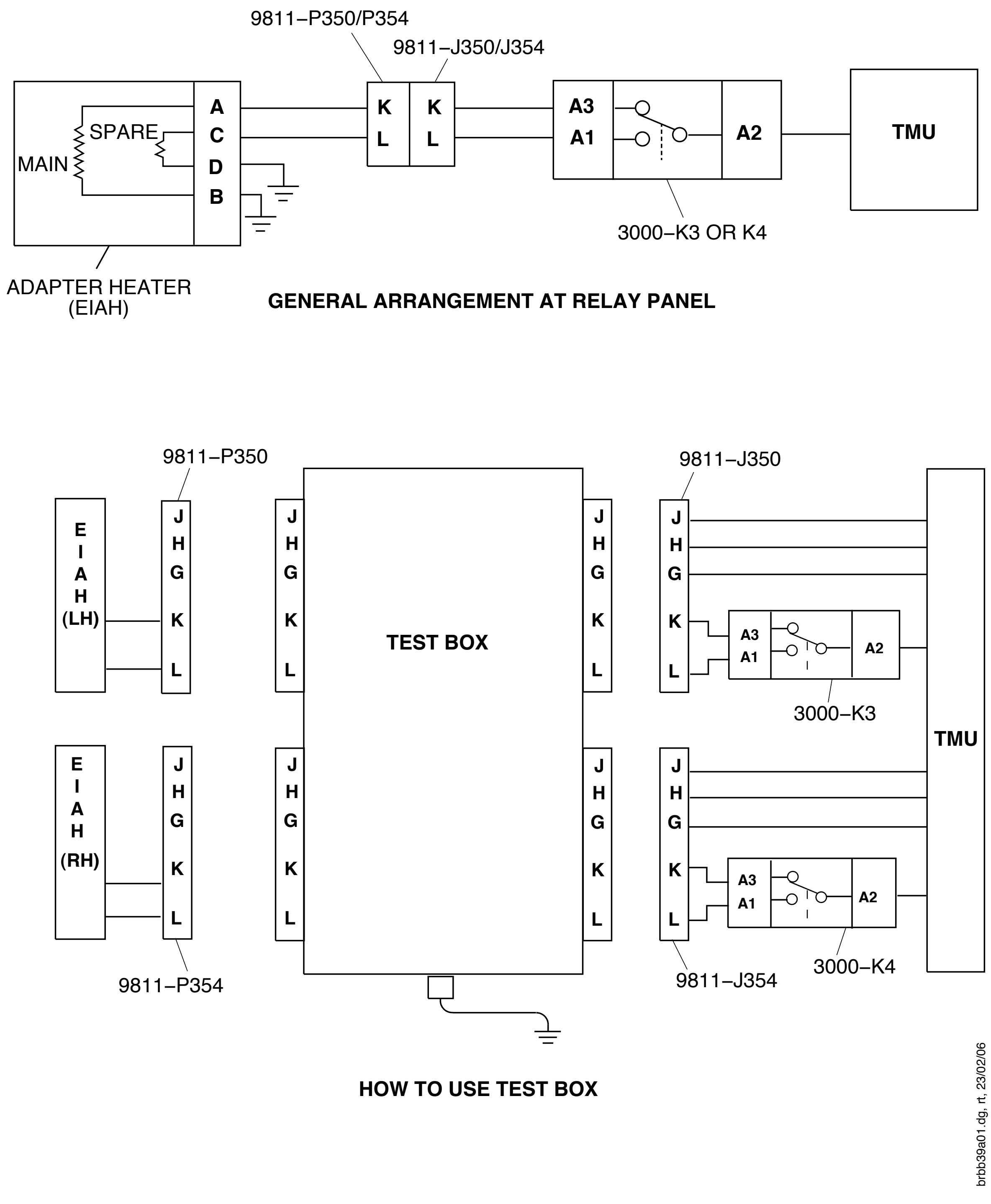
QTY	DESCRIPTION	PART NO./COMMENTS
1	Test Box	Local Manufacture
6	Neon Light With Resistor (Ne1 – Ne6)	Commercialy Available
6	100 K Ohm Resistor (R1 – R6)	Commercialy Available
2	SPST Switch (S1, S2)	Commercialy Available
1	Connector, Jam Nut Style (9811–P350)	D38999/24WF11PN
1	Connector, Jam Nut Style (9811–P354)	D38999/24WF11PA
1	Relay Panel Connector (9811–J350)	D38999/26WF11SN
1	Relay Panel Connector (9811–J354)	D38999/26WF11SA
2	Backshell	M85049/38S19N
a/r	Wire (16 gauge)	Commercialy Available
1	Banana Jack (Output A)	Commercialy Available
1	Alligator Clip	Commercialy Available

NOTES:

1. Test box to be connected inline between connectors 9811–P350/354 and 9811–J350/354 on the Relay Panel behind the wardrobe.
2. All components to be rated for 115 VAC.
3. Neon lights NE1, NE4 are for indication of 115 VAC power available to the system (depends on Relay K1/K2 position).
4. Neon lights NE2, NE5 are for indication of 115 VAC power through main heater element (depends on Relay K3/K4 position).
5. Neon lights NE3, NE6 are for indication of 115 VAC power through spare heater element (depends on Relay K3/K4 position)
6. Output A (banana jack) to be connected to a local ground by wire and alligator clip.

brbb41a02.dg, kmw, jun06/2008

OPC of the Intake Adapter Heater – Test Box Schematic and Specifications
Figure 504 (Sheet 2 of 2)



OPC of the Intake Adapter Heater − Installation of the Test Box
Figure 505

PITOT AND STATIC ANTI–ICING SYSTEM – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–30–00–710–801
Operational Test of the Pitot and Static Anti–icing System

1. General

   A. The maintenance procedure that follows is for the operational test of the pitot and static anti–icing system. The control for this system is located on the ice and rain protection panel. The pitot and static probe heaters are located in the probes. The probes are located on the external surface of the nose section of the fuselage. These procedures are applicable for systems No.1, No.2, and STBY. The procedure for each system is given.

2. Job Set–Up Information

Subtask 30–30–00–946–021

   A. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–910–801	Electrical/Electronic Safety Precautions
TASK 24–00–00–861–801	Energize Electrical System
TASK 24–00–00–861–802	De–energize Electrical System

3. Job Set–Up

Subtask 30–30–00–910–002

   A. Obey all the electrical/electronic safety precautions (Refer to TASK 24–00–00–910–801).

Subtask 30–30–00–861–023

   B. De–energize electrical system (Refer to TASK 24–00–00–861–802).

Subtask 30–30–00–865–022

   C. Open and tag the circuit breakers that follow:

CB PANEL AND CB NO	NAME
RIGHT DC ESSENTIAL, L7	PITOT HTR STBY
115VAC VAR FREQ, LEFT BUS	PITOT HEAT 1
115VAC VAR FREQ, RIGHT BUS	PITOT HEAT 2

Subtask 30–30–00–860–022

D. On the ice and rain protection panel, set the switches as follows:

(1) PITOT/STATIC PORT 1 to the OFF position.

(2) PITOT/STATIC PORT 2 to the OFF position.

(3) PITOT/STATIC PORT STBY to the OFF position.

Subtask 30–30–00–910–011

E. Remove the covers from the pitot static probes.

Subtask 30–30–00–861–025

F. Energize the electrical system (Refer to TASK 24–00–00–861–801).

Subtask 30–30–00–865–021

G. Remove the tags and close the circuit breakers that follow:

CB PANEL AND CB NO	NAME
LEFT DC, A7	DE–ICE CONT
LEFT DC, M5	DE–ICE CONT
RIGHT DC, S5	DE–ICE CONT

4. Procedure

Subtask 30–30–00–710–021

WARNING: THE PITOT/STATIC PROBES BECOME HOT VERY QUICKLY WHEN THE HEATER IS ON. THE HOT PROBES CAN CAUSE INJURY TO PERSONNEL.

CAUTION: THE LIMIT FOR THE PITOT/STATIC HEATER IS 5 MINUTES. DAMAGE TO THE HEATER WILL OCCUR IF OPERATED FOR MORE THAN THE 5 MINUTE LIMIT.

A. Do the operational test of the Pitot/Static Anti–icing System No.1 as follows:

(1) Make sure the circuit breakers for the lights that follow are closed:

(a) CAUT/WRN LTS 1

(b) CAUT/WRN LTS 2

(2) On the CAUTION LIGHTS panel, make sure that the lights that follow come on:

(a) PITOT HEAT 1

(b) PITOT HEAT 2

(c) PITOT HEAT STBY

(3) Remove the tag and close the PITOT HEAT 1 circuit breaker.

(4) On the Caution Lights panel, make sure that the lights that follow stay on:

(a) PITOT HEAT 1

(b) PITOT HEAT 2

(c) PITOT HEAT STBY

(5) On the ICE AND RAIN PROTECTION panel, set the PITOT/STATIC 1 switch to ON.

(a) Make sure that the PITOT STATIC PROBE 1 becomes warm

(6) On the Caution Lights panel, make sure that the PITOT HEAT 1 light goes out.

(7) Open the PITOT HEAT 1 circuit breaker.

(a) Make sure the PITOT HEAT light comes on

(8) On the ICE AND RAIN PROTECTION panel, set the PITOT STATIC 1 switch to OFF.

(9) On the CAUTION LIGHTS panel, make sure that the PITOT HEAT 1 light stays on.

B. Do the operational test of the pitot/static anti–icing system No. 2 as follows:

(1) Make sure the circuit breakers for the lights that follow are closed:

(a) CAUT/WRN LTS 1

(b) CAUT/WRN LTS 2

(2) On the CAUTION LIGHTS panel, make sure that the lights that follow come on:

(a) PITOT HEAT 1

(b) PITOT HEAT 2

(c) PITOT HEAT STBY

(3) Remove the tag and close the PITOT HEAT 2 circuit breaker.

(4) On the CAUTION LIGHTS panel, make sure that the lights that follow stay on:

(a) PITOT HEAT 1

(b) PITOT HEAT 2

(c) PITOT HEAT STBY

(5) On the ICE AND RAIN PROTECTION panel, set the PITOT STATIC 2 switch to ON.

(a) Make sure that the pitot static 2 probe becomes warm

(6) On the CAUTION LIGHTS panel, make sure that the PITOT HEAT 2 light goes off.

(7) Open the PITOT HEAT 2 circuit breaker.

(8) On the CAUTION LIGHTS panel, make sure that the PITOT HEAT 2 light comes on.

(9) On the ICE AND RAIN PROTECTION panel, set the PITOT STATIC 2 switch to OFF.

(10) On the CAUTION LIGHTS panel, make sure the PITOT HEAT 2 light stays on.

C. Do the operational test of the pitot /static anticing STBY system as follows:

(1) Make sure the circuit breakers for the lights that follow are closed:

(a) CAUT/WRN LTS 1

(b) CAUT/WRN LTS 2

(2) On the CAUTION LIGHTS panel, make sure that the lights that follow come on:

(a) PITOT HEAT 1

(b) PITOT HEAT 2

(c) PITOT HEAT STBY

(3) Remove the tag and close the PITOT HTR STBY circuit breaker.

(4) On the ICE AND RAIN PROTECTION panel, set the PITOT/STATIC STBY switch to ON.

(a) PITOT HEAT STBY

(5) On the CAUTION LIGHTS panel, make sure that the PITOT STBY light goes out.

(6) On the ICE and RAIN PROTECTION panel, set the PITOT STATIC STBY switch to OFF.

(7) On the CAUTION LIGHTS panel, make sure that the PITOT STBY lights are on.

5. Close Out

Subtask 30–30–00–861–026

A. De–energize the electrical system (Refer to TASK 24–00–00–861–802).

Subtask 30–30–00–865–023

B. Remove the tags and close the circuit breakers that follow:

CB PANEL AND CB NO	NAME
RIGHT DC ESSENTIAL, L7	PITOT HTR STBY
115VAC VAR FREQ, LEFT BUS	PITOT HEAT 1
115VAC VAR FREQ, RIGHT BUS	PITOT HEAT 2

Subtask 30–30–00–960–021

C. Make sure the Pitot/Static probes are cool before fitting the covers. Fit the covers.

Subtask 30–30–00–941–021

D. Remove all tools, equipment, and unwanted materials from the work area.

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WINDSHIELD AND PILOT'S WINDOWS ANTI–ICING SYSTEM – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–41–00–710–801
Operational Test of the Pilot Windshield Anti–icing System

1. General

   A. The maintenance procedure that follows is for the operational test of the Pilot Windshield Anti–icing System. Do this test to make sure that the pilot windshield anti–icing system operates correctly. This test only applies when the outside air temperature is less than 21°C (70°F) and the aircraft is not in direct sunlight.

   NOTE: If the outside air temperature (OAT) is more than 21°C (70°F) cool down the Pilot Windshield (Refer to TASK 56–10–01–880–801

2. Job Set–Up Information

Subtask 30–41–00–946–021

   A. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize Electrical System
TASK 24–00–00–861–802	De–energize Electrical System
TASK 56–10–01–880–801	Cooling of the Pilot and Copilot Windshields

3. Job Set–Up

Subtask 30–41–00–861–021

   A. Energize the electrical system (Refer to TASK 24–00–00–861–801).

   NOTE: Make sure AC power is available and connected to the busses before you start the test.

Subtask 30–41–00–865–027

   B. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT DC (ESSENTIAL), M5	DE–ICE CONT
LEFT DC (MAIN), A7	DE–ICE CONT
LEFT DC (SECONDARY), R7	PLT WS/HT CONT
RIGHT DC (SECONDARY), B6	COPLT–WS PLT–WDO HT CONT
RIGHT DC (SECONDARY), N5	PLT WDO HT RLY

CB PANEL AND CB NO	NAME
115V AC VARIABLE FREQUENCY,	L WSHLD HT

4. Procedure

Subtask 30–41–00–710–021

A. Do an operational test of the Pilot Windshield Anti–icing System as follows :

(1) On the overhead panel, set the WINDSHIELD HEAT selector to the WARM UP position.

(a) After 30 seconds, make sure there are no caution lights.

NOTE: Remove all jewelry from one of your hands.

(2) Use the bare hand to feel the windshield.

(a) Make sure windshield is cool to the touch.

(3) Set the WINDSHIELD HEAT selector to the NORM position.

(a) After 5 minutes, make sure there is an increase in the windshield temperature with the bare hand.

(b) Make sure there are no caution lights.

(4) Set the WINDSHIELD HEAT selector to the OFF position.

5. Close Out

Subtask 30–41–00–861–023

A. De–energize the electrical system (Refer to TASK 24–00–00–861–802).

Subtask 30–41–00–941–021

B. Remove all tools, equipment, and unwanted materials from the work area.

**ON A/C ALL

TASK 30–41–00–710–802
Operational Test of the Copilot Windshield and Pilot Side Window Anti–icing System

1. General

   A. The maintenance procedure that follows is for the operational test of the Copilot Windshield and Pilot Side Window Anti–icing System. Do this test to make sure that the system operates correctly. This test only applies when the outside air temperature (OAT) is less than 21°C (70°F) and the aircraft is not in direct sunlight.

   NOTE: If the outside air temperature (OAT) is more than 21°C (70°F), cool down the Pilot Side Window (Refer to TASK 56–10–06–880–801) and Copilot Windshield (Refer to TASK 56–10–01–880–801).

2. Job Set–Up Information

Subtask 30–41–00–946–022

   A. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize Electrical System
TASK 24–00–00–861–802	De–energize Electrical System
TASK 56–10–01–880–801	Cooling of the Pilot and Copilot Windshields
TASK 56–10–06–880–801	Cooling of the Pilot’s Side Window

3. Job Set–Up

Subtask 30–41–00–861–025

   A. Energize the electrical system (Refer to TASK 24–00–00–861–801).

   NOTE: Make sure AC power is available and connected to the busses before you start the test.

Subtask 30–41–00–865–028

   B. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT DC (ESSENTIAL), M5	DE–ICE CONT
LEFT DC (MAIN), A7	DE–ICE CONT
LEFT DC (SECONDARY), R7	PLT WS/HT CONT
RIGHT DC (SECONDARY), B6	COPLT–WS PLT–WDO HT CONT

CB PANEL AND CB NO	NAME
RIGHT DC (SECONDARY), N5	PLT WDO HT RLY
115V AC VARIABLE FREQUENCY,	L WDO HT
115V AC VARIABLE FREQUENCY,	R WSHLD HT

4. Procedure

Subtask 30–41–00–710–022

A. Do an operational test of the Copilot Windshield and Pilot Side Window Anti–icing System as follows:

(1) On the overhead panel, set the WINDSHIELD HEAT selector to the WARM UP position.

(a) After 30 seconds, make sure there are no caution lights.

NOTE: Remove all jewelry from one of your hands.

(2) Use the bare hand to feel the windshield.

(a) Make sure windshield is cool to the touch.

(3) Set the WINDSHIELD HEAT selector to the NORM position.

(a) After 5 minutes, make sure there is an increase in the windshield temperature with the bare hand.

(b) Make sure there are no caution lights.

(4) Set the WINDSHIELD HEAT selector to the OFF position.

(5) Use the bare hand to feel the pilot side window.

(a) Make sure the pilot side window is cool to the touch.

(6) On the overhead panel, set the PLT SIDE WDO/HT switch to the ON position.

(a) After 30 seconds, make sure there is an increase in the side window temperature with the bare hand.

(b) Make sure the SIDE WDO HOT caution light does not come on.

(7) Set the PLT SIDE WDO/HT switch to the OFF position.

5. Close Out

Subtask 30–41–00–861–027

A. De–energize the electrical system (Refer to TASK 24–00–00–861–802).

Subtask 30–41–00–941–022

B. Remove all tools, equipment, and unwanted materials from the work area.

WINDSHIELD WIPER SYSTEM – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–42–00–710–801
Operational Test of the Windshield Wiper System

1. General

   A. The maintenance procedure that follows is for the operational test of the windshield wiper system (written as the "wiper system" in this procedure). The wiper system control switch is installed on the WINDSHIELD control panel. The windshield control is installed in the flight compartment on the overhead console below the ICE PROTECTION panel. An alternative windshield wiper switch is installed on the pilot's side console.

2. Job Set–Up Information

Subtask 30–42–00–946–024

   A. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize Electrical System
TASK 24–00–00–861–802	De–energize Electrical System
TASK 24–00–00–910–801	Electrical/Electronic Safety Precautions

Subtask 30–42–00–865–022

   B. Make sure that the circuit breakers that follow are closed:

CB PANEL AND CB NO	NAME
LEFT SECONDARY, S7	PLT W/S WIPER
RIGHT SECONDARY, A6	COPLT W/S WIPER

3. Job Set–Up

Subtask 30–42–00–910–021

   A. Obey all the electrical/electronic safety precautions (Refer to TASK 24–00–00–910–801).

Subtask 30–42–00–861–005

   B. Energize the electrical system (Refer to TASK 24–00–00–861–801).

4. Procedure

**ON A/C 4001, 4006, 4008–4009, 4012–4015, 4018–4027 and ON A/C 4010, 4028–4029, 4033–4036, 4038–4039 Pre SB84–30–03

Subtask 30–42–00–710–028

Refer to Figure 501.

CAUTION: MAKE SURE THAT YOU APPLY WATER CONTINUOUSLY TO THE WINDSHIELD WHEN YOU OPERATE THE WINDSHIELD WIPERS. IF YOU DO NOT DO THIS, DAMAGE TO THE WINDSHIELD SURFACE CAN OCCUR.

A. Do the operational test of the wiper system as follows:

NOTE: Operate the wipers for a maximum of 5 seconds in each test.

(1) On the WINDSHIELD panel, set the WIPER switch (1) to the LOW position.

(a) Make sure that the wiper blades operate correctly at the low speed.

NOTE: In the LOW position, the wiper speed is 160 to 200 strokes a minute.

(2) Set the WIPER switch (1) to the HIGH position.

(a) Make sure that the wiper blades operate correctly at the high speed.

NOTE: In the HIGH position, the wiper speed is 220 to 280 strokes a minute.

(b) Make sure that the wiper blades clean the windshield.

(3) Set the WIPER switch (1) to the OFF position when the wiper is in the middle of travel.

(a) Make sure that the wipers stop travel.

(4) Set and hold the WIPER switch (1) in the PARK position.

(a) Make sure that the wiper blades align in the PARK position.

(b) Make sure that the center of each wiper blade is 1.65 to 2.15 in. (42 to 55 mm) below the bottom edge of the windshield.

(5) Release the WIPER switch (1) and make sure that the switch returns to the OFF position.

(6) On the pilot’s side console, push the ALTERNATE PILOT WIPER switch (2).

(a) Make sure that the pilot’s wiper operates in the high mode.

(b) Make sure that the green ON light comes on in the ALTERNATE PILOT WIPER switch (2) on the pilot’s side console.

(7) On the pilot’s side console, push the ALTERNATE PILOT WIPER switch (2) again.

(a) Make sure that the pilot’s wiper stops.

(b) Make sure that the green ON light goes off in the ALTERNATE PILOT WIPER switch (2) on the pilot’s side console.

**ON A/C 4003–4004, 4011, 4016–4017, 4040–4140, 4142–4199, 4201–4999 and ON A/C 4010, 4028–4039 Post SB84–30–03

Subtask 30–42–00–710–001

Refer to Figure 502.

CAUTION: MAKE SURE THAT YOU APPLY WATER CONTINUOUSLY TO THE WINDSHIELD WHEN YOU OPERATE THE WINDSHIELD WIPERS. IF YOU DO NOT DO THIS, DAMAGE TO THE WINDSHIELD SURFACE CAN OCCUR.

B. Do the operational test of the wiper system as follows:

NOTE: Operate the wipers for a maximum of 5 seconds in each test.

(1) On the WINDSHIELD panel, set the WIPER switch (1) to the LOW position.

(a) Make sure that the wiper blades operate correctly at the low speed.

NOTE: In the LOW position, the wiper speed is 160 to 200 strokes a minute.

(2) Set the WIPER switch (1) to the HIGH position.

(a) Make sure that the wiper blades operate correctly at the high speed.

NOTE: In the HIGH position, the wiper speed is 220 to 280 strokes a minute.

(b) Make sure that the wiper blades clean the windshield.

(3) Set the WIPER switch (1) to the OFF position when the wiper is in the middle of travel.

(a) Make sure that the wipers stop travel.

(4) Set and hold the WIPER switch (1) in the PARK position.

(a) Make sure that the wiper blades align in the PARK position.

(b) Make sure that the wiper blade stays on the windshield plate below the bottom of the windshield seal.

(5) Release the WIPER switch (1) and make sure that the switch returns to the OFF position.

(6) On the pilot's side console, push the ALTERNATE PILOT WIPER switch (2).

(a) Make sure that the pilot's wiper operates in the high mode.

(b) Make sure that the green ON light comes on in the ALTERNATE PILOT WIPER switch (2) on the pilot's side console.

(7) On the pilot's side console, push the ALTERNATE PILOT WIPER switch (2) again.

(a) Make sure that the pilot's wiper stops.

(b) Make sure that the green ON light goes off in the ALTERNATE PILOT WIPER switch (2) on the pilot's side console.

**ON A/C ALL

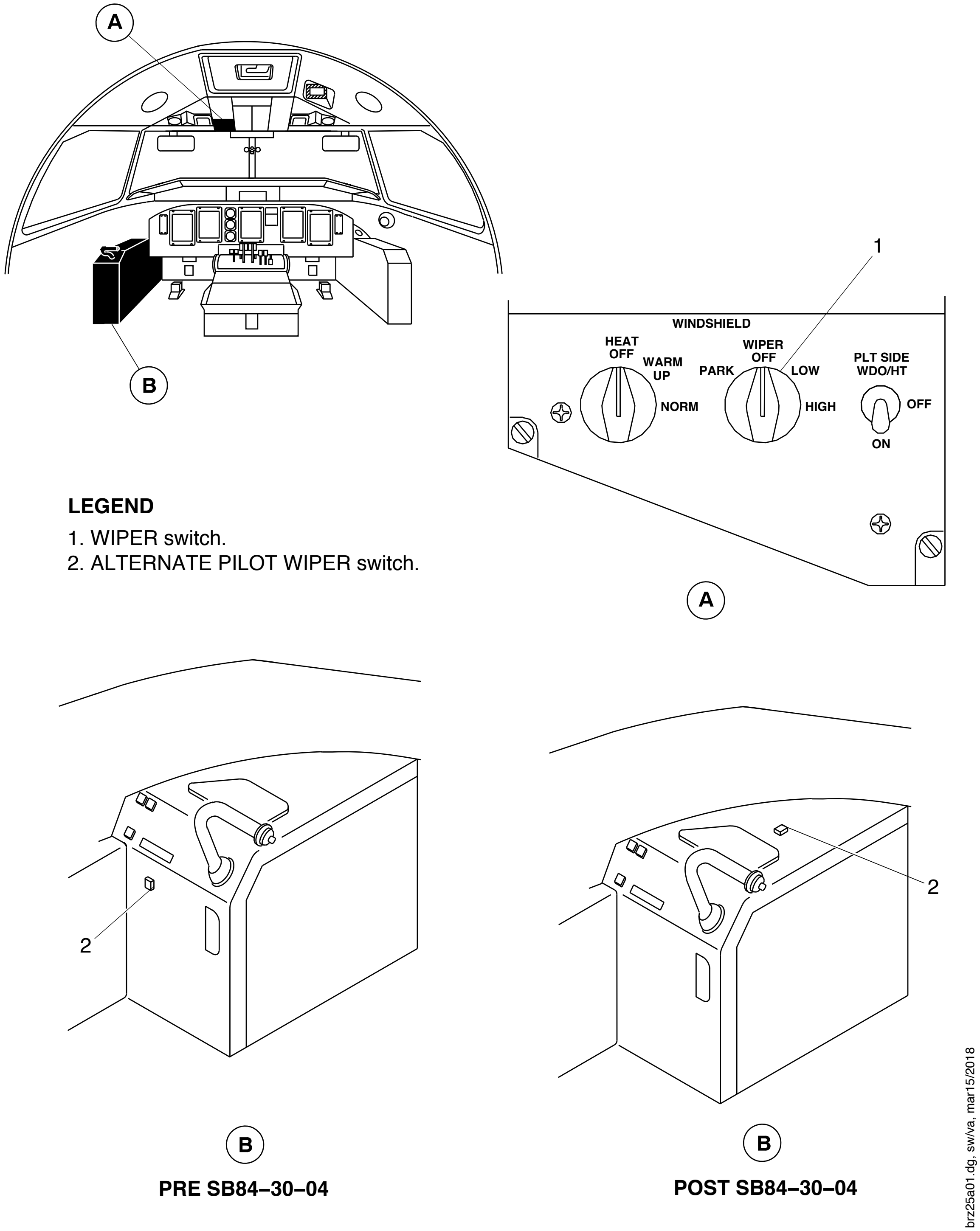
5. Close Out

Subtask 30–42–00–861–007

A. De–energize the aircraft electrical system (Refer to TASK 24–00–00–861–802).

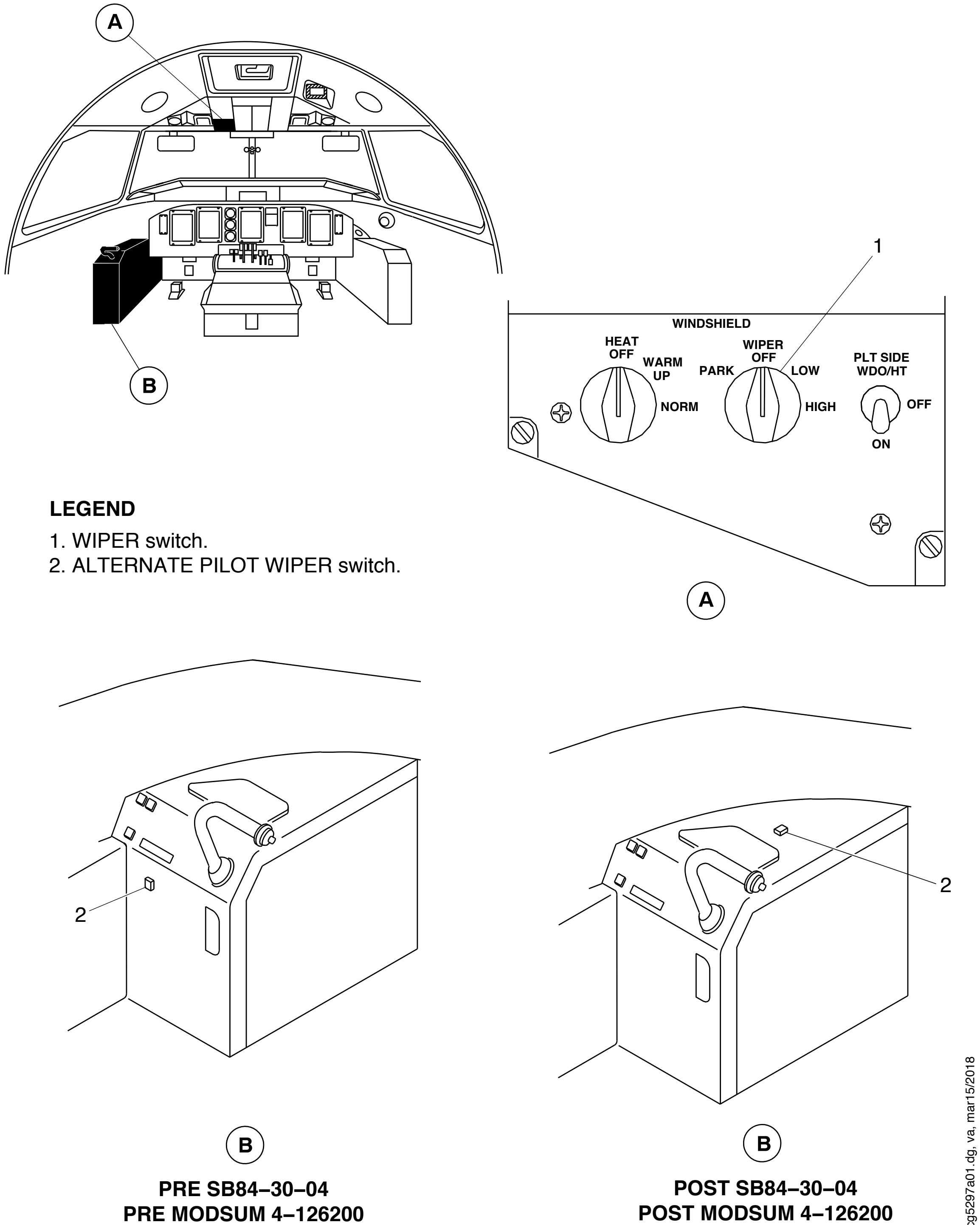
Subtask 30–42–00–940–021

B. Remove all tools, equipment, and unwanted materials from the work area.



LEGEND
1. WIPER switch.
2. ALTERNATE PILOT WIPER switch.

Windshield Wiper System – Operational Test
Figure 501



Windshield Wiper System – Operational Test
Figure 502

ELECTRICAL ANTI–ICING SYSTEMS – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–51–00–710–801
Operational Tests of the IRPS Component Heaters

1. General

   A. The maintenance procedure that follows is for the operational test of the Ice And Rain Protection System (IRPS) component heaters.

2. Job Set–Up Information

Subtask 30–51–00–944–001

   A. Consumable Material

      (1) 13–04 Mist, Freeze

Subtask 30–51–00–946–001

   B. Reference Information

REFERENCE	DESIGNATION
TASK 24–00–00–861–801	Energize the Electrical System
TASK 24–00–00–910–801	Electrical/Electronic Safety Precautions

3. Job Set–Up

Subtask 30–51–00–910–001

WARNING: OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

   A. Obey all the electrical/electronic safety precautions (Refer to TASK 24–00–00–910–801).

4. Procedure

Subtask 30–51–00–710–001

   A. Do the operational test of the IRPS component heaters as follows:

      (1) Make sure that the AIRFRAME MODE SELECT switch on the ice and rain protection panel is in the OFF position.

      (2) Make sure that the Static Air Temperature (SAT) shows more than 48.2 °F (9 °C) on Engine System Indication Display (ESID).

(3) Open the circuit breakers that follow:

Table 1 – IRPS Circuit Breakers

CB PANEL	CB NO	NAME
LEFT DC	R8	AIRFRAME DEICE CV/DRAIN HTRS
LEFT DC	P7	DR SEAL CV HTR RESTRICT
LEFT DC	P8	AIRFRAME DEICE V/HTR 1 AND 3
LEFT DC	Q8	AIRFRAME DEICE V/HTR 5
LEFT DC	M5	DEICE CONT
LEFT DC	A7	DEICE CONT
RIGHT DC	B8	AIRFRAME DEICE CV DRN HTRS
RIGHT DC	C8	AIRFRAME DEICE V/HTR #2 & #4
RIGHT DC	D8	AIRFRAME DEICE V/HTR #6
RIGHT DC	S5	DEICE CONT
RIGHT DC	S6	AFR DEICE BOOT LTS

(4) Disconnect the connectors from the components that follow:

Table 2 – DDV Connectors

COMPONENT	CONNECTOR IDENT	LOCATION
DDV 1	3000–P24	OUTBD LEFT WING
DDV 2	3000–P27	OUTBD RIGHT WING
DDV 3	3000–P25	OUTBD LEFT WING
DDV 4	3000–P28	OUTBD RIGHT WING
DDV 5	3000–P26	REAR FUSE
DDV 6	3000–P29	REAR FUSE

(5) Energize the aircraft electrical system (Refer to TASK 24–00–00–861–801).

(6) Close the circuit breakers that follow:

Table 3 – IRPS Circuit Breakers

CB PANEL	CB NO	NAME
LEFT DC	R8	AIRFRAME DEICE CV/DRAIN HTRS
LEFT DC	P7	DR SEAL CV HTR RESTRICT
LEFT DC	P8	AIRFRAME DEICE V/HTR 1 AND 3
LEFT DC	Q8	AIRFRAME DEICE V/HTR 5
LEFT DC	M5	DEICE CONT
LEFT DC	A7	DEICE CONT

Table 3 – IRPS Circuit Breakers

CB PANEL	CB NO	NAME
RIGHT DC	B8	AIRFRAME DEICE CV DRN HTRS
RIGHT DC	C8	AIRFRAME DEICE V/HTR #2 & #4
RIGHT DC	D8	AIRFRAME DEICE V/HTR #6
RIGHT DC	S5	DEICE CONT
RIGHT DC	S6	AFR DEICE BOOT LTS

(7) Make sure that the DEICE TIMER caution light on the Caution And Warning Panel (CAWP) does not come on.

(8) Make sure that SAT shows more than 48.2 °F (9 °C) on ESID.

(9) Put a test lamp on each DDV connector between the pins specified in the table that follows:

Table 4 – DDV Connectors

COMPONENT	CONNECTOR IDENT	PIN	LOCATION
DDV 1	3000–P24	4 AND 5	OUTBD LEFT WING
DDV 2	3000–P27	4 AND 5	OUTBD RIGHT WING
DDV 3	3000–P25	4 AND 5	OUTBD LEFT WING
DDV 4	3000–P28	4 AND 5	OUTBD RIGHT WING
DDV 5	3000–P26	4 AND 5	REAR FUSE
DDV 6	3000–P29	4 AND 5	REAR FUSE

(10) Make sure that the lamp on each DDV connector does not come on.

(11) Touch the components that follow with your bare hand and make sure that the temperature of the components does not increase.

Table 5 – IRPS Heater Components

COMPONENT	CONNECTOR IDENT	LOCATION
HCV 1	3000–P19	REAR FUSE
HCV 2	3000–P16	REAR FUSE
HCV 3	3000–P13	WING TO FUSE FAIRING
ADV 1	3000–P14	LEFT WING
ADV 2	3000–P15	RIGHT WING
ADV 3	3000–P17	REAR FUSE
ADV 4	3000–P18	REAR FUSE
RESTRICTOR	3000–P20	REAR FUSE

(12) Use freeze mist on the SAT sensor to decrease the SAT to less than 41 °F (5 °C).

(13) Make sure that the lamp on each DDV connector shown in Table 4 comes on.

(14) Touch the components shown in Table 5 with your bare hand and make sure that the temperature of the components increases.

(15) Open the circuit breaker that follows:

Table 6 – IRPS Circuit Breaker

CB PANEL	CB NO	NAME
LEFT DC	R8	AIRFRAME DEICE CV/DRAIN HTRS

(16) Make sure that the temperature of the check valve No. 2 (3000–P16) is decreased.

(17) Open the circuit breaker that follows:

Table 7 – IRPS Circuit Breaker

CB PANEL	CB NO	NAME
RIGHT DC	B8	AIRFRAME DEICE CV DRN HTRS

(18) Make sure that the temperature of the check valve No. 1 (3000–P19) is decreased.

(19) Open the circuit breaker that follows:

Table 8 – IRPS Circuit Breaker

CB PANEL	CB NO	NAME
LEFT DC	P7	DR SEAL CV HTR RESTRICT

(20) Make sure that the temperature of the restrictor (3000–P20) is decreased.

(21) Close the circuit breakers that follow:

Table 9 – IRPS Circuit Breakers

CB PANEL	CB NO	NAME
LEFT DC	R8	AIRFRAME DEICE CV/DRAIN HTRS
RIGHT DC	B8	AIRFRAME DEICE CV DRN HTRS
LEFT DC	P7	DR SEAL CV HTR RESTRICT

(22) Make sure that the temperature of the check valve No. 2 (3000–P16), check valve No. 1 (3000–P19) and restrictor (3000–P20) is increased.

(23) Increase the SAT to more than 48.2 °F (9 °C).

(24) Keep the temperature more than 48.2 °F (9 °C).

(25) Make sure that the lamp on each DDV connector shown in Table 4 goes off.

(26) Touch the components shown in Table 5 with your bare hand and make sure that the temperature is decreased.

(27) Open the circuit breakers that follow:

Table 10 – ADU Circuit Breakers

CB PANEL	CB NO	NAME
AVIONICS CB PANEL	A6	ADU 2
AVIONICS CB PANEL	A7	ADU1

(28) Make sure that the lamp on each DDV connector shown in Table 4 comes on.

(29) Touch the components shown in Table 5 with your bare hand and make sure that the temperature is increased.

(30) Close the circuit breakers that follow:

Table 11 – ADU Circuit Breakers

CB PANEL	CB NO	NAME
AVIONICS CB PANEL	A6	ADU 2
AVIONICS CB PANEL	A7	ADU 1

(31) Make sure that the lamp on each DDV connector shown in Table 4 goes off.

(32) Touch the components shown in Table 5 with your bare hand and make sure that the temperature is decreased.

(33) On the ice and rain protection panel, turn the AIRFRAME MODE SELECT switch to MANUAL position.

(34) Make sure that the lamp on each DDV connector shown in Table 4 comes on.

(35) Touch HCV 1 (3000–P19) and HCV 2 (3000–P16) with your bare hand make sure that the temperature is increased.

(36) Touch the components that follow with your bare hand and make sure that the temperature is not increased.

Table 12 – IRPS Heater Cmponents

COMPONENT	CONNECTOR IDENT	LOCATION
ADV 1	3000–P14	LEFT WING
ADV 2	3000–P15	RIGHT WING
ADV 3	3000–P17	REAR FUSE
ADV 4	3000–P18	REAR FUSE
HDV 3	3000–P13	WING TO FUSE FAIRING
RESTRICTOR	3000–P20	REAR FUSE

(37) On the ice and rain protection panel, turn the AIRFRAME MODE SELECT switch to OFF position.

(38) Make sure that the lamp on each DDV connector shown in Table 4 goes off.

(39) Touch the components shown in Table 5 with your bare hand and make sure that the temperature is not increased.

(40) Remove all the test lamps. Connect the connectors to their related components.

5. Close Out

Subtask 30–51–00–941–001

A. Remove all the tools, equipment, and unwanted materials from the work area.

CONTROL, DE–ICING TIMER MONITOR – ADJUSTMENT/TEST

**ON A/C ALL

TASK 30–61–06–710–801
Operational Test of the Timer Monitor Control Unit (TMCU)

1. General

   A. The maintenance procedure that follows is for the operational check of the new Timer Monitor Control Unit. These procedures are applicable to the No. 1 and No.2 systems. The procedures for the No. 1 system is given. The differences for the No. 2 system are identified.

2. Job Set–Up Information

Subtask 30–61–06–946–005

   A. Reference Information

REFERENCE	DESIGNATION
TASK 12–00–06–861–801	Connection of External DC Electrical Power
TASK 12–00–06–861–802	Removal of External DC Electrical Power

3. Procedure

Subtask 30–61–06–710–001

   A. Do the operational check of the new TMCU as follows:

      (1) Do the built–in test function for the timer monitor control unit (TMCU) as follows:

         (a) Connect the external electrical power (Refer to TASK 12–00–06–861–801).

            NOTE: With a 28V dc supply available, the TMCU will automatically do a self test. The two ‘PROP’ indicators (on the ice and rain panel) and the ‘PROP DEICE’ caution indicator will come on momentarily to show this. The TMCU self test takes approximately 12 seconds to complete. Operation of the TMCU is prevented for 45 seconds after this test to let the ARINC data bus to come on. No more functions can be selected for 60 seconds.

         (b) Make sure that the ‘PROP DEICE’ caution indicator does not comes on.

            NOTE: It is normal for the ‘PROP DEICE’ caution indicator to come on and then go off at the start of the test.

      (2) Remove the external electrical power (Refer to TASK 12–00–06–861–802).